



SPEAKING TRUTH TO POWER

ANTON
EBERHARD

A selection from two decades of newspaper op-eds



It has been sobering re-reading this selection of my op-eds, written mostly for non-expert audiences. Many of the issues I raised decades ago persist and constrain our ambitions to develop an energy system that powers economic and social development reliably, at affordable prices, and in an environmentally sustainable way. While some identified challenges have been resolved, many have not.

Nevertheless, the power sector in South Africa today is very different to that at the turn of the century. Private sector investments in the form of Independent Power Producers (IPPs) are now prevalent. The energy minister and the National Energy Regulator of South Africa are no longer gatekeepers to the power market; the requirement for approving generation licences has been removed. Effectively, there is now competition for the market in the form of tenders or auctions.

Attempts to conclude an expensive nuclear energy deal with Russia were thwarted, although the energy ministry still retains ambitions to revive a nuclear build programme. Efforts to procure coal power IPPs also failed when it became clear that finance institutions would not fund their construction. In contrast, solar and wind energy investments have multiplied and they will contribute a growing share of our electricity supply at competitive prices.

25 years after our first attempts to restructure South Africa's electricity sector, to break up Eskom, and to create a power market, we are on the cusp of fundamental reform. As long advocated, legislation is now finally before parliament that will take the Transmission and System Operator out of Eskom and will create a market operator that will provide a fair and transparent platform for contracting and trading power, thus accelerating investment in new power generation and transmission.

Unfortunately, the political establishment has been too slow in recognizing pending power shortages, despite ample warning and advice. Their responses to power cuts that are crippling economic growth have been dilatory. Power plans have not been updated, neither have they resulted in timely or adequate procurements of new power. Tariffs are still not cost-reflective and inhibit new investment. Critically, and for too long, the ruling party has resisted structural reforms that would increase competition and private sector participation.

In reflecting on my experience of undertaking energy policy research, advocacy and advice, it is clear that there have been times where we have enjoyed success, and others where we have not made sufficient headway. We need to understand better the broader political-economy surrounding policy development and implementation processes and the ways in which policy actors advance their political or material interests, how they access *de jure* or *de facto* power, or use formal institutions or informal networks, and the ways in which they employ ideology, ideas and narratives to advance their positions or to discredit opposing views.

These are the issues that my colleagues and I at the University of Cape Town's Power Futures Lab will explore in more depth as we continue to undertake energy research that informs effective evidence-based policy development and hastens the radical energy transition to a more sustainable future.

Anton Eberhard

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ESKOM RESTRUCTURING POSES POWERFUL DILEMMAS

Anton Eberhard & Fiona Thompson

BUSINESS DAY
18 October 2001

A study by investment strategy group BusinessMap has brought into sharp focus the range of options available to government in restructuring the electricity supply industry. Cheap, reliable power is one of South Africa's key competitive advantages in the global economy, but no matter what, wholesale electricity prices will rise.

Eskom can partly take credit for present low prices resulting from productivity improvements. But massive over-investment by Eskom in electricity generation plant in the 1970s and 1980s is the main reason. Little new investment has been required in recent years and so financing costs have fallen sharply. Eskom's debt to equity ratio has declined from 2.93 in 1986 to 0.63 in 2000.

This apartheid-era inheritance is running out. South Africa will soon run out of surplus generating capacity (between 2007 and 2011) and because of the long lead times in generation investments, decisions on new investments will need to be made soon. Generation capacity will have to more or less double in the next 25 years to meet estimated future demand. This will require an investment in new electricity generation plant of around R100bn over the period.

Electricity prices are below the level required to support new investment. Prices will rise no matter how the electricity industry is organised in the future. Two immediate challenges arise: first, how do we manage prices to a level which will support new investment without causing price shocks when capacity runs out and, second, how do we create an industry structure which will promote new investments most efficiently and at the least cost?

The introduction of competitive markets should improve efficiency in allocating capital and in operating plant and

hence lead to low costs. Low costs will lead to low prices if these benefits are passed to consumers. A revolution in the electricity industry internationally has demonstrated that competition is possible in the generation and supply of electricity. Wholesale and retail electricity prices, when deregulated, have fallen in many countries; recent examples are Germany and New Zealand.

Yet the BusinessMap study shows that a competitive market structure in South Africa might actually result in higher prices in the medium term than if the present regulated and integrated Eskom monopoly continues.

This is because the calculation of regulated prices is based on historical asset values and average costs in the industry. Much of the existing asset base is old and therefore heavily depreciated. Original asset values have also been eroded by inflation. The low asset values of older plant serve to keep average costs, and hence average prices, low. Once new investment takes place and the mix of assets changes from old to new assets, so average costs will tend to rise.

In a competitive market, prices are unregulated and essentially based on the interaction of supply and demand. Once surplus generating capacity has been eroded by increases in demand, prices will move quickly to around the cost for new entrants (i.e. the long-run marginal cost) and this attracts new investment.

In regulated markets, new investment in plant is planned, based on demand forecasts and consumers therefore bear the risks of mistakes. A key element of competitive markets is that since prices trigger new investments, investors bear the risks of this investment. This is the feature of competition that should lead to efficient investment decision-making.

The modelling in the study assumes international benchmark prices for new generation investments and that the operational efficiency gains in a competitive market are likely to be modest, given our low fuel costs and high productivity levels.

Two trends could bring regulated prices closer to competitive market prices. Firstly, the pricing methodology now used by the National Electricity Regulator could change to incorporate higher rates of return on an optimised replacement value of assets. In this case, regulated prices might approach those necessary to support new investment. Secondly, in the long run, unregulated competitive markets should lead to lower costs than regulated ones. Competitive pressures should keep operating costs low, and market risks should encourage investment at optimal levels. Ultimately this should lead to lower prices, but not necessarily during a "transition" period as older capacity is phased out and new capacity is phased in.

Estimated future electricity prices affect valuations of assets. The BusinessMap study indicates some remarkable results in the potential valuation of Eskom generating assets. Values vary between R24bn and R140bn, depending on market structure and the level of competition.

In unregulated markets prices will rise to a level until they provide an adequate return to incentivise new investment. At these prices, older fully depreciated assets still in operation will earn extremely high returns and consequently, towards the end of their operating lives will become very profitable. This in effect represents a transfer from consumers (paying the market price) to the owners of this low-cost, older plant.

Mechanisms to avoid this transfer have been quite common in other countries that have restructured their electricity

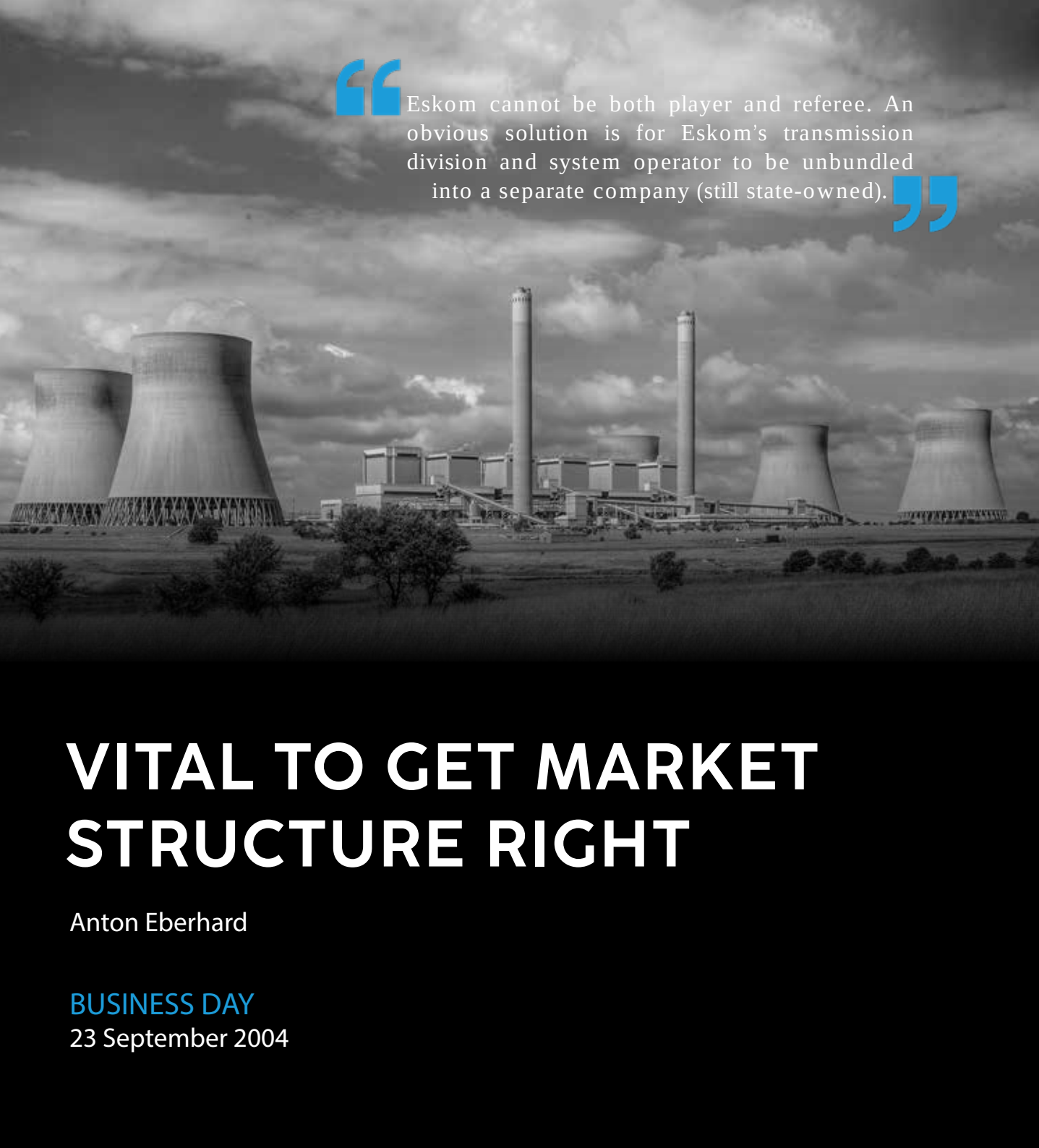
industries. They typically take the form of contracts between existing plant and power purchasers, vested at the time of reform and fixing prices based on a formula.

Thus South Africa might begin to move to an industry structure where market prices begin to signal the marginal cost of new generating plant and therefore promote efficient additional investment but in the transition period vested contracts can protect consumers from sharp price increases.

The BusinessMap study also indicates the scope for restructuring to meet other government objectives such as black economic empowerment, attracting foreign direct investment and realising privatisation receipts to reduce public debt. Lower interest payments on this debt could free additional fiscal resources for social investment in electrification as well as for restructuring local government finances to replace the municipal surcharge on electricity sales. In this way the upward pressure on electricity prices for consumers could be reduced.

The restructuring of Eskom presents government with opportunities to meet a wide spectrum of economic and social policy goals. Government will have to manage the restructuring process carefully though to maintain competitive electricity prices and at the same time lay the groundwork for efficient new investment in the industry.

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“Eskom cannot be both player and referee. An obvious solution is for Eskom's transmission division and system operator to be unbundled into a separate company (still state-owned).”

VITAL TO GET MARKET STRUCTURE RIGHT

Anton Eberhard

BUSINESS DAY
23 September 2004

South Africa's looming power shortages will probably be averted, but at what cost to the economy?

Eskom is starting to re-commission old mothballed coal-fired power stations. It now appears that government will authorise Eskom to procure further power generation capacity to meet peak electricity demand in the period 2006-2007.

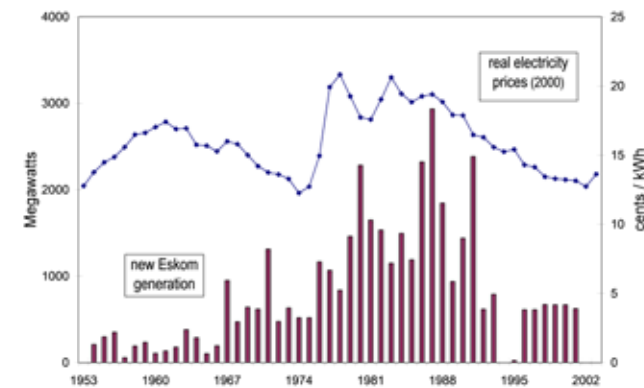
Meanwhile, the Department of Minerals and Energy has also initiated a private sector tender to procure new capacity in 2008.

South Africa's growing electricity demand will require capacity additions in each subsequent year and because of the long-lead times, many of the investment decisions need to be made soon. Over the next 10 years, approximately R100 billion will have to be committed to new power generation projects.

The magnitude of these new investments creates both opportunity and risk. If capital is allocated efficiently then the electricity industry will contribute to increased economic growth and competitiveness.

If poor decisions are made around market structure, technology and the timing of investments, then the resulting cost increases will be passed to consumers with negative consequences for economic performance and social welfare. An analysis of the performance of Eskom over the past 50 years indicates that this is exactly what happened during the last investment cycle.

The graph below shows that electricity consumers faced a real increase of 70% between 1974 and 1978 and high prices continued through the 1980s – because Eskom ordered power stations that were not needed. The graph also demonstrates that electricity prices are no cheaper today, in real terms, than they were in the early 1950s or mid-1970s. This implies that, despite massive new investment, Eskom has not achieved any lasting or dramatic efficiency improvements. And electricity prices are again increasing.



Source: Eskom Annual Reports and Statistical Yearbooks

One could argue that Eskom today is a different organisation. It's commercial management has improved since the 1980s. It operates as a company with a (government) shareholder agreement specifying performance expectations. And the National Electricity Regulator is progressively imposing a more rigorous cost-of-service regulatory regime.

Nevertheless, the basic institutional model of a publicly-owned monopoly industry means that risks are shifted disproportionately to consumers or taxpayers. And because of information asymmetries, the regulator will never have a perfect picture of costs and potential efficiency gains.

What are the alternatives? Two broad electricity market models have emerged over the past two decades. The first is competition for access to the electricity market. The private sector is encouraged to invest in new power generation capacity through competitive tenders or auctions. If well managed, competition can result in improved efficiencies in technology and investment choices.

The second electricity market model involves ongoing competition between electricity generators and suppliers

to dispatch and sell electricity to consumers. Competition is managed through a power exchange or bilateral markets or both, and risks may be managed through derivative financial markets.

What is now emerging in South Africa is a hybrid model. The incumbent, Eskom, which generates 96% of our electricity, has been given responsibility for immediate new investment. Eskom is also actively exploring a range of new options, including joint-venture investments. At the same time, government is proceeding with a tender for new private IPPs. Meanwhile, Cabinet decisions in 2001, and subsequent design work on a power exchange and bilateral contract electricity market, appear to be on hold.

Ministers now make speeches indicating that competition is not a priority and Eskom's unbundling or privatisation in the short term is unlikely. However, no new or comprehensive electricity policy has been announced officially.

As has been pointed out by analysts in this newspaper and elsewhere, the current situation creates a great deal of uncertainty for private investment – and uncertainty means higher risks and higher costs.

What is the sensible way forward? Clearly the first step is for government to spell-out the institutional and market structure within which new power investments will be made. If the preferred model for the next few years is for IPPs to come into the market – then the procurement of these multi-billion investments will need to be managed by an institution with robust financial, technical and contracting capability. Government departments and the regulator are not well positioned to fulfil this function.

This leaves Eskom as a possible candidate. However, Eskom cannot be both player and referee. An obvious solution is for Eskom's transmission division and system operator to be unbundled into a separate company (still state-owned).

A new, single-buyer office could be established within this company which would be responsible for system planning and capacity procurement.

This is a relatively uncomplicated, no-regrets option that could be accomplished within 18 months. In the medium to long-term, it makes sense anyway to separate potentially competitive elements of the industry (generation and supply) from non-competitive, natural monopoly components (transmission). Cabinet, in 2001, has already agreed to this. There is every reason now to proceed. It solves the immediate need for creating a robust, transparent and fair process for new generation procurement, while establishing a platform for possible competition in the future.

It is equally clear what NOT to do. It would be disastrous to consider privatising Eskom or selling a strategic equity stake (as was initially done with Telkom) without first unbundling Eskom and creating the conditions for competition. Privatisation within the existing industry structure would severely restrict government options for creating a more competitive structure for the industry in the future.

The above approach also has many potential advantages for Eskom. Its business model could shift from a defensive stance of trying to protect domestic market-share and obstructing market reform, to being a forward-looking company, confident that it can succeed in competitive markets, not only in South Africa, but also internationally in emerging markets.

Government is turning its attention to the large infrastructure industries that underpin our economy. It is committed to ensuring that these industries receive substantial new investment and that efficiencies are improved. Getting the structure of the electricity market right will be an important first step.

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POWER POLICY AT A CROSSROADS

Anton Eberhard

BUSINESS DAY
17 June 2005

“Let’s hope that sense will prevail and that cabinet and Parliament will consolidate the progress being made... on the rationalisation and regulation of the entire distribution industry in order to secure reliable and affordable electricity services for all.”

In the coming weeks, cabinet and parliament will make decisions with far-reaching consequences for the reliability and affordability of electricity services for all of us. These decisions will either lay a foundation for improved services or increase the risks of black-outs and rising prices as currently experienced in Johannesburg and many other municipalities around the country.

At the heart of these decisions is the role of municipalities in the distribution of electricity services. A cabinet memorandum is being prepared by the Department of Minerals and Energy that will report on progress with sorting out the festering problems in this sector and will make certain policy proposals. At the same time, Parliament will consider a new Electricity Regulation Bill. Both initiatives may take the wrong turn.

The solution to the country’s electricity distribution problems has been debated for over a decade. During this time municipal electricity providers have grown less reliable due to insufficient investment in the maintenance of municipal infrastructure – notwithstanding the existence of large trading surpluses due to rising consumer tariffs. The slow progress with industry restructuring has failed to achieve economies of scale and has led to a steady loss of skills and capacity.

Residents in Johannesburg now take it for granted that some traffic lights and areas of the city will be without power on a daily basis. City Power spokespersons urge residents, institutions and businesses to invest in back-up power supplies. They continue to remind us that many components of the electricity network are more than 30 years old - but they do not explain why there has been insufficient investment in renewing these assets.

This problem has deep structural roots. Municipalities face contradictory incentives in managing and regulating electricity distribution services. On the one hand municipal councillors face pressure to lower electricity prices, sometimes even below the cost of provision. On the other hand they have an incentive to maximize revenue from their electricity assets. Unfortunately surpluses have been diverted and investment in extending electricity assets has suffered at the expense of other capital priorities. Sadly, the conflict between these governance roles is seldom understood and has yet to be resolved.

So what should be done? Over the past 12 years, the ANC government has engaged all stakeholders and won approval that the electricity distribution assets and staff of all municipalities and Eskom should be consolidated into six Regional Electricity Distribution companies (REDs) that cover the entire country. The intention was that the REDs would be publicly-owned and would be regulated by the National Electricity Regulator in order to achieve an appropriate balance between financial viability, efficient operation and affordable tariffs for customers.

However, a draft cabinet memo now appears to take a different position altogether. Through the intervention of National Treasury officials the version that may be taken to Cabinet says that the future should be more of the same: i.e. a few more municipal entities like City Power in Johannesburg (plus the takeover of some Eskom distribution assets) in each of the five largest metro areas. And the remaining small municipalities scattered around the country will somehow be combined into a sixth RED – a proposal that negates the rationale for creating viable, efficient distribution companies.

At the same time, National Treasury officials have rewritten chapters in the draft Electricity Regulation Bill that effectively remove the National Electricity Regulator’s (NER) powers to regulate municipal electricity reticulation services, i.e. the majority of electricity consumers in South Africa! Are these proposals serious? Unfortunately, yes.

Perhaps there is insufficient understanding that electricity distribution is unlike other municipal services such as garbage removal or paving roads. Electricity investments are huge, often beyond the means of smaller municipalities. The industry demonstrates clear economies of scale – the key rationale behind the six RED model.

Elements of the electricity industry are natural monopolies and therefore need to be regulated to protect consumers. The NER has restrained Eskom’s monopoly pricing in the generation and transmission sectors – but the draft Electricity Regulation Bill will effectively prevent the NER from regulating the final distribution stage in the value chain. Upstream gains may therefore never reach the end consumer. Clearly, this does not make sense.

Why do officials from National Treasury appear to be acting against the national interest and sabotaging a decade’s worth of policy consensus? The answer appears to lie in their interpretation of the Constitutional provision that grants municipalities executive and administrative powers over electricity reticulation within their municipal areas. To the extent that municipalities represent local interests they have an important planning role in service delivery. They can set targets and monitor progress. But this does not mean that they have to actually operate electricity distribution businesses or

that they have exclusive rights to set tariffs and expropriate surpluses.

National Treasury officials are taking a fundamentalist view of the Constitution; one that gives municipalities unfettered powers over all aspects of electricity reticulation. This is a peculiar and myopic view. It ignores the problems explained above. It ignores the fact that most counties around the world do not allocate electricity distribution to municipalities (just as they do not allocate other national infrastructure services such as telecommunications). It is not even clear that these officials represent the view of their minister who is surely committed to a solution that guarantees reliable and competitively priced electricity services to all businesses and households.

Let’s hope that sense will prevail and that cabinet and Parliament will consolidate the progress being made by the Department of Minerals and Energy, the EDI Holdings Company and the National Electricity Regulator, who are committed to the rationalisation and regulation of the entire distribution industry in order to secure reliable and affordable electricity services for all.

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POWER CRISIS IDEAL TIME FOR RETHINK

Anton Eberhard

CAPE TIMES

2 May 2006

“This experience raises a number of important issues, including rethinking our electricity market structure, how to better manage investment and uncertainty, improved planning systems, tighter regulatory oversight, better management and the design of innovative programmes to achieve sustainable market transformation in energy efficiency – subjects perhaps for another article.”

As residents in the Western Cape are enjoined by Eskom spokespersons to avoid electricity blackouts by switching off electricity appliances and to wrap in blankets to keep warm, we might ask: *why do we, who pay our electricity bills, have to bear the cost of Eskom not providing a reliable service? It seems we lose both ways: either we switch off appliances this winter or Eskom will cut off our supply.*

Everyone- electricity consumers, and I would assume, Eskom's management and employees - would agree that this is unacceptable. We cannot, and should not, be resigned to blackouts. Eskom has a dominant market position. It generates 96 percent of our electricity. It is regulated by the National Energy Regulator of South Africa (NERSA) and its licence conditions require it to adhere to national quality of supply and customer service standards

NERSA has investigated six serious failures, between November 2005 and March 2006, in the two main power supplies to the Western Cape: the Koeberg nuclear power plant and the high voltage transmission line from the north. NERSA's report will soon be made public. It is likely to conclude that Eskom has transgressed its licence conditions. It will also raise serious concerns around the integrity and robustness of Eskom's operation and maintenance systems.

Eskom has known since Christmas 2005 (when the stray bolt caused catastrophic damage to the generator linked to one of the two Koeberg's nuclear reactors) that it would be short of about 400 megawatts (MW) of electricity supply this winter. Eskom has published on its website (www.eskom.co.za) an Electricity Recovery Plan for the Western Cape. In general it makes sense. The focus is on restoring power at Koeberg, improving the reliability of the electricity transmission line from the north, incentivizing energy conservation and efficiency by industry and consumers in the Western Cape and utilizing emergency back-up generators.

But what progress has been made in realising this recovery plan? Eskom (with the help of the Presidency) has been

successful in persuading the national electricity utility in France, EdF, to release the only spare generator rotor of its kind in the world. Our navy has transported it to Cape Town and it has arrived at Koeberg. Eskom has been repairing the other damaged part, the generator stator, and power should be generated by Unit 1 at Koeberg by the third week of May.

In the meantime, fuel is running out in the second nuclear reactor at Koeberg and power output is slowly deteriorating. As soon as the first unit is back in operation, Unit 2 will have to be taken out for two months for refuelling and maintenance. In other words, between now and the end of July we shall have only one of Koeberg's units producing power at any given time. To make matters worse, in April and May, the single functioning unit will be operating at reduced output. The transmission line from the north, plus the hydro-electric pumped storage schemes at Palmiet, in the Overberg, and Steenbras, behind Gordon's Bay, are unlikely to make up the gap between supply and demand this winter.

Eskom's recovery plan thus relies on persuading electricity consumers to switch to more efficient energy appliances and technologies or to generate their own power or to selectively switch off electricity consuming devices. They aim to persuade households to substitute compact fluorescent lights (CFLs) for less efficient incandescent light bulbs; to subsidize energy efficient technologies (such as geyser blankets or ceiling insulation); to get consumers to switch to other energy supplies (such as gas or solar water heating); to incentivize industrial energy efficiency and to compensate companies for self-generation.

If these plans are successful, then electricity blackouts will be avoided over the Western Cape. However, the Western Cape Stakeholder Status Report, dated 17-21 April and available on the Eskom website, indicates that only 132 MW of a targeted 270MW has been achieved in load reduction relief from industry through its demand market participation scheme (i.e. industry is being paid to switch off machines at certain times of the day). Eskom has also managed to contract 27

industrial customers to provide a total of 42MW of self-generation. No information is given on whether Eskom has managed to procure sufficient additional emergency power. (The targeted 66MW in its plan seems far to low). And half a million CFLs have been distributed, yielding a further 17MW of savings. Eskom is working hard to make up the anticipated shortfall, but it has a long way to go before it reaches the required 400MW.

While a range of options are suggested in Eskom's Electricity Recovery Plan, there is very little information on the incentives available to consumers for switching to more energy efficient technologies and appliances. I phoned local Eskom numbers to find out how I could switch to energy efficient lights, or access subsidies to install insulation around my hot water geyser, or put insulation in my ceiling to minimise space heating requirements, or install a solar water heater to reduce to the need for electricity water heating. None of the share-call operators had any relevant information. To their credit, they transferred my call to other Eskom contacts. After the fifth telephone call I was put in touch with someone who was running a door-to-door campaign on the Cape flats, offering free compact fluorescent light in exchange for old incandescent lamps. I asked whether a similar campaign was available in my suburb. The answer was no – but perhaps there might be an exchange scheme at my local supermarket. There is none. After a further three calls I was put in touch with a private company that had a contract with Eskom to implement a scheme for hot water geyser blankets. Again, I tried to find out how I could access this subsidy. Approved suppliers were still being identified. The scheme was still not fully operational. I tried further to see whether I could get a subsidy for ceiling insulation or a solar water heating. The answer was no.

We are at the end of April. The energy efficiency programmes are still not fully operational. Sufficient emergency power does not seem to be in place. Meanwhile, the power output of Unit 2 at Koeberg diminishes inexorably as its fuel runs out. Winter approaches.

This is not a trivial situation. The social and economic costs over the winter for the Western Cape are likely to be great. Residents will be cold. Those without gas or paraffin cookers will have uncooked meals. There will be insufficient light for reading and studying. The FA cup final on TV may be missed. Trains may not function. Traffic gridlock will worsen. Industries will lose millions again in idle production while continuing to have to pay fixed costs. And the reputation of the Western Cape as a desirable investment location will suffer further damage.

This experience raises a number of important issues, including rethinking our electricity market structure, how to better manage investment and uncertainty, improved planning systems, tighter regulatory oversight, better management and the design of innovative programmes to achieve sustainable market transformation in energy efficiency – subjects perhaps= for another article.

The Western Cape Provincial Government has set up an Energy Risk Management Committee. Thankfully, load-shedding is now being co-ordinated and there is much better information available from Eskom and the City of Cape Town on which areas will be affected and when. But we also need to track progress in restoring supply security and avoiding blackouts. And we need to know how we can access opportunities to move to more efficient and sustainable energy use patterns. There is nothing quite like unanticipated inconvenience or a supply crisis to focus minds and energy on policy changes and the initiation or adoption of new programmes. It's a marvellous opportunity to imbed long-lasting shifts to more sustainable electricity supply and use patterns in the Western Cape.

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“Eskom will hopefully learn the lessons from the earlier blackouts by improving its planning, operations and maintenance systems. An additional lesson for South Africa is the need to launch systematic market transformation initiatives.”

ENERGY SECURITY AND THE MARKET

Anton Eberhard

BUSINESS DAY

23 May 2006

Despite dire predictions of blackouts this winter in the Western Cape (including by this writer), the light have generally stayed on. By now forecasted electricity demand should have exceeded available supply capacity. In practice it has not. Eskom's electricity recovery plan is clearly making a difference.

Eskom's plan rests on four pillars. First, power has to be restored at the Koeberg nuclear plant, 30 kilometres north of Cape Town. Progress appears to be on track. Unit one has been repaired and is producing power. Unit two is now being refuelled and is scheduled to be back online by the end of July.

Second, Eskom has a program to minimize Cape transmission line failures by spray-washing pollution and replacing below-spec insulators.

Third, Eskom has implemented an unprecedented energy efficiency program. It has contracted a number of companies to go door-to-door, in mainly low income housing areas, to replace incandescent light bulbs with more efficient compact fluorescent lights (CFLs). Eskom has air-freighted 5 million CFLs from China. Already 1.5 million have been installed. Eskom is confident that the remaining stock will be distributed by the end of July.

Eskom has also purchased 140,000 hot water geyser blankets and is working through energy service companies to distribute these, free-of-charge, to mostly low and middle income households. It is also offering to swap-out electricity cookers for gas appliances.

As cold fronts sweep repeatedly across the Western Cape, power consumption rises and the demand/supply balance becomes tighter and tighter. The critical period is between 6.30 and 8.30 in the evening when households switch on

lights, cook their evening meals, heat their living spaces and utilise hot water. Yet actual demand during this period is now remarkably flat. Energy efficiency programs appear to be working.

Eskom is also targeting commercial, industrial and agricultural enterprises and is offering to pay the full cost of energy saving investments. Although there is a relatively high transaction cost to these interventions, significant savings have been achieved. Eskom is also paying industries to activate self-generation and fuel-switching options rather than taking a supply from Eskom.

Together, these energy efficiency programs have delivered tens of megawatts (MW) of savings. While further energy conservation gains will be made, it is, nevertheless, unlikely that they will fully mitigate the total predicted power shortfall of 360MW in June and July.

The fourth pillar of Eskom's electricity recovery plan thus involves a number of contingency programs. It is initiating a high-profile media programme, including direct television and radio alerts, which try to persuade electricity consumers to switch off electricity appliances during tight demand/supply periods.

Eskom has also contracted 66MW of emergency diesel-powered generators. The City of Cape Town is running its pumped storage generation units at Steenbras, above Gordon's Bay, and its Roggebaai gas turbines, at the Waterfront, during peak periods. And finally, Eskom is incentivizing "demand market participation" whereby industries are paid to switch-off electricity-using processes briefly during periods of emergency shortages. This program is probably the most successful of all – yielding more than 100 MW of potential savings to date.

Despite this impressive progress, electricity system operators confirm that the supply/demand situation in the Western Cape (indeed also on the national power grid) is extremely tight. Brief outages remain a possibility.

A number of lessons can be learned from these emergency programs. First, and perhaps this is obvious, emergency measures to restore electricity security are very, very expensive. Hundreds of millions are being spent on the programs described above. Payments for self-generation, demand market participation, and emergency generation are five to ten times more expensive than Eskom's standard generation costs and full subsidies for CFLs, geyser blankets, fuel switching and industrial energy efficiency investments are almost certainly unsustainable. Electricity consumers, as a whole, are ultimately paying for these programs through higher Eskom costs and tariffs.

While these expenditures may be justified in the short term (as power outages can be even more costly to the economy) attention has to shift to more sustainable and cost-effective programs in the future.

Clearly South Africa needs to improve the efficacy of electricity supply planning and investments, but we also need to develop sustainable demand-side management programmes. Many energy efficient technologies are close to commercial viability – but various market barriers prevent more widespread application. For example, on a life-cycle cost-basis, CFLs are more cost-effective than ordinary incandescent lamps: while they cost more initially, they last longer and use less electricity. But consumers are reluctant to pay higher initial purchase costs.

Market barriers need to be tackled systematically. First, there needs to be an enabling policy and regulatory environment

that facilitates energy efficiency programs. Second and third, appropriate finance and business models need to be in place. Fourth, consumers need adequate information. And fifth, the requisite technology and capacity to deliver needs to be in place.

Government could establish energy efficiency standards, testing, certification and labelling systems. The electricity regulator could require electricity distributors to achieve energy conservation targets and could regulate the use of system benefit charges to fund initial efficiency programs. Banks could be encouraged, though partial risk guarantees, to develop dedicated financial products and instruments to fund energy efficiency businesses and programs. Or utilities could finance approved customer energy efficiency investments through repayments on electricity bills. Energy service companies could be established to assist utilities in reaching consumers and offering specialised energy efficiency products and services. Government, utilities and companies could run information campaigns. And there could be special support programs for technology and skills development.

Eskom will hopefully learn the lessons from the earlier blackouts by improving its planning, operations and maintenance systems. An additional lesson for South Africa is the need to launch systematic market transformation initiatives in a range of niche energy efficiency market segments.

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PLUGGING INTO SOURCE OF THE FAILURES

Anton Eberhard

BUSINESS DAY

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We now know that South Africa's security of electricity supply is inadequate. The Western Cape blackouts between November 2005 and March 2006, the national load-shedding on 18 January 2007, and numerous distribution failures, have caused costly damage to the economy and loss of welfare to electricity consumers.

The proximate causes of these failures are well understood. Reserve margins are too low and the supply/demand situation is too tight. During winter, when peak demand is at its highest, all available plant is being run to its maximum. In summer, when planned maintenance shutdowns are implemented, any excess demand for electricity or unplanned plant outages (beyond those for which contingencies exist) severely stress the system and can force rolling load-shedding.

Furthermore, there are parts of country where transmission capacity is insufficient to meet security standards. And the reliability of distribution networks in many municipal areas is threatened by inadequate maintenance and inadequate investment in system strengthening.

The root causes of these failures are sometimes misunderstood. Many have argued that electricity consumption and demand growth have been higher than expected because of higher economic growth rates. Let's look at the facts. Annual peak demand has grown on average by just over 3.6 % per annum since 2000. Current peak electricity demand is actually lower than that predicted in Eskom's Integrated Strategic Electricity Plans that were prepared in 2001, 2003 and 2005. Electricity forecasts have been reasonably accurate.

Furthermore, as far back as 1998, the Energy Policy White Paper warned of supply shortages around 2007. The claim that electricity demand has grown faster than predicted – and that this accounts for current supply deficits – is not supported by the data.

So what are the ultimate causes of recent supply failures? There are at least four main causes. First, policy uncertainty between 1998 and 2004 inhibited and slowed investment. During this time, consideration was given to breaking up Eskom and introducing competition and private investment. In 2001 Eskom was actually prohibited by government from building new generation capacity. Despite this embargo on Eskom, no concrete steps were taken for alternatives.

Investment planning and decision-making fell 18 months behind schedule. Some consequences of the delays are that the commissioning of base-load generating plant (in the form of return-to-service coal stations that were previously mothballed) and new peaking plant (open-cycle gas-fired turbines) has been too late. In addition, policy uncertainty delayed the re-establishment of dedicated project teams and departments in Eskom to manage the new capacity expansion programs.

Second, poor coordination has caused further set backs. The lack of co-ordination and integration of the different electricity planning, investment decision-making, approval and procurement processes between Eskom, the National Energy Regulator (NERSA), the Department of Minerals and Energy (DME) and the Department of Public Enterprises (DPE) has created dangerous risks in terms of contradictory and badly-timed decisions being made, and procurement processes that might result in costly and late investments. An example is the delay in finalising the DME bids for private independent power plants.

Emerging risks are also evident in the licensing delays by NERSA and the legal requirement for ministerial approval for any deviation from the official NERSA electricity plan – which is already out of date.

Third, while there has been a great deal of planning, some of the earlier planning assumptions were wrong. The estimates for existing generation plant availability were too optimistic. It was assumed that municipalities would be able to contribute more generation capacity than is currently available. The assumed cost of un-served energy was unrealistically low. And planning was constrained by applying too low a reserve margin (10% compared to a more acceptable 15%). Together these faulty assumptions have resulted in planned capacity additions that are 18 months to two years too late.

Fourth, inadequate maintenance or negligence may have played a hand. An investigation by the National Energy Regulator concluded that (in the case of the Western Cape outages) there was negligence on the part of responsible Eskom personnel, maintenance procedures and remedial actions were inadequate, protection systems had been operated incorrectly and licence conditions as stipulated in the Grid Code were breached. Eskom disagrees with these findings but has not lodged any formal court appeal, as it is entitled to.

These root causes help explain how we have arrived at where we are today. What of the future? The national electricity/demand situation will remain extremely tight over the next five years and may continue to be so in the period after 2011 unless investment is accelerated. In the meantime, new peaking plant will provide the system operator with useful new resources. However, the urgent need is now for more base-load capacity.

Investment decisions are being taken, huge capital budgets are being allocated and every effort is being made by Eskom to ensure that plant will be built and commissioned on time. The constraint now is not a lack of planning or investment. The primary constraint is time and the risks that the global

supply squeeze in critical power equipment and engineering capability will result in deadlines slipping. Further risks may arise from higher unplanned outages, higher demand growth, coal delivery constraints, problems in the fuel supply to the new peaking plant, delays in environmental approvals and transmission right-of-way expropriations and the ongoing scarcity of skills.

There is also a renewed focus in Eskom on maintenance and the security and quality of supply which appears to be bearing fruit: for example, the serious transmission failures of early 2006 have not been repeated. Nevertheless, serious challenges exist in strengthening the grid and capital refurbishment (including transformers) in high risk areas.

Probably, the most serious threat to the security of electricity supply in the future resides in the distribution sector. Policy uncertainty persists and there is still no clear road map to establishing the Regional Electricity Distributors (REDs). Unless remaining policy issues are resolved (including the question of local governments' constitutional rights over electricity reticulation) backlogs in maintenance and strengthening of distribution systems will continue to grow and the risks of supply failures will be exacerbated. Single-area distribution failures do not have the same impact as generation and transmission failures, but cumulatively they are eroding South Africa's ambition to lift economic growth rates and improve the welfare of its people.

Anton Eberhard is a Professor at the University of Cape Town's Graduate School of Business. This article has examined the root causes of electricity supply failures. A second article will examine the adequacy of plans to restore supply security.



“While attention is focused on the generation sector, it is sometimes forgotten that the greatest threats to supply security in the future probably reside in the distribution sector, where policy uncertainty and the lack of progress in reforming the sector is inhibiting”

FAULTS IN PLANS FOR SUPPLY SECURITY

Anton Eberhard

BUSINESS DAY
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Despite a massive effort by Eskom, involving over R150 billion of capital expenditure over the next five years, electricity supply security will remain inadequate and reserve margins will be paper thin.

Eskom is probably doing all it can to restore supply security. Its current installed net generation capacity of about 37000 MW will be boosted by the return to service of about 3500MW of old coal-fired stations, about 2000MW of new open-cycle gas turbines, a small 100MW wind farm and a massive new coal plant of about 4200MW (the so-called Alpha and Charlie projects). A new pumped storage scheme of 1300MW is also being built.

The Minister of Public Enterprises has also announced that additional investment decisions will be made on further coal-fired stations and two new conventional nuclear plants. Eskom has confirmed that its future expansion plans are built around “big coal”, “big nuclear” and “big networks”.

But big, new base-load plants take time to plan, procure, build and commission. None will be operating before 2011. By Eskom’s own admission, reserve margins will remain below the recommended 15%. Any unexpected increase in electricity demand or higher than planned plant outages will stress the system and further load-shedding might be necessary.

Plans to restore electricity supply security have not been aided by the painfully slow progress in contracting private independent power producers (IPPs). The competitive bid-process is way behind schedule and unless the request for proposals (RfP) is finalised immediately, and contracts placed with the winning bidder, the target date of 2009 for these vital peaking plants will be missed.

A further worry is the plan to procure private investment in a Combined Cycle Gas Turbine plant in the Eastern Cape by 2009. This base-load or mid-merit plant will be essential for

meeting demand in the short to medium term. The risks of this plant not being built on time are high. And delays in procuring IPPs are clouding the investment climate for increased private participation in the sector.

Government and Eskom are acutely aware of the tightness of demand/supply over the next five years. Eskom has embarked on a “stretch” programme to liberate more capacity through improvements to plant availability and the utility is exploring cogeneration contracts, an expanded energy efficiency and demand-side management programme, possible acceleration of commissioning dates and the possibility of converting open cycle gas turbines to more efficient combined cycle technology.

How do we avoid getting into such a tight spot in the future and how do we resolve remaining policy and planning impediments. Duplicate electricity planning systems and parallel and non-transparent investment decision-making, approval and procurement processes are complicating efforts to restore electricity supply security.

Recent experiences illustrate these problems. The Minister made a bold public announcement of a big new generation plant. Eskom’s Project Development Department was still busy with the preparation of a full business case to be taken to Eskom’s Investment Committee and finally to its Board for approval. Under the provisions of the Public Finance Management Act, approval also has to be obtained from the Department of Public Enterprises (DPE).

Eskom then still has to obtain a licence from the regulator who might be miffed that the investment decision appears to have already been made. If the proposed plant is not in the official (outdated) plan of the regulator then, by law, approval for this deviation has to be sought from the Minister of Minerals and Energy, who also has the power to determine what kind of capacity, and of what type, should be built when.

Meanwhile, the Department of Minerals and Energy (DME) is running its own competitive bidding and procurement process for private IPPs on timelines that are not always consistent with original plans. And the processes whereby Cabinet decides on which new IPP DME should procure are not transparent.

There is clearly much room to streamline policy, planning and procurement processes. Following international best practice, government needs to set and publish a clear electricity supply standard. The System Operator should be charged with responsibility for reporting and publishing actual performance against this standard. And the Minister of Minerals and Energy should delegate responsibility to the National Energy Regulator (NERSA) for monitoring security of supply and for providing early warning of any future capacity deficiencies.

Electricity planning should be co-ordinated and integrated by combining the planning processes of NERSA and Eskom. These could be transferred to a new Central Electricity Planning office located in a secure institutional environment, possibly next to the System Operator but under appropriate and transparent governance. National electricity plans and investment opportunities should be published on an annual basis, as is common practice internationally.

The processes whereby new generation capacity opportunities are allocated to either Eskom or the private sector should be transparent and should be co-ordinated through the proposed new Central Electricity Planning Office. Investment approval and licensing processes for new generation capacity should be streamlined through better co-ordination between electricity planning, the allocation of new build opportunities, approval processes in Eskom, DPE, DME and NERSA licensing.

Procurement of new IPPs should follow international best practice through institutionalising procurement in a new

Single-Buyer Office, perhaps attached to the System Operator (which might become independent over time). Government also needs to establish a clear policy that sets the maximum level of electricity imports that is consonant with security of supply standards. Competitive regional options need to be evaluated fairly against Eskom expansion plans.

In the meantime, there are some exciting new regional projects. One example is the proposed Mmamabula coal-fired plant in Botswana. Given the extent to which Eskom is stretched in its current activities, it may well make sense to fast-track private projects such as these in order to bolster supply security.

While attention is focused on the generation sector, it is sometimes forgotten that the greatest threats to supply security in the future probably reside in the distribution sector, where policy uncertainty and the lack of progress in reforming the sector is inhibiting investment. Networks are crumbling and skilled and experienced staff are leaving the sector. Area-specific supply outages will increase unless government finds a way through the current impasse.

Anton Eberhard is a Professor at the University of Cape Town’s Graduate School of Business. A previous article by the author in Business Day has examined the root causes of electricity supply failures. The third article in this series will examine the most serious threat to electricity security – the lack of progress in reforming the electricity distribution industry.



ROSIER FUTURE WITHOUT THE REDS

Anton Eberhard

BUSINESS DAY

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While the Western Cape blackouts of last year, and national load-shedding in January this year, provoked deep concern around the security of South Africa's electricity generation, frequent area-specific distribution failures induce irritation and sometimes resignation. Such complacency is dangerous. Cumulatively, these supply interruptions are eroding South Africa's ambition to lift economic growth rates and improve the welfare of its people. The greatest threats to supply security in South Africa in the future may derive, not from inadequate generation capacity, but rather from a deteriorating electricity distribution system.

Recently the National Energy Regulator commissioned independent technical audits of eleven of the largest electricity distributors (nine municipalities and two Eskom regions). These audits paint a disturbing picture of deteriorating supply security amongst municipal distributors (however, some exceptions were also noted, such as eThekweni).

The proximate causes of these failures are inadequate investment in physical and human capital. The electricity distribution networks of smaller municipalities, in particular, were observed to be in a poor state of repair and there are instances where basic contingency requirements are not met. Staff are de-motivated and often under-skilled for the requirements of the job. Very few formal systems are in place for managing maintenance processes or for training and developing new staff.

By contrast, Eskom distribution regions were observed to be performing relatively well. They have competent management and maintenance systems, adequate funding for refurbishment and access to sound technical expertise. However, Eskom accounts for less than half of electricity distribution; municipalities are responsible for supplying the majority of customers.

The audits also noted that the lack of municipal investment was related to the hiatus in government's plans to restructure the electricity distribution industry. The drivers for reform and restructuring were, and still are, inefficient operations, lack of economies of scale and scope through industry

fragmentation, financial instability, inequitable and multiple tariffs and ineffective electrification.

Reform of the sector has been discussed seriously since an ANC Electricity Conference in 1992, which for the first time brought all the relevant stakeholders together. Subsequently, the National Electrification Forum was established and, in later years, the Electricity Working Group and the Electricity Restructuring Inter-Departmental Committee. This work culminated in the PWC Restructuring Blueprint Report and a number of Cabinet decisions to amalgamate more than 180 municipal distributors and Eskom's distribution regions into six, adequately-resourced, Regional Electricity Distribution Companies (REDs).

In 2004, government established the EDI Holding Company to implement these mergers. Despite nearly 15 years of talk, studies, Cabinet decisions and the expenditure of more than R100 million, very little progress has been made towards the establishment of the REDs. RED1, which was established on paper in July 2005 in the Western Cape, is to be disbanded after a failure to transfer staff and assets from local utilities.

This author recently interviewed major stakeholders in the sector. With the exception of one government department, all stakeholders privately expressed misgivings about the road map for setting up the REDs and dismay that so many key issues remained unresolved. They also expressed scepticism whether any real progress would be made without dealing with the Constitutional provisions that currently give municipalities sole rights over the reticulation of electricity.

The focus of the Department of Minerals and Energy is now on drafting legislation that will govern the restructuring process. However, legislation takes time and there could be another 18 months delay before key uncertainties are clarified. And without changing the Constitution, the movement of municipal distributors into the new REDs will remain voluntary. Cape Town has announced that it will not participate in a RED.

However, it is not only opposition party-led councils that are opposed to the plan. Most of the metros and many ANC

led councils are deeply opposed to losing their electricity distribution businesses, particularly if Eskom will also participate in the REDs. Eskom, itself, is holding back until its future role in the restructuring process is clarified. Given this level of resistance in the industry it is not surprising the EDI restructuring process is going nowhere.

While the voices of those who call for the removal of municipalities' right to reticulate electricity are increasingly been heard, a constitutional amendment remains politically unlikely. Even within the ANC, there are serious differences between national and local government representatives, who argue that it is critical that they maintain a developmental responsibility to provide these services to their people.

How do we break this impasse? It seems obvious to me that any successful restructuring process should be built on at least the following three foundations: first, any restructuring plan should enjoy the genuine endorsement of the majority of stakeholders in the industry such that they become willing and active participants. Second, because Eskom's distribution businesses are performing better than many municipalities, Eskom should play an integral part in the restructuring process. And, third, any restructuring process of this magnitude (in 2004 total electricity distribution sales were R57 billion and combined staff numbers were 27,000) must be undertaken extremely carefully so as not to undermine the security of supply.

So how do we get the support of the majority of the industry? How do we maximise Eskom's involvement? And how do we manage the merger securely?

Twelve of the largest municipalities account for about 80 per cent of electricity distributed by all municipalities. Allowing these large municipalities to retain their electricity businesses – but providing intensive and sustained support for them to develop ring-fenced, corporatised, effectively regulated and well managed utilities, with adequate investment in physical and human capital – will be an important and practical first step.

Second, while Eskom should be required to transfer, on a proportional basis, those networks, staff and systems that serve customers within the boundaries of these twelve large municipalities, Eskom should be allowed to retain the balance of its distribution business: i.e. in rural areas, small towns and to qualifying large customers.

Third, Eskom should be allowed to take over failing electricity businesses from smaller municipalities. These municipalities are, in many instances, not making any money from electricity sales, and attractive financial incentives could be offered for them to transfer their electricity divisions.

Fourth, those medium sized municipalities that are providing a reasonable level of service to their electricity customers (and there are some) should be left alone.

These proposals involve abandoning the REDs model. Politically, a major policy rethink is difficult, as recent decisions by Parliament's Portfolio Committee on Minerals and Energy, as well as Cabinet, to stick with the six REDs model demonstrate.

However, the above proposals have the best chance of winning the support of the most important and powerful service providers (the large municipalities and Eskom) and hence stand the best chance of being implemented successfully. These proposals will also involve the least disruption to the sector, while promoting performance improvements and service delivery. By creating policy certainty, the industry will at last be encouraged to make the necessary investments in maintenance, system strengthening and skilled people to secure electricity supply such that it supports accelerated economic growth and improved welfare for all our people.

Anton Eberhard is a professor at the University of Cape Town's Graduate School of Business. He was a Board member of the National Electricity Regulator and later a Board member of RED1. This is the third in a series of articles published in Business Day that have analysed the root causes of electricity supply failures and the adequacy of measures to restore electricity supply security



ERWIN HAS SWITCHED US ON TO A TIMELY DISCUSSION ROLE

Anton Eberhard

BUSINESS DAY
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Minster Alec Erwin's reported statement last week that government would have to intervene, if the National Energy Regulator (NERSA) did not raise Eskom's electricity price sufficiently, raises some interesting questions.

One can see where the Minister is coming from. Current electricity prices are too low to fund new investment necessary to restore supply security. Average generation prices of around 12 c/kWh are less than half the cost of new coal-fired generation plant. The recently installed turbines, which are keeping the lights on over the Cape winter, produce power at a cost much higher than even the highest industrial tariff.

A recent international survey by NUS Consulting found that Eskom's industrial tariffs are so cheap that they are only half the level of the next cheapest country (Canada) and only 15 per cent of the most expensive country (Denmark). In other words, South Africa could more than double its electricity prices and still be competitive internationally.

How did such a situation arise? Electricity prices are low because Eskom did not have to build new electricity generation capacity in recent decades as it over-invested in the 1970s and 1980s. The capital cost of existing plant has largely been amortised. At present, electricity prices reflect mostly operating and fuel costs. But the situation has now changed and Eskom has a massive investment plan to meet growing electricity demand.

Someone has to pay for Eskom's capital expansion plan: either electricity consumers or taxpayers. Getting electricity

consumers to pay for their service is clearly more efficient and desirable than using tax income that is required for education, health and other pressing needs. And if electricity consumers have to pay for new investment, then electricity prices will have to rise substantially. Failure to hike prices will place Eskom's balance sheet under severe pressure. Debt/equity and interest cover ratios will deteriorate alarmingly.

Already the credit rating agency Fitch has changed the outlook on Eskom's long-term local debt to negative with the implication that it could be downgraded in the near future, which in turn would raise the cost of Eskom's debt financing and ultimately the cost of electricity. Cost pressures are also coming from power equipment manufacturers who are struggling to meet global demand for new power sector investment.

So the Minister is right to express concern. However, his statement that government may have to intervene challenges directly the notion of independent regulation. Like many countries in the 1990s, South Africa established independent regulators in its infrastructure sectors. The logic was clear: electricity, telecommunications, water and transport are all network industries, parts of which have natural monopoly characteristics. Consumers need protection and utilities need incentives to provide reliable and competitively priced services.

The *"independent"* regulator model grants considerable decision-making discretion to regulators. The idea was to insulate price-setting and licence approvals from opportunistic political behaviour that could undermine

economic efficiency as well as the attainment of social and developmental objectives.

This has been the standard model adopted around the world, including developing countries. Over half of Africa's countries have independent electricity regulators and about three-quarters have telecommunication regulators. Even Lesotho, with barely 0.2 per cent of South Africa's electricity generation capacity, has an independent electricity regulator.

However, what has yet to be fully grasped is that the *"independent"* regulator model implies two pre-conditions. First, it presumes an adequate level of regulatory commitment. In other words, governments believe in the importance of regulation and put in place robust legislation that protects their independence and stakeholders can appeal to the courts if they disagree with their decisions.

The second presumption is that regulators will be skilled, knowledgeable, experienced and competent and will make credible, fair, transparent and reliable decisions. Unfortunately, in many cases, neither precondition is in place: governments often exert pressure on regulators (overtly or subtly) and in many countries regulators face challenges in terms of skills and experience. Regulatory institutions in developing countries are also new and often fragile. Much time is therefore spent on governance, management and process issues and not enough on substantive decision-making.

Increasingly there is international recognition of some of these problems and alternative, hybrid or transitional regulatory models are being proposed that initially limit the

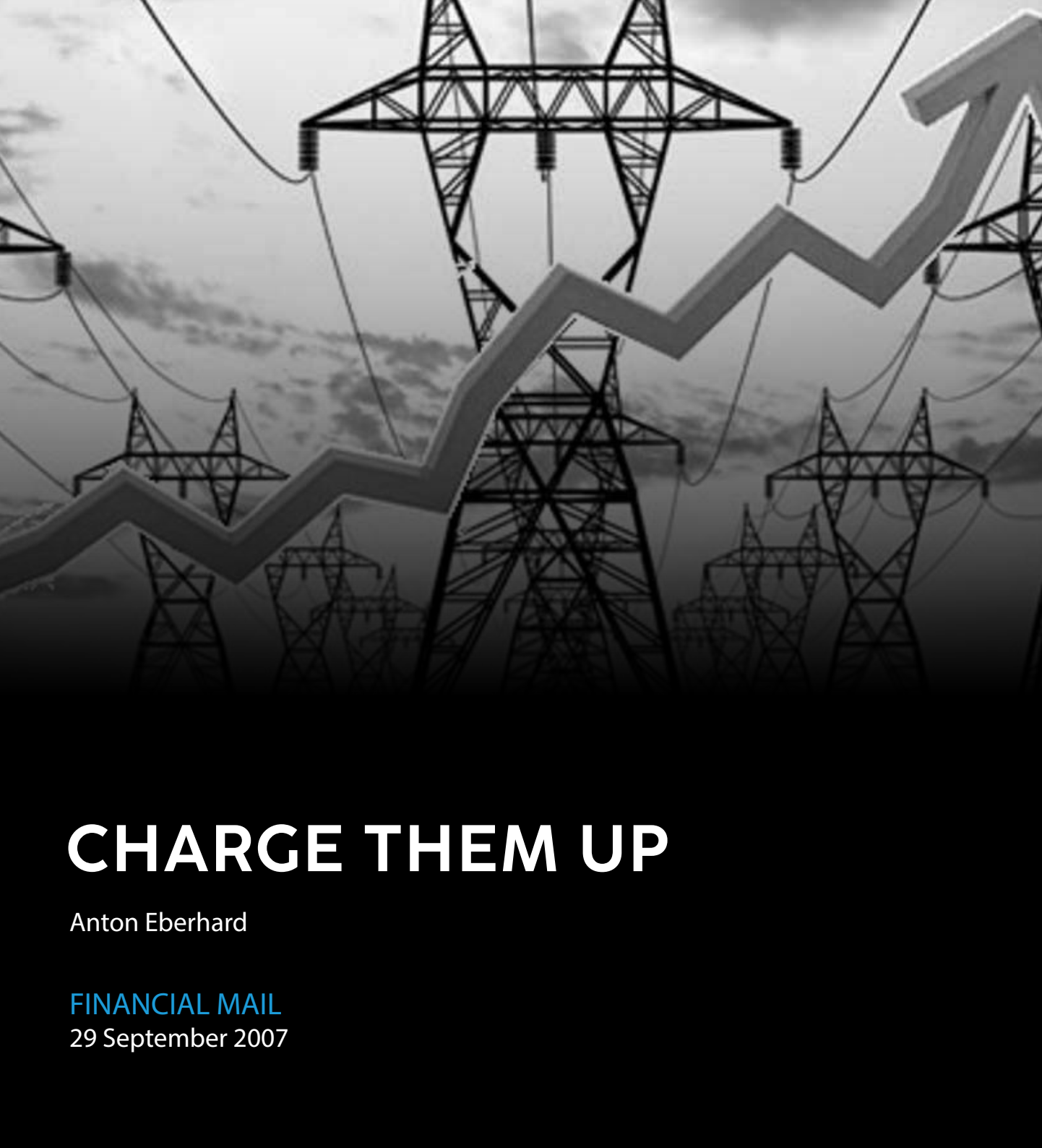
decision-making discretion of regulators as experience and skills are built. These alternative or hybrid models incorporate regulatory contracts (that specify in detail, for example, how prices are set) or involve outsourcing of certain regulatory functions to third parties, such as expert panels.

Examples of regulatory contracts may be found in many countries. For example, electricity concession agreements in West Africa and in Latin America are in effect regulatory contracts that govern the price and quality of electricity supply. And examples of expert panels may be found in Chile and in Romania.

While there are obvious advantages in building strong, competent and experienced infrastructure regulators that make credible and reliable decisions around prices and competitive market entry, this may take time. In instances where there are either challenges in regulatory commitment or in institutional and human resource capacity, it may make sense to limit the discretion of regulators and to put in place hybrid or transitional regulatory mechanisms such as regulatory contracts and mandated use of expert panels.

Erwin's comments may be controversial, but they could also provoke a worthwhile policy debate about how infrastructure regulation could or should evolve in South Africa.

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CHARGE THEM UP

Anton Eberhard

FINANCIAL MAIL
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There is a growing understanding in South Africa that electricity prices will have to rise to fund investment in new generating capacity. But how high should prices rise and what would be an efficient price level?

Large South African consumers currently enjoy among the world's lowest prices for electricity. The government views this as good for industry and a means for attracting energy-intensive investments. Part of the reason for the low electricity price is very cheap coal that cannot be profitably exported because of its location and quality.

But the larger part of the reason for the low electricity price is that Eskom's power stations were built over 20 years ago when their cost in money of that day was much lower, financed out of debt now largely repaid, and whose book value is a tiny fraction of their current replacement value. Eskom's 2007 balance sheet shows a book value for its power stations just under R 30 billion - about R700 per kilowatt of installed capacity, or \$US 100/kW. A new coal-fired power station will cost more than \$US 1,500/kW.

Most existing power stations have more than 20 years of remaining life, so their economic value is much higher than their book value. If Eskom were required to sell electricity at a

price that covered the real value of its assets, rather than their excessively depreciated book value, then prices would more than double. Electricity from a new coal fired power station would cost close to 30c/kWh, whereas Eskom's reported average generation cost is only about 12c/kWh. Furthermore, Eskom is adding 50 per cent more capacity in the next ten years and the value of its assets will multiply. Prices, in any event, will need to rise sharply to fund this new investment. If the Government offers a development tariff at or below the 2007 Eskom average industrial price, then it is effectively providing a massive subsidy to new industries that would be paying generation costs less than one-third of Eskom's extra costs of supply.

One of the consequences of not charging large industrial users the right economic price for electricity is that energy-intensive industries, such as aluminium producers, have chosen to locate in South Africa rather than where electricity costs are truly low. This situation is unlikely to continue. Electricity from new base load coal stations in South Africa cannot compete with surplus, and low-cost, hydro or gas resources in countries such as Iceland or those in the Gulf. And South Africa's coal-fired power stations generate additional costs to the planet in terms of high greenhouse gas emissions.

Eskom is clearly undercharging for its existing supply. If Eskom and the Government were behaving in a commercial way, the regulator would have to set a wholesale price of electricity high enough to reflect the economic value of its assets and to fund new investment.

The usual objections to raising electricity prices are that it would hit the poor and vulnerable, it would increase inflation, and deter foreign investment and development. All these arguments are bogus. If the poor receive a certain number of units each month for free, and prices were doubled, they would still receive these units at no higher cost - twice nothing is still nothing. Nor does raising the industrial price necessarily mean raising domestic prices.

If electricity prices rise to large users and import duties or other purchase taxes fall, then there is no impact on the price level. More important, if the Government acts as a sound shareholder in its companies, requiring a normal (but not excessive) rate of return, and its income goes up, it can reduce borrowing or taxes and reduce inflationary pressures. It could increase investments in other infrastructure sectors and lower prices in telecoms and ports (which, by international standards, are currently very expensive).

South African electricity prices are presently so cheap, that even large price hikes would not affect the competitiveness of existing South African industry. More economically efficient pricing would also lead to more efficient electricity consumption patterns and a less energy and capital intensive, more labour-absorbing, development path.

There is now a growing acceptance that electricity prices in South Africans will have to rise to fund Eskom's new capacity expansion programme. But there is an additional argument that needs to be addressed: the importance of raising electricity prices further to economic levels that will encourage allocative efficiency. Government and the National Energy Regulator will soon face these critical choices with not insignificant consequences for the nature and pace of economic development in South Africa.

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Applied Economics at Cambridge University.*

POWER INQUIRY NEEDED

Anton Eberhard

CAPE TIMES
13 March 2008

“The depth of the power crisis requires us to understand not simply immediate, but also root, causes. An evidence-based inquiry is now required that will provide the basis for sound policy, institutional and managerial reforms and action plans (2008).”

The depth of the power crisis requires us to understand not simply immediate, but also root, causes. An evidence-based inquiry is now required that will provide the basis for sound policy, institutional and managerial reforms and action plans.

Government's national response plan is understandably focused on restoring security of supply. That is our first priority and key stakeholders from the coal industry and other mining and industrial sectors are working closely with government to facilitate Eskom's efforts to stabilise the national power system. However, we also need to develop a common understanding of how we got into this crisis and what needs to be done to minimise a recurrence of power shortages in the future.

The immediate causes of the failures are obvious: Eskom does not have enough generating capacity to meet current demand, its existing plant is not operating adequately and transmission and distribution networks are not reliable enough. But these are proximate causes: more interesting and revealing are ultimate causes.

Six reasons are generally given for why South Africa has not maintained an adequate generation reserve, not all of them valid. The most convenient excuse is that electricity growth has been higher than expected. But surely we planned for

higher economic growth. I have also looked at the forecasts that Eskom has made over the years and while it is true that electricity demand growth has been high over the last two years, it is still at a level that fits within earlier planning scenarios.

Second, it is claimed that our generation planning has been inadequate. There has certainly not been a lack of planning and the Energy Policy White Paper published in 1998 stated quite explicitly that we would run out of generation capacity by 2007 and that investment decisions had to be taken by the end of 1999!

A third possible reason is that the regulator's low price increases prevented Eskom from investing in new plant. However, an examination of Eskom's financial statements indicates that the utility's profits increased markedly over this period and it was able to reduce its net debt to almost zero. Eskom was well positioned to finance new investment from retained earnings and bond issues.

A fourth reason given for current capacity shortages is the most convincing. Government prohibited Eskom from building new generators between 2001-2004 when a new competitive electricity market was being designed and

private participation was expected. Critically, this moratorium has put us three to four years behind where we should be in terms of new investments.

Two further reasons have also contributed to power scarcities. Government's plans to procure Independent Power Producers have still not materialised. Finally, Eskom's new build programme has slipped beyond original target dates.

These are the reasons for inadequate generation capacity. But more important now is to understand the failures in existing generation plant and networks. Eskom has been experiencing unprecedented breakdowns and there have been times when more than a quarter of its plant has been out of service. Extraordinarily, Eskom has allowed its coal stocks to decline to levels that have prejudiced power output. There is much anecdotal information on inadequate maintenance systems, skill shortages, etc. The seriousness of the blackouts and their consequent economic damage provide the imperative for understanding ultimate, root causes.

We need an independent inquiry (or at least a private sector peer review process) into these systemic investment and operational failures as well as the reasons we still have duplicate and contradictory power planning processes, why

the single-buyer model has still not been designed and implemented, why IPPs have still not been contracted, how we can rapidly accelerate industrial cogeneration investments and whether the proposed power conservation programme will be properly targeted and make adequate use of market-based mechanisms.

Ultimately the power crisis is a reflection of a fundamental governance problem and the challenges of ensuring state-owned enterprises operate efficiently and effectively. While major institutional restructuring is probably inadvisable in the midst of power scarcities, the questions of market structure, institutional design, and effective governance remain pertinent in the future.

Eberhard is a professor at UCT's Graduate School of Business



SHINING A CRITICAL LIGHT ON PLANS FOR REPOWERING SA

Anton Eberhard

BUSINESS DAY
7 May 2008

Last year, when South Africans for the most part still enjoyed cheap and reliable electricity supply, the minerals and energy department produced a Master Plan for Electricity Security, which was approved by the cabinet. The plan did not, however, prevent the most extensive and costly failure in the history of the electricity supply industry in SA.

The plan failed to anticipate or address the immediate causes of the current blackouts: Eskom's deficiencies in coal contracting and its inability to keep its existing generation plant operating at acceptable levels. And while the plan recognised the need to restore generation reserve margins to at least 19%, its proposed investment programme has already slipped. For example, Eskom is not on track in bringing online either its next diesel-fired peaking units scheduled for this year or its next big coal-fired station originally slated for 2010. The master plan also anticipated that an independent power project (IPP) would start operation next year, yet this investment has been cancelled. Furthermore, barely a year later, the plan's assumptions about inflation, economic growth, electricity demand and power conservation may be seen to be way off beam.

There are clear lessons from this experience. First, planning in a key sector such as electricity has to be much more robust and dynamic. Rigid, poorly crafted plans are probably more dangerous than no plans at all. Second, plans need to be executed accurately. It is this area that requires the most attention as we manage the crisis.

In essence, there are now five main thrusts to repowering the country: rebuilding Eskom's coal stockpiles; improving maintenance of Eskom's existing power stations so unplanned outages are reduced; ensuring Eskom's investment programme in new generation capacity does not slip; fast-tracking the contracting of private supplies through co-generation and IPPs; and demand-side management, power conservation and energy efficiency. There is also a sixth, cross-

cutting imperative - increasing the price of electricity - that will enable and facilitate each of the above programmes. After months of blackouts, it is fair to ask how well the effort to restore supply security is being managed.

Eskom's new executive director, responsible for primary energy and generation, has brought a welcome urgency in confronting the extraordinary negligence that saw the utility's coal stocks reduced to levels that prejudiced power generation. But there are still many unanswered questions around Eskom's primary energy contracting practices, its information and risk management systems, accountability and responsibility for failures, as well as the financial incentives that saw more value placed on bottom-line profits than on national energy security.

Eskom is committed to expanding its coal stockpiles, but this will take time and it will be extremely costly. Criminally, Eskom has not maximised its old cost-plus or fixed-price coal contracts with the major suppliers, favouring instead a host of small contractors who supply via congested and deteriorating roads. In the past, Eskom had access to the cheapest coal in the world: less than R100 a ton while international prices soared to \$100 a ton. No longer; now it is increasingly exposed to spot prices and will pay dearly for new coal.

Indeed, most of the 60% price hike Eskom now seeks from the regulator is for this coal.

How is Eskom doing in improving the availability and reliability of its existing power stations? In years past, Eskom aimed at, and exceeded, a target ratio of 90 to 7 to 3, corresponding to the percentage of time power stations were available to generate electricity versus planned maintenance versus unplanned outages or breakdowns. More recently, that ratio has fallen to 86 to 9 to 5. And in the months since October last year, plant availability dropped even further, to 77%. Power supply will remain vulnerable unless Eskom is able to reverse the recent trend of deteriorating plant performance.

Once again, Eskom's new executive director, now in charge not only of new investment and coal but also existing generation plant, is asking probing questions. For example, the frequency of boiler tube failures will not be accepted as a given. The top German utility, RWE, has been contracted to undertake technical audits of each of Eskom's power stations. Hopefully, penetrating analysis and recommendations will follow.

What is noticeable is that the RWE audit has not been widely publicised - reflecting perhaps a misguided sensitivity around exposing Eskom's skills and competencies to external scrutiny. Far too often, questions around the adequacy of Eskom's management, contracting and maintenance systems are seen as criticisms of its affirmative action and transformation policies. But the depth of the failures and the seriousness of their consequences must, and surely will, result in a deracialisation of the skills debate. The skills shortage is now so acute that anyone with competence and experience will be given the opportunity to contribute.

The main effort to restore supply security remains Eskom's massive investment programme in new generation capacity. Eskom has been rebuilding its project development, engineering and contracting divisions, and management and consulting firms have been engaged to assist Eskom staff.

However the challenges are formidable. For example, the international power equipment market is extremely tight, manufacturing slots are difficult to secure, and prices are increasing. Eskom will need to demonstrate that it can contract efficiently and deliver on time.

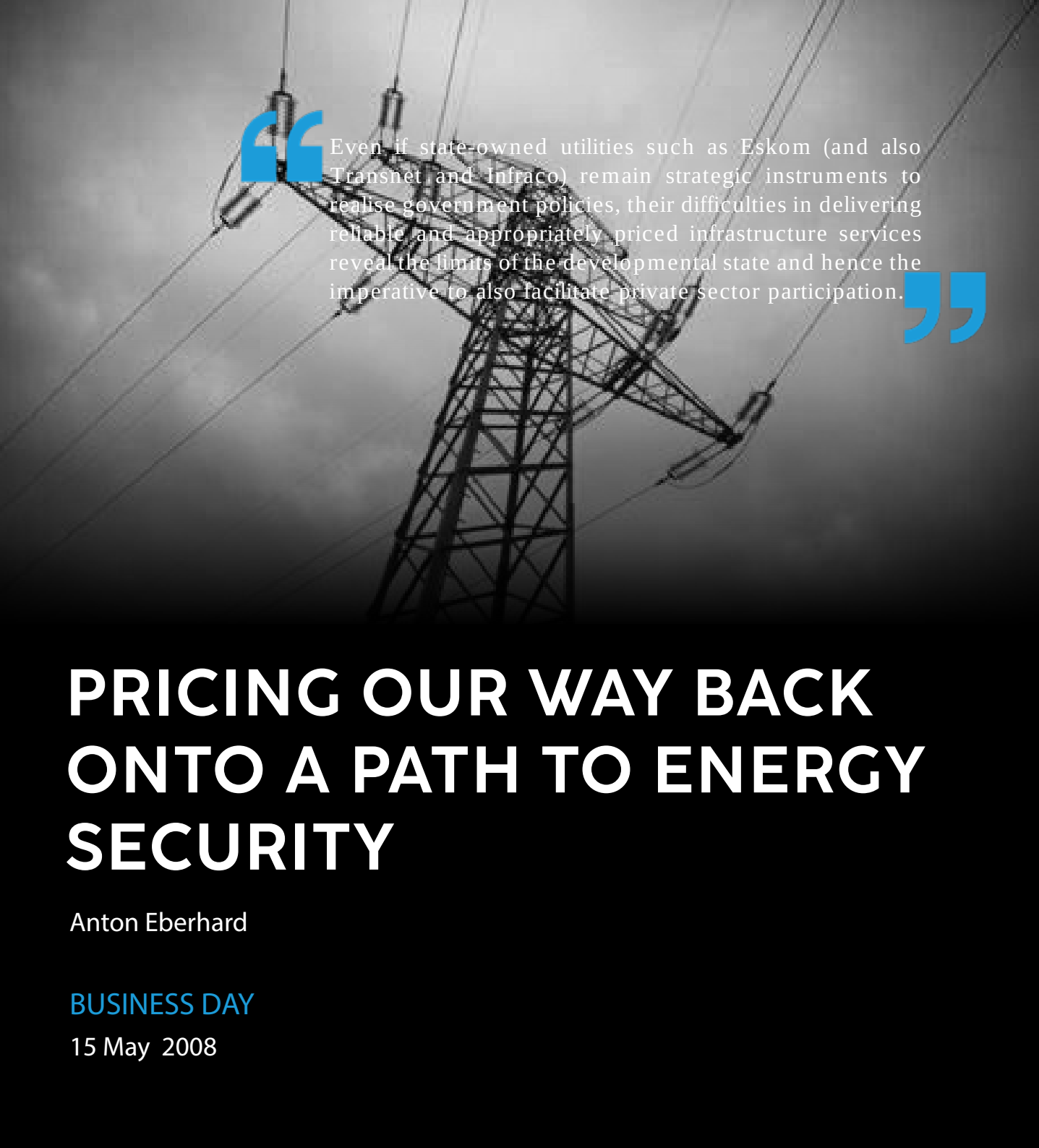
But it is now evident that Eskom's investment programme on its own will not restore supply security in the next five to seven years, as it takes many years to build new base-load capacity. In the meantime, we shall have to rely also on private sector participation through co-generation, IPPs and power conservation. There is certainly a great deal of interest from the private sector. The key challenge is for Eskom to show it can fast-track these contracts.

In an ideal world, the most efficient route would have been a competitive bidding process, in which Eskom evaluated each bid on its merits, stacked bids in terms of price and then contracted the most attractive offers. However, this process would take months, perhaps years. What is now needed is a standard offer. Eskom needs to disclose its avoided cost and to publish prices at which it is prepared to contract co-generation, IPPs or demand-side options. Simplicity and certainty are essential if we are to attract the necessary investment to plug the medium-term supply gap.

The price of electricity is crucial to the success of each of the above programmes. A future price path needs to be agreed on by key stakeholders - one that pays for the costs of restoring coal stocks, allows adequate maintenance, funds new investment, attracts private participation and induces the required energy savings.

A pervasive criticism from industry and other stakeholders is that there is insufficient "*felt leadership*" to drive, co-ordinate and communicate actions to restore supply security. Certainly, we do not lack in the number of response teams, committees and work streams. But real problems have arisen around lack of alignment and slow implementation. New proposals for more visible presidential leadership and a project management office will address some of these concerns. Investors, industrialists and households will be watching and tracking progress in each of the above areas to see whether, and how soon, we repower SA.

Eberhard is a professor at the University of Cape Town's Graduate School of Business. This is the first in a series of articles assessing progress in restoring electricity supply security.



PRICING OUR WAY BACK ONTO A PATH TO ENERGY SECURITY

Anton Eberhard

BUSINESS DAY

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“Even if state-owned utilities such as Eskom (and also Transnet and Infraco) remain strategic instruments to realise government policies, their difficulties in delivering reliable and appropriately priced infrastructure services reveal the limits of the developmental state and hence the imperative to also facilitate private sector participation.”

The single most important agenda item for this week's national energy summit will be the price of electricity. Just as well; for if there is one policy instrument that will eventually rid us of blackouts, it is ensuring that electricity prices rise to economic levels that will fund new investment while at the same time inducing more efficient consumption.

Few would now contest the logic of raising electricity prices. Current price levels are clearly sub-economic. Over much of the past two decades, prices have declined in real terms. Only in recent years have we seen above-inflation increases. If the 1990 electricity price had been allowed to rise each year at the rate of inflation, electricity would now cost at least 50 per cent more than current prices.

Two further examples illustrate the un-sustainability of current prices. Eskom's average generation price is less than half the cost of new supply. And the regulatory formula which determines Eskom's price, is based, in part, on a rate of return on the historical value of Eskom's productive assets. Yet this book value represents only five per cent of the current replacement value of Eskom's generation plant!

Thus the question is no longer whether electricity prices have to rise; it is rather by how much and how quickly. The ANC has taken the lead in calling for a national energy summit and Nedlac will help bring government, business and civil society together to debate these issues. I anticipate that sensible pricing proposals will be made: Eskom's 60 per cent price hike is likely to be rejected – but perhaps a 30 per cent increase this year and 20 per cent in ensuing years will be palatable. What would this price path mean for Eskom's financial sustainability and its new build programme?

Eskom's cash flow, income statement and balance sheet are under severe pressure. Interest rate cover in the next few years may even become negative. The price path suggested

above would almost certainly imply the need for additional shareholder (i.e. government) injections, or asset sales, if pre-tax interest cover is to recover to more acceptable levels of around three and if Eskom's debt: equity ratio is to stay below two. The involvement of National Treasury and the Department of Public Enterprises is thus critical to any agreement around a future price path.

This raises a crucial dilemma. Current legislation gives sole responsibility to NERSA to set Eskom's electricity price. It has established complicated multi-year price determination (MYPD) rules. But are these rules still appropriate? Will they provide an outcome that responds to the above challenges? I suspect that they will not.

What is now needed is for the regulator to reach an agreement on a desirable price path with key stakeholders including government, industry and consumers. This agreement should be informed by robust financial modeling on Eskom's funding requirements as well as the price level that is necessary to induce sufficient electricity conservation.

It is instructive to reflect on the consequences of not reaching such an agreement. A few years ago, the regulator, confronting the inevitability of electricity price increases, began to debate the need to agree a future price path with Eskom's shareholder (i.e. DPE) and National Treasury. But no agreement was reached. Indeed, the regulator even struggled to obtain a copy of the shareholder compact between DPE and Eskom which contained relevant financial targets. At that time, Eskom had very low debt levels and was not yet investing in new capacity. Above-inflation electricity price increases would have resulted in politically-unacceptable free cash flows and profits for Eskom, which could have been ameliorated only by government extracting additional dividends or creating a capital development fund for future investments. But there was no policy alignment on this issue and prices stagnated.

Now we face a similar but opposite challenge. The problem is not excessive profits. Now we need increased revenue to fund new investment. But, as in the past, we also need a common understanding and approach on a future electricity price path between the regulator, Eskom's shareholder, DPE, and National Treasury. This is no time for either turf wars or slavish adherence to regulatory methodologies which may no longer be appropriate. The mere fact that Eskom has twice requested the re-opening of the original price determination for 2008 and that its latest application is ten times the original increase, makes a mockery of the current MYPD. The motivation of moving to the current multi-year pricing formula was to create pricing certainty and to incentivise Eskom efficiency improvements. It has failed spectacularly to do either. That provides some cause to consider alternative pricing approaches.

Electricity price increases are necessary to fund new investment; they are also vital for encouraging energy efficiency. International data suggests that a price elasticity of minus 0.2 might be expected; i.e. a ten per cent price increase would result in a two per cent reduction in electricity demand; or a 50 per cent price hike would result in a ten per cent saving. There has been a great deal of contestation and unhappiness around Eskom's imposed load curtailment and pre-emptive load-shedding. A much more efficient and effective mechanism to attain the necessary savings will be a predictable price path that reflects the real economic value of electricity.

Yet there remain populist arguments that government should subsidise the price of electricity. Unfortunately these arguments often seem to imply that the overall price should be subsidized – rather than just the price for poorer households. The latter is obviously desirable. Poor households are struggling to cope with increased inflation and high food prices. Targeted electricity subsidies for poor households can be easily implemented through expanding the current free

basic electricity provisions or through introducing a new social tariff. The proportion of electricity consumed by low-income households is small and is easily cross-subsidized by other electricity consumers.

It is far less obvious why we would want to subsidise electricity for commercial and industrial users, or middle or higher income households. Considering the range of social services that government has to provide, electricity should at least be one sector that can, as a whole, pay for itself. What is not often appreciated is that a supposedly once-off National Treasury subsidy, which results in electricity being priced below its economic cost, either simply delays the inevitability of prices rising in the future or implies an ongoing and increasingly expensive subsidy in ensuing years.

Failure to raise electricity prices adequately will have two disastrous consequences: Eskom will not be able to afford investment in new capacity and profligate electricity consumption will continue. The ultimate consequence will be continued blackouts as supply fails to meet demand. Economic production, growth, employment and exports will be constrained, current account deficits will increase, the rand will depreciate, imported inflation and pressure on interest rates will increase, further depressing growth prospects. Hopefully the national energy summit will take bold decisions that will empower and motivate the regulator to move beyond its narrow price determination rules so that it raises electricity prices quickly to economic levels. Prices are surely our most effective instrument to restore electricity supply security.

Anton Eberhard is a professor at the University of Cape Town's Graduate School of Business. This is the second in a series of three articles assessing progress in restoring electricity supply security.



ON THE PRIVATE ROAD BACK TO POWER SUPPLY SECURITY

Anton Eberhard

BUSINESS DAY

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It is now inevitable that the private sector will have an expanded role in restrung electricity supply security in SA. This may seem a surprising assertion, given Public Enterprises Minister Alec Erwin's recommitment in Parliament last week to the *"vanguard role of state-owned enterprises (SOEs) in powering a developmental state"*. Yet serious initiatives to secure increased private investment in electricity generation have already begun. Eskom is running three separate processes to contract private power.

Erwin does not, of course, rule out private participation. He talks of the advantages of SOEs accessing private capital markets and also of entering into partnerships with local and global companies that can provide access to new strategic technologies. Yet he remains deeply committed to building a developmental portfolio of SOEs in areas such as network infrastructure (electricity, freight transport and broadband) in order to pursue public and strategic objectives.

Eskom remains one of the government's foremost champions in achieving the vision. Yet, as Allister Sparks argued on these pages a few weeks ago, Eskom's failure to keep the lights on reveals the limits of the developmental state. Eskom used to be regarded as SA's premier SOE. Since 1994, Eskom has provided electricity to more than 3,5-million new households. In 2001, the Financial Times of London named Eskom *"Power Company of the Year"*. In 2004, it received the Markinor Sunday Times award for SA's *"most admired brand"*. Such awards are now inconceivable.

Yet many within the African National Congress and the government still hold a deep antipathy to the role of the private sector in infrastructure. In part this stems from an old ideological stance around the role of the state. But it is also informed by unfortunate experiences with private sector participation. The strategic equity sale of part of Telkom resulted in neither improved services or lower costs. This was

not surprising given the fact that the government did little to mitigate the market power of this new private monopolist. Yet the enduring memory of many in government is that we were screwed by the private sector.

Some government ministers still believe that the power crisis is a result of the private sector not investing in 2001-04, when a competitive market was being designed. A more recent example is the collapse of negotiations with a private investor to build 1 000MW of peaking plant. In December 2003, the cabinet mandated the minerals and energy department to run a competitive tender for independent power producers (IPPs). Five years later, the process is in disarray, in part because the preferred bidder demanded changes that were so far from the original tender that even the project's senior lenders walked away because of the risk of legal challenges to the legitimacy of the process.

But the ultimate lesson from these examples is not the futility of private sector investment in infrastructure – too many successful international examples could be cited. Rather, it is the importance of state institutions learning how to contract the private sector more effectively. The department's IPP bid process was excruciatingly slow and complex. Thousands of pages of clarifications were issued along the way to bidders. Key issues remained unresolved for long periods. No reserve bidder was retained. It is sobering to note that even if the department had reached financial closure, the plant would only have come in line in 2010, seven years after the process started. A similar tender in Jordan, also for an open-cycle gas turbine plant, commenced later and the same bidder, AES, is already producing power. Across Africa, there are more than 40 IPPs, many of them in countries with investment climates much more challenging than ours.

Prior to the recent tender, there was no way for the private sector to enter the South African power market. The

government had abandoned plans for a power exchange where private generators could sell electricity and there as no contracting framework for IPPs. With Eskom's average electricity price less than half the price of new power, no consumer would contract directly with IPPs. The only way for IPPs to enter the market would be to sign a long-term power purchase agreement with Eskom who would then average out prices for consumers.

And that is what will now happen, not because of any ideological shift, but rather because there is no alternative. Eskom is quietly getting on with the process of contracting private generators. It is doing so because it understands that even if it restores its coal stockpiles and gets its existing generators to operate more reliably, and even if the price of electricity rises to more economic levels and consumers save electricity, and even if its own investment programme in new generation capacity delivers on schedule, we will still be short of power over the next five to seven years. As one Eskom executive put it, Eskom's "big coal, big nuclear and big networks" investment strategy will take time to materialise; multibillion-rand base-load power plants take years to plan and build.

In the meantime, the private sector can potentially offer smaller, more flexible and quicker solutions in the form of IPPs and industrial co-generation plants (using waste combustible materials and heat to also generate electricity, which can be fed back into the grid).

Crucial to the success of these programmes will be a robust, transparent and fair contracting framework. Esko staff are starting to learn valuable lessons in negotiating with potential private suppliers. They have also engaged experienced international legal and financial transaction advisors. It will be critical that contracts are as clear and as simple as possible so certainty is provided and investment decisions are made as

soon as possible. It is encouraging that, in its medium-term power purchase programme, Eskom appears to have revealed its avoided cost and has indicated it is willing to contract at prices of between 65c/kWh and 105c/kWh between 2009 and 2013, falling progressively from 2014 to an eventual level of 35c/kWh in 2018. These prices are very much higher than its current sub-economic tariffs.

But the current procedures for contracting the private sector also create potential contradictions. In effect, Eskom will be acting as a single-buyer of power produced by IPPs, while at the same time also investing in its own generation plant. Ideally the planning, contracting, system operation and transmission functions of Eskom should be in a separate independent institution, but major restructuring will be unwise and is unlikely in the midst of a power crisis. Nevertheless, careful thought should be given to an institutional design and government mechanism that gives the regulator, investors and consumers confidence that Eskom is procuring and contracting transparently and fairly.

Even if state-owned utilities such as Eskom (and also Transnet and Infraco) remain strategic instruments to realise government policies, their difficulties in delivering reliable and appropriately priced infrastructure services reveal the limits of the developmental state and hence the imperative to also facilitate private sector participation. Therein lies the core challenge for the government. We shall be able to provide reliable and competitively priced electricity (and transport and telecommunications) only to the extent that the government learns to contract the private sector effectively, or introduces private competition where that is possible.

Eberhard is a professor at UCT's Graduate School of Business. This is the final article in a series assessing progress towards restoring power supply security.



IF USERS DON'T FOOT ESKOM'S BILL, THE STATE WILL HAVE TO

Anton Eberhard

BUSINESS DAY

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Over the next few months, the government, the ruling party and its alliance partners will face one of their most challenging economic tests. Decisions will have to be made around electricity price increases that will have far-reaching implications, not only for the financial viability of Eskom, but for the security of power supplies and consequent effects on economic development and growth.

The magnitude of price increases will also have a direct effect on the rating, and hence cost, of Eskom's debt and whether there will be a requirement for sovereign guarantees and/or further national treasury transfers to Eskom, and hence fewer resources for other social programmes.

The decision around Eskom's price increase is not simply *"business as usual"*. Gone are the days of price increases below or just above inflation. Even the step change in electricity prices of 27,5% this year, and the envisaged 20% in each of the next three years, will be insufficient to pay for required new generation capacity or to keep Eskom solvent.

Eskom's price increase application last year of 60% provoked a heated national debate and led to a national energy summit, at which the government, business and civil society became directly involved in policy discussions on a desirable future price path for electricity. Eventually a compromise was proposed that would smooth price increases over the next few years.

But costs have continued to escalate. A rule-of-thumb estimate for the costs of coal-fired power stations used to be about \$1000 a kilowatt (kW) of installed capacity. With escalations in input costs, a global scarcity of engineering skills and tight equipment supply markets, those costs have escalated to more than \$2500/kW. Nuclear costs have escalated even more: a recent report by Moody's suggests that all-in capital costs could exceed \$7000/kW.

Primary energy costs are also rising. Eskom has been paying about a tenth of global coal prices, mainly due to favourable long-term contracts with mines adjacent to its power stations.

But as it expands power production, it will increasingly be exposed to coal prices that will escalate at a rate very much higher than inflation. Eskom financing will also become more expensive after the downgrading of its debt.

Eskom's investment requirements are gargantuan. The previous five-year estimate of R343bn will most likely be revised to above R400bn and could even rise to R500bn, with the fall in the rand and higher costs. Historically, this is the largest investment by a single enterprise in SA. Eskom had hoped to fund its capacity expansion programme partially through retained earnings and partially through local and international private capital markets. But with the global credit crunch, the latter option becomes more difficult and expensive and Eskom has recently cancelled a proposed bond issue.

Eskom also faces serious challenges in funding required investments through retained earnings. For the first time in decades, Eskom will sustain a loss in this financial year. It may also record a loss next year, depending on what price increase is agreed on. Its interest cover will be negative and its debt-to-equity ratio will rocket from close to zero to a figure of nearly four within a few years.

Eskom's finances are under severe strain. It may have to negotiate a World Bank loan, provided the government is prepared to offer a sovereign guarantee. This will be the first significant loan from the World Bank to SA after years of keeping the institution, and its policy prescriptions, at bay.

Electricity price increases at least double this year's record increase will be necessary in each of the next three years. Current regulated prices are clearly sub-economic. Eskom's average electricity price in the 2007-08 financial year was

19,45c a kilowatt hour (kWh), among the cheapest in the world. The generation component accounted for 13c/kWh (the rest consisted of transmission and distribution costs). Yet the cost of new coal-fired power generation is now approaching 60c/kWh and nuclear energy will be higher than 90c/kWh and perhaps even more than R1. It is obvious that prices need to rise sharply.

But electricity prices are not set in the market. The electricity sector is entirely government-owned and prices are regulated by the National Energy Regulator of SA.

The system of independent regulation has worked well in country contexts where governments have been prepared to delegate responsibility for price-setting, where competent and experienced regulatory institutions have developed, and where price determinations are either below inflation or, at worse, not much higher. Not all of these conditions prevail in SA and it is likely that the regulator will come under intense political pressure over the next months.

There is thus an urgent requirement to reach a political consensus on a future electricity price path. This consensus cannot simply be based on a crude negotiation in which Eskom asks for a certain price increase, which is then negotiated downwards to a level acceptable by the different social partners. The basis for the price increase has to be the revenue requirement calculation by the regulator supplemented by an analysis of Eskom's financial viability and financing requirements. Ultimately, this is a political decision that has to be based on robust financial modelling.

Political arguments against sharp electricity price increases will mostly revolve around affordability issues for poor households. But it should be remembered that poor households comprise only a tiny fraction of total electricity demand and that there is much scope for targeted cross-subsidies from other customer categories.

Mines, industries and businesses may not like the fact of higher electricity prices, but they understand the need for these increases to pay for new capacity. Their primary concern is security of supply and many are prepared to pay more to guarantee their supply.

If the government, the African National Congress and its alliance partners do not rise to this challenge the consequences are clear: Eskom will not earn sufficient revenue to pay for new generation capacity, it will have to borrow more at higher cost, or (more likely) it will require further cash injections from the government, which will mean less money for other priority social programmes.

And remember the numbers: Eskom has to invest R400bn-R500bn over five years. Do we really want the government to spend nearly as much on Eskom as it spends on education or health or safety and security?

The worst scenario is if the decision on much higher electricity prices is ducked and the government refuses, or is unable, to inject further cash into Eskom. Then Eskom will be unable to build sufficient new generation capacity. The lights certainly will go out, with devastating consequences for economic development and growth.

But if brave and informed decisions are made to raise prices to adequate levels, not only will we be able to fund investments in new generation capacity, there will be the added benefit of reducing electricity demand through accelerated investments in energy-efficient equipment. It will take time for new power stations to be built and commissioned. But energy savings can be made now and will be essential if SA is to survive this period of power scarcity.

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PRICE HIKES INEVITABLE TO KEEP LIGHTS ON

Anton Eberhard

BUSINESS DAY

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Once again, Eskom's application to the regulator for a steep hike in electricity prices will provoke a national debate around the affordability of electricity services and the financial sustainability of the power sector. We shall also have to confront implications for the security of electricity supply and questions around a more appropriate electricity market structure.

The regulator's current three year price determination for Eskom expires at the end of March. Remarkably, Eskom has delayed its price application for the next period until the last moment. Clearly there has been a great deal of behind-the-scenes lobbying. Ordinarily the regulator would need a few months to consider the application, allow for public hearings and make its decision. Now Eskom will enter its new financial year without clarity on the electricity price path and municipalities, whose finances depend in large measure on electricity sales, do not know what to budget for the next years.

Eskom will most likely apply for an electricity price increase of more than 80 per cent this year and more than fifty per cent next year. Prices could increase more than three fold over the next three years. Serious stuff! But Eskom faces extremely serious financing challenges in the short and medium term. For the first time in decades, it will make a loss in this financial year.

The core problem is the current sub-economic price of electricity generation. The average Eskom electricity tariff in the 2007-8 financial year was 19.45 cents a kilowatt hour (c/kWh). The power generation component accounted for about 13 c/kWh (the remainder comprises power transmission and distribution costs). Yet the cost of new coal-fired power generation is now above 60c/kWh and nuclear and renewable energy costs are even higher. It is obvious that electricity prices need to rise substantially.

But will the regulator have the political courage to make this decision? Like most countries, South Africa has a legally independent electricity regulator which was established to deal with exactly this kind of challenge. Cost-reflective tariffs are necessary to secure the financial viability of the utility, but at the same time there is intense political pressure to constrain price increases. This is precisely the role of the regulator. It needs to protect consumers (and it can do this through targeted pro-poor tariffs) but it also needs to ensure that the utility has the resources to provide reliable electricity services.

In determining an appropriate electricity price, the regulator will use an internationally acceptable methodology for calculating Eskom's revenue requirement based on its operating expenses plus depreciation and a return on assets which allows it to finance new investments in power generation, transmission and distribution. Hopefully it will value Eskom's assets appropriately and adopt a more commercial rate of return. The regulator will also examine vital financial indicators such as Eskom's debt to equity ratio and its interest cover.

Eskom's balance sheet faces severe pressures. After many years of not investing in new generating capacity, it has now embarked on a massive new capex programme to cope with power scarcities. The destruction of demand in the wake of the global financial crisis has provided unexpected, but temporary, relief to South Africa's strained power infrastructure. It takes many years to build new power generation capacity and we cannot delay any of the investments necessary to power South Africa's economy in the future.

Eskom's investments requirements are gargantuan. The previous five year estimate of R343bn will now be very much higher. Eskom funds its investments partially through retained earnings and partially through capital markets, including bond issues. The global credit crunch and the downgrading of Eskom's credit rating have made this finance more expensive.

Government has sought to strengthen Eskom's balance sheet through a R60bn long-term subordinated loan (effectively equity) and through debt guarantees amounting to R176bn. But it is now clear that the state is approaching its limit in support for Eskom. It has other social priorities, including health, education, safety and security and social grants for poverty alleviation. Should tax payers be subsidizing Eskom, or should electricity consumers pay the true cost of power? Getting this equation right is crucial for ensuring that South Africa has adequate power for its economy and its people. Getting it wrong would mean an unsustainable drain on the fiscus or an inexorable drift into inadequate investment and maintenance and ultimately more blackouts.

This is not an alarmist scenario. It has happened with depressing regularity across many developing countries whose economic potential is being constrained through lack of power. We cannot afford to go down that path. We need to take the hard decisions now around economic pricing. We also need to think about alternative ways of raising finance for the sector. And those alternatives unavoidably involve private sector participation and market reform.

If Eskom and government are struggling to finance much needed investment in new power generation capacity, then it is obvious that we need to reconsider possible roles for the private sector. Wholesale privatisation of Eskom, or the sale of a strategic equity stake, is probably not a good idea. It would simply entrench a private monopoly. The experience of the partial privatization of Telkom, without confronting competition issues directly, provides a sobering example: there were few improvements in service quality or prices.

But other options could be considered. Eskom owns and operates ten large coal-fired power stations. Some of these could be sold off and could compete in a wholesale power market or could be given competitive long-term power contracts. Or private investors could be encouraged to invest as independent power producers (IPPs) with off-take

agreements with the System Operator, ideally separate from Eskom.

There is no doubt that private investment and participation in the power sector is a great deal more difficult and expensive in the context of a global credit squeeze. Yet it is still possible. IPP developers are still active across Africa. What they need is policy and regulatory clarity.

Unfortunately the latest draft regulations from the Department of Minerals and Energy, regarding the entry of IPPs, do exactly the opposite. They create additional bureaucratic hurdles that will inhibit investment. One of the challenges of the new post-April government will be to provide leadership and clarity on an appropriate market structure for the power sector.

The shock of Eskom's price increase application thus focuses our minds on two key challenges: we shall have to accept higher regulated prices and we shall have to encourage wider private participation in our power sector. Both ultimately will be good for South Africa. If Eskom is financially sustainable it can provide reliable electricity services for our economy and our people. And if private investors are encouraged (in Eskom debt, the purchase of Eskom power stations and in new IPPs), then we have a better chance of meeting future growth imperatives. Higher prices will support new investment. They have the added benefit of encouraging more efficient use of electricity.

There has been much discussion behind closed doors around these issues. Now South Africans need to confront the hard choices. Either we accept the inevitability of much higher electricity prices plus increased private finance and participation or we become resigned to a gradually deteriorating power system with disastrous consequences for our economy and welfare. The choices are clear.

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“ If we are to avoid costly blackouts in the future, then clear responsibility needs to be allocated, and appropriate institutional capacity developed, to ensure that new power is procured in time and at the least cost ”

TEN WAYS TO PROVIDE SA WITH MORE AND CLEANER POWER

Anton Eberhard

BUSINESS DAY
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It is the hope of all South Africans, and the belief of many, that we shall be better governed in the next five years than we were in the past five. There is the prospect of new leadership, policy renewal and a greater commitment to implement policy more effectively.

Most of us could point to sectors of government which could be improved. My area of expertise is energy and there is no doubt that we could have done a lot better. South Africa experienced unprecedented blackouts which curtailed economic growth and caused widespread inconvenience. The distribution of petrol, diesel and gas has not been as reliable as it should have been. Far too many people in our country continue to suffer inadequate, unsafe or unaffordable energy services. And environmental impacts remain problematic.

A comprehensive energy policy was published as a White Paper in 1998. Its overall policy objectives were sensible: an energy sector which promotes economic competitiveness, social equity and environmental sustainability. Energy security and improved sector governance were seen as important elements in achieving these goals. There was a refreshing demand-side emphasis. However, many of the detailed policies in the White Paper need updating or more effective implementation.

I would suggest ten important policy priorities for the new Energy Minister.

First, it is striking that government has no integrated coal policy. More than 70% of our primary energy and more than 90% of electricity derives from coal. South Africa ranks fifth internationally as a producer and exporter. Coal has been the fastest growing fuel globally over the past decade, yet government has no clear export strategy and there is no integrated development of mining, rail and port infrastructure to facilitate either exports or anticipated increases in local production and consumption, within acceptable environmental constraints.

Second, South Africa's continued dependence on coal implies that our carbon emissions are disproportionately large. International climate change negotiations will inevitably include countries such as China, India and South Africa in new commitments to reduce greenhouse gas emissions. Trade penalties on carbon-intensive exports are also being considered.

While our environment ministry has developed long-term mitigation scenarios and strategies they are not yet having any direct impact on actual investment decisions. Eskom is building two more coal-fired power stations and will probably contract a third and fourth. Sasol is planning another coal-to-liquids plant. Clearly government needs to develop a more consistent and integrated climate change policy.

Third, while DME has published a paper on the security of supply of liquid fuels, there is little evidence that it is providing leadership around a range of key issues, including new refinery, pipeline, storage and distribution capacity, as well as more efficient transportation options. In the absence of clear policy pronouncements and purposeful action, industry players appear to be acting separately, sometimes with opposing and costly outcomes.

Fourth, electricity supply security remains important despite the temporary supply cushion as a result of the fall in demand linked to the global recession. If we are to avoid costly blackouts in the future, then clear responsibility needs to be allocated, and appropriate institutional capacity developed, to ensure that new power is procured in time and at the least cost.

Three different agencies are still undertaking electricity expansion planning, procurement initiatives for new private power appear to be ad hoc, procedures for dealing with unsolicited bids are not transparent, and contracting and dispatch arrangements for independent power producers (IPPs) are exposed to potential conflicts of interest within Eskom. The original electricity sector recommendations in the

1998 White Paper have been superseded by piecemeal, and sometimes contradictory, measures. In short, what is needed is a new Electricity Sector Policy.

Fifth, while a reliable electricity supply depends on having sufficient, operating, power stations, it requires also reliable grid networks to transport the electricity to users. Municipalities distribute approximately half of our electricity and many are not investing adequately in maintaining or strengthening these networks.

Government's solution has been to transfer these networks into regional electricity distributors but the only way that will happen is if the constitutional rights of local government are curtailed. The proposed constitutional amendment will be opposed. A more pragmatic solution would be to invest in human and physical capital in the twelve largest distributors which account for 80 per cent of the electricity distributed by local government. Eskom could take-over smaller, poorly performing distributors, whilst some of the medium-sized municipalities that are doing a reasonable job could be left alone.

Sixth, while the electricity regulator has made a bold decision to approve feed-in tariffs to Eskom that could support private investment in renewable energy sources, clarity and progress is needed around a range of issues before these investments become a reality. Further work is needed on robust power purchase agreements, institutional capacity for contracting effectively, and transmission connection arrangement.

Seventh, sharp electricity price increases are necessary to finance Eskom's investment in new capacity. Government has probably reached the limit of its support for Eskom in the form a R60 billion subordinated loan and debt guarantees totaling R176 billion. Eskom's access to private debt is becoming more difficult and expensive. Failure to raise electricity prices to cost reflective levels will mean Eskom making a loss, maintenance and investment programmes being delayed, and the lights going out. Devising and targeting special tariffs for low-income families will be an additional challenge.

Eighth, South Africans need to be encouraged to use energy more efficiently. We are the 16th most electricity-intensive economy in the world, which prejudices our competitiveness and environmental sustainability. Increased prices will probably be the most effective policy measure to shift consumption patterns, but these need to be supplemented by more effective demand-side management incentives.

Ninth, the energy needs of poor households are still inadequately met. Nearly a third of South Africans still have no access to the grid. The electrification programme has slowed and there is no chance that the goal of universal access by 2014 will be met. If a political backlash is to be avoided, the programme will need a thorough review in terms of targets, planning, technology choices, funding and implementation.

Even those with access to electricity can afford to use only modest amounts and rely also on paraffin, gas and fuelwood. A great deal of useful work was done in South Africa in the 1990s around household energy; we have yet to see an integrated and adequately resourced programme to tackle energy poverty. In rural areas, little progress has been made in sustainable production of fuelwood and its safe combustion in efficient stoves.

The final, tenth, energy policy priority relates to nuclear energy. South Africa continues to invest in the experimental pebble-bed nuclear reactor and fuel fabrication. Government is also considering a future fleet of conventional nuclear power stations. South Africa needs a national debate on the future development and use of nuclear energy, including its potential costs, safety, environmental benefits and dangers of weapons proliferation.

The above list of ten policy priorities may seem daunting – but real progress can be made if there is greater policy clarity and a renewal of institutional capacity to enable effective implementation.

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REFORM OF STATE-OWNED ENTERPRISES STILL NECESSARY

Anton Eberhard

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Minister Hogan's recent about-turn on the possible sale of non-performing state-owned enterprises (SOEs) is revealing of the new constraints that are being imposed on policy formulation within government. She backed down in the face of strident censure from some ANC, SACP and union leaders. Was this really because she contradicted ANC policies? Or are we seeing an ongoing battle for hegemony in discourse around contested areas of economic policy and management?

A re-reading of the ANC's 52nd National Conference resolutions taken in December 2007 at Polokwane, or its more recent 2009 Election Manifesto, reveals that there is no explicitly stated opposition to the sale of state assets or of private sector participation.

While these policy documents promote a developmental state and argue that state-owned enterprises should be strengthened, they also recognise the reality of a mixed economy where the state and the private sector complement each other in order to foster shared economic growth and development.

Indeed, in all these conference resolutions, there are just three brief references to state-owned enterprises: first, that there should be more integrated planning; second, that they should respond to a clearly defined public mandate and act in terms of overarching industrial policy and economic transformation objectives; and, third, that their role in skills development and training should be revived, in partnership with the private sector. These policies also assert that private capital should be engaged strategically and that public-private partnerships should be built.

Not too long ago, the Department of Public Enterprises published an Accelerated Agenda for the Restructuring of State Owned Enterprises. In his foreword to the document, Minister Radebe explained that restructuring refers to the matrix of options that include redesign of business management principles within state enterprises, the attraction of strategic equity partnerships, and the divestment of equity where appropriate. He went on to argue that restructuring involves improving the efficiency of the enterprise, mobilising private

sector capital and expertise, and the creation of effective market structures in sectors currently dominated by SOEs.

While some of the policy prescriptions in this document, such as the break-up of Eskom, were later repudiated by Radebe's successor, Minister Erwin, many of the core concerns around improving SOE performance remain on the agenda.

The same people who publicly criticised Hogan, and assert the importance of creating and maintaining jobs, or alleviating poverty, would have to agree that most SOEs have not met these objectives. Eskom had 66,000 employees in 1985; today that number is around 35,000. Transnet and Telkom (still majority-owned by government and the PIC) have also shed many jobs.

The impact of SOEs on development and poverty alleviation is at best mixed. Eskom made rapid progress with electrification in the 1990s, but that programme has slowed; a third of South Africans remain without access to electricity and the monthly allocation of 50 free kilowatt hours to low-income families remains poorly targeted. In the telecommunications sector, Telkom has made minimal progress in extending telephony and data services to poor and rural households despite a dedicated Universal Service Fund. In contrast, private cell-phone providers have nearly saturated the market.

South Africa's largest SOEs – Eskom, Transnet, Telkom – are all in the infrastructure sector. Government recognised that accelerated and shared economic growth was being constrained by infrastructure bottlenecks and initiated substantial new public investments in these sectors. The impact of the global recession on South Africa's economy would be very much worse without this ongoing infrastructure investment programme. However, we have yet to see significant efficiency improvements in infrastructure service quality and prices.

A few years ago, National Treasury and the Department of Public Enterprises commissioned reviews of the electricity, transport and telecommunications sectors. On the review team were leading international experts, including

respected academics David Newbery from the University of Cambridge and Roger Noll from Stanford University. The consistent message from these studies was that South Africa's infrastructure sectors significantly under-perform those in competitor countries.

Eskom, Transnet and Telkom might wish to dispute some of the data or benchmarks quoted in these studies, but consumers and market participants in South Africa, particularly those who have travelled or worked abroad, know that Telkom's voice and data, and Transnet's port, rail and pipeline services, are expensive and slow. Eskom's power failures are still fresh in our memories.

Eskom will soon report record financial losses. And while Telkom and Transnet are earning profits, these derive from monopoly rents which significantly add to the cost of doing business in South Africa.

It is a striking fact that Transnet is the only integrated port, rail and pipeline monopoly in the world. It is equally striking that we continue to protect Telkom's dominant market position at the expense of private sector investors or operators who are desperately keen to offer superior services. Also surprising is that, despite our power shortages, we have not facilitated private generators which could serve as benchmarks to measure Eskom's performance.

Structural reform and competition are perhaps the best ways of improving infrastructure performance. More difficult, but still possible, is the adoption of an integrated set of SOE reform measures, some of which we have already embarked on – such as clarification of roles and responsibilities through public-entity legislation, corporatisation and performance contracts. However, for these measures to be effective, government has to create professional supervisory and monitoring units that are prepared to institute credible incentives and sanctions for SOE managers.

Another set of measures to improve performance involves, what the Harvard academic Gomez-Ibanez refers to as, changing the political-economy of state-owned enterprises.

By this he means permanently altering the political forces that surround the firm through strengthening the role of interest groups with a stake in more commercial behaviour. Examples of such interest groups are taxpayers, customers and private investors. This can be done by reducing information asymmetry, transparent benchmarking, selective contracting out to the private sector and mixed capital enterprises where bond-holders, investors and rating agencies encourage a new financial discipline amongst SOE managers.

We shall also need a renewed commitment to independent regulation of monopoly infrastructure services. If independent regulatory agencies are to make credible and transparent decisions around prices, service quality and market entry, then they need to be supported by adequate regulatory commitment by government and investment in institution building and human resource development. Strong regulators are also important for promoting effective pro-poor access policies and tariffs.

Unfortunately, constant undermining of ICASA, the telecoms regulator, has held the sector back and condemned our economy to restricted and costly bandwidth. In the electricity sector, we have recently seen Eskom undermining NERSA through a tariff application which eschews the carefully constructed multi-year regulatory framework and which provides scant information for the regulator to do its job. The ports regulator has yet to facilitate the entry of new operators and competition between port terminals.

These are not trivial examples. Cumulatively they constrain investment, make us less competitive as an economy, and prejudice the welfare and development of South Africans. Hegemonic and ideological interventions that close down public debate on these issues do nothing to help us achieve an alternative outcome – one that is characterised by a competitive and efficient infrastructure sector that can set us on the path to economic recovery, higher growth, more jobs and shared benefits.

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ESKOM'S FUNDING CRISIS WILL SPARK NEW ENERGY DEAL

Anton Eberhard

MAIL & GAURDIAN

31 August 2009

“The issue of power sector market restructuring is thus once again on the table. It is clear that Eskom cannot do everything, nor does it wish to. The associated pressures of climate change mitigation and financing will gradually erode barriers to private sector entry and South Africa's power sector will migrate inevitably to a market structure that encompasses more diversity and cleaner technologies”

Eskom is facing an acute funding crisis which is likely to persist into the next decade. As climate-friendly technology options are considered, its generation mix will become even more capital-intensive, making it more difficult to finance these off its balance sheet. The combination of Eskom's financial woes and pressures to mitigate climate change will more than likely lead to a new electricity market structure in South Africa where the private sector plays a much bigger role.

Eskom has just made a R12,98-billion, before tax, loss. Over the next five years its capital expansion programme will approach R400 billion (nearly ten times the budget for the arms deal). Despite government assistance, in the form of a R60 billion long-term subordinated loan and debt guarantees totalling R177 billion, Eskom faces an R80 billion funding gap.

In the past, Eskom funded its investment programme through private debt and retained earnings. Adequate tariff levels made this possible. This year, inexplicably, Eskom ducked the imperative of applying for increases sufficient to fund investment in new capacity. It is mind-boggling that Eskom is in the middle of a massive multi-year investment programme without any clarity on its funding model.

What is to be done? There are limits on how much Eskom can borrow, although Eskom is being more aggressive in accessing capital markets through a range of financing instruments, including local and international bonds, export credit, and loans from development finance institutions such as the World Bank and the African Development Bank.

It is unlikely that government can provide more in terms of equity for Eskom, given the recession, reduced tax income and increased demands for social expenditure. Perhaps a special tax-payer levy, further sub-ordinated loans or sovereign guarantees can be explored, but none of these will be easy in the current economic context.

Ultimately debt has to be repaid and there is only one sustainable source of revenue – electricity tariffs. As Public Enterprises Minister, Barbara Hogan, made clear in her recent budget speech, prices need to cover costs, including the cost of capital. Steep tariff hikes are thus likely to continue into the future.

In addition to these financing challenges, Eskom will have to start confronting the imperative of mitigating climate change. More than 90% of South Africa's electricity is generated from coal. South Africa and other emerging economies argue that they still need to invest in least-cost power generation technologies in order to grow their economies and meet socio-economic needs. Rather than limiting emissions at below those of industrialised countries they seek a more just outcome where the world moves to equal emission rights.

The problem is that South Africa's carbon dioxide emissions are already way above the global average in both per capita terms.

South Africa's wiggle room regarding greenhouse gas emissions is diminishing. Last year, Cabinet noted the recommendations from the Long Term Mitigation Scenario

modelling exercise which would permit South Africa's emissions rising to a plateau in the early 2020s and then declining. A policy process was announced in March this year which aims to produce a White Paper on a National Climate Change Response Policy by the end of 2010.

Eskom's Board are aware of these developments and are considering alternative investment paths which would limit carbon emissions after 2025. Such a scenario almost certainly includes building a third mega coal-fired power station after the two currently being constructed – Medupi and Kusile – and, perhaps, also a fourth.

The big question is what will replace coal? Front runners in planning scenarios are nuclear energy and concentrated solar plants (CSP). But, as intimated earlier, these technologies are much more capital intensive than coal and will cost two to three times more per installed megawatt. Indeed CSP, with its lower capacity factor, will be even more expensive per effective megawatt. Wind energy will also play a role, but not as large as the other base-load technologies.

And so we are back to the challenge of how we shall finance future power generation capacity. Eskom's funding crunch is most acute over the next three years. Large tariff increases will help. But the next wave of investment decisions, that will soon have to be made, will once again create a financing crisis which will probably only be solved by facilitating the entry of more players into the electricity industry.

This is already government policy: Cabinet decided long ago that up to 30% of new power generation could come from

private independent power producers (IPPs). Eskom has not yet signed any power purchase agreements with IPPs. That will have to change.

Electricity sector planning has now been shifted to Eskom's System Operator who is producing a national plan, including low carbon scenarios. These will soon be published. What is still missing is a governance mechanism which will encourage more transparency and confidence in the impartiality of the planning process.

The Department of Energy is talking about an Independent System Operator which could create a more objective institutional base for contracting the private sector. That would involve a profound restructuring of Eskom and might not happen in the short term. In the meantime, the regulator is trying to implement feed-in tariffs for renewable energy technologies, but much work still needs to be done on acceptable power purchase and transmission access agreements.

The issue of power sector market restructuring is thus once again on the table. It is clear that Eskom cannot do everything, nor does it wish to. The associated pressures of climate change mitigation and financing will gradually erode barriers to private sector entry and South Africa's power sector will migrate inevitably to a market structure that encompasses more diversity and cleaner technologies.

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CHALLENGE NOW IS TO LESSON SHOCK OF PRICE HIKES FOR THE POOR

Anton Eberhard

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“Electricity prices are currently uneconomic and insufficient to support new investment by either Eskom or the private sector.... But steep electricity price hikes will force even lower consumption, reduce welfare and deepen poverty. Designing and implementing an effective and targeted subsidy system for poor households seems both right and possible.”

The electricity regulator will soon make a necessary, but unpopular, decision. It will allow Eskom tariffs to double over the next three years.

The decision is necessary because electricity prices are currently uneconomic and insufficient to support new investment by either Eskom or the private sector. It will be unpopular because each of us will be affected negatively. Inflation will increase, economic growth will be blunted and welfare diminished. How do we best mitigate the negative impacts of much higher electricity prices?

Our economy is likely to experience structural changes. We shall have less energy-intensive industries. Mines and businesses will invest in more energy efficient equipment and processes. Some will be able to generate their own power using waste heat. Middle and higher income households will reduce electricity consumption by investing in compact fluorescent light bulbs, insulation in their ceilings, solar water heaters and other smart energy-management devices.

But what about the poor who consume only basic amounts of energy and have limited access to relevant information and investment opportunities to save energy? Already, energy constitutes a significant proportion of their disposable income. Steep electricity price hikes will force even lower consumption, reduce welfare and deepen poverty. Designing and implementing an effective and targeted subsidy system for such households seems both right and possible.

What is not widely appreciated is that South Africa already has extensive electricity subsidies, amounting to more than R8 billion per annum. There are fiscal subsidies for connecting low-income households (R2.6 bn) as well as a monthly free basic consumption allowance of 50 units (R1bn). There are also extensive cross-subsidies from Eskom's large industrial

consumers to its small household consumers (R2.2 billion) and non-payment, or theft, amounts to an additional free subsidy (R2.5bn), at the expense of other paying consumers, some of them poorer than those not paying.

Of Eskom's four million registered household consumers, an astonishing one third recorded no electricity consumption over the past year. What's happening? Either consumers have switched back to other fuels such as paraffin or, more likely, many of them have worked out how to by-pass the meter and are consuming electricity without paying. Municipalities, who supply the other half of electricity consumers are, in all likelihood, experiencing also high levels of "non-technical" losses. Data is scarce.

So subsidies (either explicit or hidden) are already extensive. But they, by no means, reach all who deserve them. One in four South Africans still do not have access to electricity. They are the poorest of the poor, living in informal urban settlements or remote rural communities. Dealing with this service delivery backlog would mean an investment of about R6 billion per annum over ten years. And extending the free basic electricity allowance of 50 kilowatt hours per month to all low-income households would imply an annual subsidy of R4 billion in three years time, when electricity prices double. Furthermore, Eskom's current cross-subsidies to its Homelight customers would balloon to R5 billion, as would the value of stolen electricity.

Eight billion rand in electricity subsidies now, in three years time, could easily become R20 billion. Clearly the question arises as to how affordable will electricity subsidies be in the future, either fiscally or in terms of cross-subsidies from other electricity consumers? Some have proposed that more electricity is provided free to poor households each month. But a more sensible approach may be to look at whether

existing subsidies are well enough targeted: i.e. are all those who need them actually receiving them.

Following international best practice, it makes sense to continue with subsidies, first, for the capital needed to connect those who still do not have access to electricity and, second, for a basic level of consumption for poor households who can afford only small amounts of electricity. Eskom's Homelight consumers fall into this category. Connection costs are fully subsidized and a free basic amount is provided each month. The supply is limited to 20 Amps – i.e. households can use lights, a small cooker, kettle and other small appliances, but not a hot water geyser or full space heating.

This may seem a patronizing option: we'll give you a subsidy, but only if you limit your consumption. But the reality is that this supply/tariff option best matches the consumption patterns of these households. And the application of the subsidy is simple. Those households who need and want the subsidy self-select this supply option

However, not enough consumers are actually receiving these subsidies. Only 150,000 new connections are being made each year and Eskom disburses only R270 million in free basic electricity subsidies. Data on municipal disbursements of this subsidy are hard to come by, but it would be fair to assume that they also experience challenges.

One of the problems with the Eskom tariff that is applied to these consumers is that after the free basic amount, the consumption charge is higher than the marginal rate of energy production. Eskom is thus trying to recuperate some of its fixed monthly costs of supplying these customers through hiking the consumption tariff. A better subsidy would be to cover all fixed costs, offer the same level of free basic electricity each month, and then to charge subsequent energy use at the marginal cost of production.

So we face really difficult choices. Electricity is becoming unaffordable for the poor. But the scale of subsidies might also be unaffordable fiscally or in terms of cross-subsidies. Any new proposals for subsidies needs to calculate their full cost if properly applied, how they might grow in the future and whether they are affordable.

There are a number of additional subsidy challenges. What do we do about the not-so-poor, i.e. those who do not want to accept a demand-limited supply but who consume modest amounts of electricity and cannot easily access information or investment to become more energy-efficient? In principle, one can think of subsidy options, but they are more difficult to implement and would need more careful study before finalizing.

Further ways of mitigating electricity price increases would be to limit the profits municipalities make from electricity sales. There is no reason why municipal customers' electricity bills should double over the next three years. While the cost of power purchases from Eskom clearly needs to be passed on to municipal consumers, the balance of the tariff-their own distribution and customer service charges-should rise only by the rate of inflation.

So the challenge is to understand what is affordable in terms of subsidies and how best to target and implement them. Hopefully the electricity regulator will consider these issues carefully before announcing the inevitable electricity price increases.

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REFORMING THE GOVERNANCE AND POLITICAL-ECONOMY OF STATE-OWNED UTILITIES

Anton Eberhard & Grové Steyn

BUSINESS DAY

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“Key areas that will need attention in improving SOE governance are clarifying roles and responsibilities, smarter performance contracts, more effective monitoring and regulation, more transparent and relevant information and fostering of mixed-capital enterprises in a way that strengthens stakeholders with an interest in seeing improved utility performance”

South Africa's major infrastructure industries in electricity, transport and telecommunications are dominated by state-owned enterprises (SOEs). Few would express satisfaction with the prices, reliability and quality of these infrastructure services. What can be done to improve performance and to position these sectors to contribute more effectively to economic growth and development?

Electricity prices will double in three years and the risk of further blackouts in the medium term is real. Improvements in rail and port costs and efficiencies are urgently required to boost trade and improve macro-economic indicators. And the costs and limitations of information and communication technologies such as broadband frustrate businesses and households alike.

Much of the focus in recent months has been on the contest around leadership and management of SOEs. These are billion rand industries and large employers: it is not surprising to find contestation around political patronage, procurement practices and economic empowerment. However, there are more fundamental challenges which have to do with how these enterprises are governed and how they can be positioned to serve their customers and the South African economy more effectively.

Some have suggested that Eskom and Transnet should no longer function as corporate companies with independent boards, but should be folded back into their line ministries with more direct control by Ministers and their senior bureaucrats. There is little international evidence that such an approach would reap positive benefits. The opposite is more likely: non-transparent interventions in staffing and procurement practices will be easier, political pressures will result in non-economic price levels, rates of return will fall below the cost of capital, assets will not be able to be maintained or refurbished and there will be inadequate investment resources for system expansion with the inevitable consequence of a deterioration in infrastructure services. These circumstances are unlikely to attract the top notch executives that are required to run these organizations.

Furthermore, there are good reasons why SOEs such as Eskom and Transnet are corporatised entities and report to the Department of Public Enterprises (DPE) rather than the Departments of Energy or Transport. DPE as the sole shareholder in Eskom and Transnet is responsible for ensuring that they are financially healthy. In the case of Eskom the Department of Energy (DOE) is responsible for overall energy policy as well as for the sector regulator, NERSA.

The respective roles of these two government departments gives rise to a healthy tension: promoting financial sustainability of utilities, while protecting consumers. Confusing these roles, or sublimating either, is potentially disastrous. When state-owned infrastructure companies fall within one sector ministry, the likelihood of running down assets to meet short term political pressures becomes dangerously more likely.

What this discussion points to is the need for a deeper understanding and consensus around how roles and responsibilities should be clarified, delegated and co-ordinated. The roles of government as shareholder, policy maker and regulator need to be clearly distinguished.

In managing its role as shareholder, corporatisation is an obvious choice. Government as owner appoints a board which then oversees the management of the utility and the hiring and firing of executives. As Minister Barbara Hogan affirmed in her recent budget speech, government should not interfere directly in the management of SOEs. Government's focus is rather on desired outcomes, specified in general sector policy statements (legislation and white papers) and performance contracts with specific financial and technical performance requirements.

Deputy Minister Enoch Godongwana argues, rightly in our view, that the overall governance framework for SOEs in South Africa is broadly acceptable. The Public Finance Management Act provides for a number of checks and balances. Utilities are subject to the Companies Act, and international reporting and accounting standards apply. Performance contracts

are in place. And commercial imperatives for managers are mostly not confused with social objectives, as funding for the latter generally takes the form of separate and transparent fiscal flows. There are some exceptions to this, such as non-transparent subsidies between tariff categories. But there are also good examples where National Treasury funds directly the costs of providing infrastructure for poor households.

However, the application of this general governance framework is less than perfect. The appointment of CEOs (and sometimes even senior executives) is often still subject to political approval and Boards are not exercising sufficient leadership in dealing with ineffective managers. Performance contracts are often poorly drafted, incomplete and not effectively monitored. A classic case was that of Eskom, where short-term financial performance measures trumped the requirement to keep the lights on, while longer term sustainability indicators were ignored.

Performance contracts have inherent difficulties. There is the classic principal-agent, information asymmetry and moral hazard problem. Managers enjoy an information advantage over owners and can negotiate targets that are hard for outsiders to evaluate or easy for utilities to achieve. Incentives and penalties may not be credible or sufficient. And political owners may not always be committed to financial viability or improved efficiencies when faced with short-term constituency demands, especially when elections are pending.

This is not to say that performance contracts are useless; but we do need to build on international experience in designing these contracts more effectively. Governments and regulators need to deal purposefully with the challenges of information asymmetry, smart incentives and credible commitments. Effective supervisory and monitoring units in the ministry, and empowered regulators, can strengthen government's oversight of these utilities.

Clarification of roles and responsibilities is clearly important. But further governance reforms are necessary. Harvard

academic, Gomez-Ibanez, refers to these additional interventions as changing the political-economy of state-owned utilities. By this he means permanently altering the political forces around utilities by strengthening the role of stakeholders who have an interest in more commercial behaviour and improved performance by the utility.

An obvious first step is to improve transparency and information in order to empower stakeholders. Timely publication of comprehensive annual reports and financial statements is an obvious requirement. Government should also make available its performance contracts with SOEs and publish actual performance measures achieved. Benchmarking and customer surveys should also be published so that consumers are better able to hold their service providers to account.

But better information alone will not guarantee actions to improve performance; there are often obstacles to effective collective action. An additional intervention is the promotion of mixed-capital enterprises. The state may still choose to hold a controlling shareholding, but it could encourage SOEs to sell debt to private investors. Bondholders typically insist on debt covenants and, supported by credit rating agencies, keep a close watch on the performance of these utilities. South Africa's large infrastructure SOEs already make extensive use of private capital markets.

The governance of South Africa's SOEs is currently being reviewed. Key areas that will need attention are clarifying roles and responsibilities, smarter performance contracts, more effective monitoring and regulation, more transparent and relevant information and fostering of mixed-capital enterprises in a way that strengthens stakeholders with an interest in seeing improved utility performance.

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“The unbundling of Eskom is not only desirable, it is now inevitable. Government must release an unambiguous policy statement on the future market structure and translate this into legislation as swiftly as possible”

URGENTLY UNBUNDLE ESKOM TO UNLOCK PRIVATE INVESTMENT

Anton Eberhard & Mark Pickering

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As South Africa's latest integrated electricity resource plan (IRP) nears completion, two key questions arise: what is the optimal mix of power generation sources to meet future demand and who will build and finance these power stations?

The power plan will indicate how much generation capacity is required, by when, and which technologies should be used. However, the plan will probably not indicate whether Eskom or the private sector should build particular plant. Nor will it solve Eskom's most pressing challenge, namely how to raise sufficient finance for these capacity expansions.

Eskom has yet to secure the necessary funding to complete its second new coal power station, Kusile, and has halted development of further plants. Is the stage finally set for Independent Power Producers (IPPs) to enter the market? In contrast to much of the developing world, South Africa has failed to attract IPP investments to date. Other African countries have done much better: some 50 grid-connected IPPs are already operating across the continent and IPPs generate the bulk of electricity in Latin America.

Government has had positive intentions for IPPs for some time. The 1998 White Paper on Energy promised a gradual liberalisation of the power sector. In 2001 Cabinet announced that IPPs would account for 30% of generation. And the energy regulator proposed attractive feed-in tariffs for renewable energy power providers nearly two years ago.

Despite these intentions progress has been painfully slow with requests for proposals halted mid-stream, standard power purchase agreements still in the making, and uncertainty about grid access. To be fair Eskom recently signed power purchase agreements with industrial cogenerators for some 200 MW, but in contrast to the utility's own build programme of over 10,000 MW this is a paltry amount. The procurement

process took nearly five years to reach this point and it will be some time before the power is actually available. So why does South Africa have such a poor record, and what should be done to fix it?

Investors point out that Eskom has had few incentives, and indeed no obligation, to contract with IPPs. This is understandable to a degree. Dominant monopolies usually believe that they can do a better job than new entrants. But time has now run out and government must kick-start private investment if we are to sustain social and economic development.

The experience of other countries points the way. Take, for instance, Kenya which already has five IPPs operating, recently tendered for three more and is negotiating an off-take agreement for Africa's largest wind farm. What enabled Kenya to progress so much faster than South Africa?

In 1997, Kenya unbundled its national utility into a generation company (Kengen) and a separate grid company (KPLC). Besides being responsible for system operations KPLC was mandated to prepare long term generation plans (under the supervision of the regulator) and to procure new generation capacity from either Kengen or IPPs. Initially KPLC relied on transaction advisors but, over time, has become an effective independent entity capable of running competitive international tenders and negotiating contracts with winning bidders.

Earlier this year President Zuma and the Minister of Energy announced that an Independent System Operator (ISO) would be established to address Eskom's conflict of interest and attract IPPs. Not surprisingly the proposal has been met with some scepticism from Eskom and its shareholding ministry, the Department of Public Enterprises. The argument has been raised that a major restructuring could jeopardise

Eskom's credit ratings at a time when the utility is desperately trying to raise debt from international capital markets.

But a cursory examination of Eskom's balance sheet should alleviate this concern. The carrying value of Eskom's transmission assets is a mere R10.5 billion, compared to the utility's total asset base of R246 billion. This fraction will reduce further as new generation assets are added. Vertical unbundling is not an unusual event for international lenders and should not be an obstacle, as long as it is well managed.

In the meantime, Eskom has promoted the concept of an "Interim Independent System and Market Operator (ISMO)" as a ring-fenced entity within its Systems Operations and Planning Division. Whilst some form of interim arrangement will be necessary, policy makers must not be diverted from the main debate. Our immediate priority is to attract IPP investment.

In our view the current debates around the ISO or ISMO do not fully capture what needs to be done. We believe that Eskom's System Operations and Transmission divisions should be unbundled and a new utility (perhaps called Central Power) should be established with six main functions. Firstly it must prepare regular plans for the future generation mix and the evolution of the transmission system to transport power to where it is needed. Secondly, it must allocate new-build opportunities between Eskom and IPPs, either on the basis of government policy or through competitive bidding. Thirdly, it must procure and contract with successful bidders to ensure a secure power supply for the country. In doing so it must treat supply-side and demand-side resources on an equal basis. Fourthly, it must blend or aggregate historic and new power purchase agreements to achieve an average wholesale electricity price to electricity distributors. Fifthly, it must operate the system in real time to balance supply and demand. And, finally, it should own and operate the

national transmission system, to ensure that no generator has preferential access to this critical infrastructure.

The governance of this new state-owned enterprise must be independent and ensure that no generator, consumer or other interested party has undue influence over these important functions. The governing board must have a reasonable degree of authority to evolve market rules over time, within the constraints of government policy and the oversight of the regulator.

An important market design decision will be whether to establish the new entity as a mandatory single buyer or to allow for bilateral trades from the outset – as is permitted by current legislation. The economic benefits of older, depreciated power stations must obviously be reserved for the general consumer base. Narrow interests should not be allowed to capture this benefit. That said, there is no reason why consumers should not be allowed to contract directly with future generators for their specific power requirements – be these for green power, more secure power and possibly even imported power. The network operator should wheel this power from the producer to the consumer on an open-access and non-discriminatory basis, within the physical constraints of the system.

The unbundling of Eskom is not only desirable, it is now inevitable. Government must release an unambiguous policy statement on the future market structure and translate this into legislation as swiftly as possible. The more certain the policy, the quicker the six functions can be transferred out of Eskom and into the Central Power Company. This will attract IPPs, diversify our power mix and ensure security of supply.

Eberhard is a professor at the University of Cape Town's Graduate School of Business. Pickering is a partner at Meridian Economics



NATIONAL SUSTAINABLE DEVELOPMENT PLAN 2030

Better. Ever-Stronger. Together

NINE CHALLENGES TO OVERCOME TO MAKE SA BETTER PLACE

Anton Eberhard

BUSINESS DAY

13 June 2011

Too few South Africans are employed and our educational outcomes are extremely poor. These are the core challenges in fighting poverty and inequality highlighted by the National Planning Commission (NPC) in its recently published diagnostic report.

At face value, this is an unsurprising and noncontroversial analysis. Is there anything new in what the NPC is saying? What value will the NPC add? President Zuma, in the NPC's inaugural meeting, emphasized that the mandate of the commission is to take a broad, cross-cutting, independent and critical view of South Africa, to help define the South Africa we seek to achieve in 20 years' time and to map out a path to achieve those objectives. The commission is expected to put forward solid research, sound evidence and clear recommendations for government.

The NPC thus has a longer time perspective than the day-to-day business of government. We are also independent and have been tasked with holding up the mirror to South Africa so that we confront the underlying causes and seriousness of our economic, social and environmental challenges as a basis for crafting a realistic and sustainable long term development path. While the commission acknowledges what has been achieved in the past 17 years of our democracy, the diagnostic overview report, and the more detailed chapters on our website, are frank and direct in describing our condition. What we see in the mirror is not always attractive; it may not be unfamiliar, but hopefully the image we present is sharper and clearer and will be a spur to action.

The commission distilled its diagnostic down to nine main challenges. First, as mentioned above, South Africa's extremely high rates of unemployment and low levels of economic participation are a contributor to widespread poverty. Only 41 per cent of South Africans between the age of 18 and 60 work. This ratio is very low by international

standards where the norm is closer to two-thirds. Positive and sustained growth between 1997 and 2008 made inroads into unemployment, falling to 23 per cent, but a million jobs have since been lost. Unemployment amongst our youth is highest with 51 per cent of those between 15 and 24 without a job. Almost 60 per cent of all unemployed have never worked.

Second, while South Africa has made impressive gains in school enrolment (98 per cent for grades 1-7 and 85 per cent for grades 8-12), numeracy and literacy scores are extremely low compared with most other countries, not only amongst pupils but, shockingly, also teachers. Only 33 per cent of Grade 4-6 teachers scored more than the minimum expected from the average learner in their own subjects. 65 per cent of learners leave school without achieving a grade 12 certificate. While the pass rate for grade 12 was 68 per cent, only 15 per cent achieved an average mark of 40 per cent or more. The diagnostic identifies some of the underlying causes for our poor educational attainment, including teacher performance and the quality of school leadership.

Third, inadequate and inefficient infrastructure limits social inclusion and faster economic growth. Infrastructure policy, regulation and economics are my own speciality and it's an area that I have concentrated on in the commission. It is clear that we have underinvested in energy, transport, water and ICT infrastructure. Net capital formation is only half the level it should be. Consumers want reliable and competitively priced services and yet have been let down on both counts. We shall have to consider governance and regulatory reforms as well as new market structures in our network industries.

Fourth, the spatial legacy of apartheid continues to weigh on the entire country. In general, the poorest people live in rural areas. In the cities, the poorest live far from places of work and economic activity. A fundamental concern is the failure to coordinate delivery of household infrastructure as part of a broader process of building vibrant and viable

human settlements, including more efficient public transport systems.

Fifth, South Africa's growth path is highly resource-intensive and could be unsustainable. We are amongst the most energy-intensive economies, chiefly because of the prominence of coal in our energy mix, historically cheap electricity and a large minerals extraction and beneficiation sector. We are blessed with the most valuable minerals endowment of any country. A critical future challenge will be to balance the potential benefits from further development of our mineral resources with a less energy intensive development path that mitigates carbon emissions and climate change.

Sixth, our ailing public health system confronts a massive disease burden. South Africa has 0.6 per cent of the world's population, 17 per cent of the world's HIV infections and 11 per cent of the world's tuberculosis cases. There is a scourge of trauma cases resulting from violence and road accidents – nearly twice the global average. Infant and maternal mortality rates, at 43 and 6 per 1000 births respectively, are much higher than other middle income countries. Non-communicable diseases such as diabetes and cardiovascular problems are rising sharply. While our country's disease burden is worsening, the health system is collapsing, in part because of policy, institutional, management and system failures.

Seventh, the performance of the public service is uneven. While some institutions such as SARS have been turned around, others, and especially those related directly to service delivery, are failing. Causes include poor leadership and management, skills deficits, an erosion of accountability and authority structures, poor organisational and systems design and low staff morale.

Eighth, corruption undermines state legitimacy and service delivery. The commission's interactions with the agencies tasked with investigating cases of corruption provide

incontrovertible and worrying evidence that levels of corruption are high and growing.

Ninth, while we have made significant progress since 1994, South Africa remains a divided society where race, gender, education, skills and income levels define opportunity and world views. Social strife is a real risk unless we improve educational standards, get more people into work, increase income and reduce inequality.

This is our diagnosis. It is not new. But hopefully, in its totality, it is starkly clear. In the public engagements over the next few months we look forward to hearing the views of South Africans so that we can sharpen the analysis of our key challenges and begin the process of developing a long term and sustainable development plan.

Reflecting on the experience of being a commissioner over the past year, it has been both a stimulating and sobering experience. It has been a privilege to have had access to research and data spanning all major sectors of our economy and society. We have participated in multiple engagements with stakeholders and expert groups. We have been exposed to the severity of the problems and challenges that we face if we are to reach the aspirations that were so eloquently expressed in our Constitution and Bill of Rights. While we have differed amongst ourselves on the interpretation of these challenges and what needs to be done, we are, I believe, united in our belief that we can make real progress towards the kind of South Africa that we would all like to live and work in. Other countries have made remarkable progress. We can too.

Anton Eberhard is a professor at the University of Cape Town's Graduate School of Business and is one of 25 National Planning Commissioners. The NPC is chaired by Minister Trevor Manuel and the Deputy Chair is Cyril Ramaphosa. The NPC's Diagnostic overview and supporting chapters are available on www.npconline.co.za.

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Ultimately our greenhouse gas mitigation targets can only be met if we transform our power sector. Eskom's coal-fired power stations contribute about half of our carbon emissions. However, we are embarking on a radical shift in our energy mix. Half of our planned investment in electricity generation capacity over the next 20 years will be in renewable energy

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TOUGH CHOICES IN SOUTH AFRICA'S DEVELOPMENT PATH TO 2030

Anton Eberhard

BUSINESS DAY
14 November 2011

By 2030 poverty will have been eliminated and inequality reduced. All citizens will have the capabilities to grasp ever-broadening opportunities available to them and the life chances of millions of South Africans, especially the youth, will have changed for the better. This is the essence of the vision that frames the National Development Plan released last Friday.

This may seem a utopian dream, but the National Planning Commission (NPC) has laid out, in substantial detail, how these objectives may be achieved. The plan focuses on addressing the nine challenges identified in the NPC's Diagnostic published in June this year: creating jobs, expanding infrastructure, transitioning to a low-carbon economy, transforming spatial development patterns, improving education and training, providing quality health care, building a capable state, fighting corruption and uniting the nation.

President Jacob Zuma, in accepting the report from the NPC, emphasized the importance of government being open to independent, frank and critical analysis of the challenges we face and to what needs to change. Taken together, the proposals from the NPC constitute a powerful, integrated and in inspiring development agenda not only for the state but for all South Africa.

The NPC has responded to this mandate: while many of its proposals are reminders of the obvious policies, strategies, programmes and projects we all agree need to be implemented, others step into contested arenas where tough choices need to be made.

For example, there are no surprises in our macro-economic proposals: we need stable prices, and prudent and counter-cyclical fiscal and monetary policies. Sustainable growth and development will require higher savings, investment and export growth. Proposals include diversifying our trade towards emerging economies; revitalising logistics and transport links; promoting manufacturing in areas of

competitive advantage; packaging regional tourism offerings; and lowering the costs of living and of doing business.

Nothing new there - but we also need to confront special interests that frustrate our development and the attainment of higher employment and poverty reduction. The NPC puts forward a number of proposals around improving the functioning of labour markets to make it easier for young work seekers to get jobs and for dealing with dispute resolution and discipline.

One of the contested areas of our economy is the future of our mining and minerals beneficiation sector. Some argue the sector's contribution to GDP and jobs has declined and that we should curtail further growth if we are to mitigate greenhouse gas emissions. Yet we are blessed with an extraordinarily valuable minerals endowment which the world's growing economies wish to access. The sector generates important indirect benefits in other industries and makes a crucial contribution to our balance of payments.

How do we balance the imperative of growing our economy, creating jobs and eliminating poverty with the transition to a low-carbon economy? Our modelling shows that it will be difficult to meet our Copenhagen climate mitigation pledges without massive financial and technology transfers from the international community. Even if we assume modest growth in mining and minerals beneficiation, lower energy intensity and increased use of renewable energy, as envisaged in the Integrated Resource Plan 2010-2030, greenhouse gases from this sector will exceed the carbon budget implicit in the National Climate Change Response White Paper.

Through highlighting these contradictions, the NPC does not seek to undermine South Africa's ambition to act responsibly in global efforts to mitigate global warming. Rather we wish to interrogate the numbers, bring a sense of reality to the debate, and highlight the tough trade-offs that will need to be made.

It makes no sense to discard our minerals riches. Yet we also need to become competitive both in the transition to and in a low carbon future. Inevitably that means some sectors will be winners and others losers. Why, for example, do we still import bauxite from abroad and transform it into aluminium with cheap electricity? All we are doing is exporting under-priced electricity that is badly needed in other beneficiation sectors linked to local minerals.

Ultimately our greenhouse gas mitigation targets can only be met if we transform our power sector. Eskom's coal-fired power stations contribute about half of our carbon emissions. However, we are embarking on a radical shift in our energy mix. Half of our planned investment in electricity generation capacity over the next 20 years will be in renewable energy. We shall see increased use of solar and wind energy. This emergent industry will create jobs, as will investments in energy efficiency. But we also need to be mindful of the wider economy and the imperative of securing reliable and competitively priced energy. We face tough choices around nuclear energy, greater use of gas (including imports of liquefied natural gas, coal bed methane and shale gas) and need to explore more carefully associated challenges of financing, energy price paths, institutional capabilities and environment impacts.

Adequate and efficient infrastructure is vital for unlocking faster economic growth. Investment spending will need to reach 30 per cent of GDP by 2030 (up from a low of around 16 per cent in the early 2000s). The NPC makes a number of detailed recommendations for investments in energy, water, telecommunications and transport. Again, many of these are not new – but the plan does urge faster and deeper reforms in the governance of state-owned utilities, infrastructure regulation, competition and private sector participation.

The National Development Plan also makes a number of frank recommendations in key areas such as education, health and the public sector. These are complex areas and

there is no silver bullet that will resolve poor performance. A set of integrated actions are proposed which could make a difference. In order to improve educational outcomes, the plan confronts head-on issues such as teacher performance, appointment procedures and accountability. The challenges in public health management are also addressed. And in building a capable state, detailed recommendations are made on professionalising the public service and making it a career of choice. Some areas of government have improved, mostly SARS (and also in pilots in other departments), through improving operations management and delivery systems. These lessons and practices need to be rolled out to more areas of government.

To make meaningful, rapid and sustained progress in reducing poverty and inequality over the next two decades, South Africa needs to write a new story. At the core of the NPC's proposals is a new development paradigm that focuses on creating the conditions, opportunities and capabilities that enable people to lead the lives they desire. We need an active citizenry coupled with effective government and strong leadership that seeks to develop people's capabilities to be able to improve their lives through education and skills development, health care, better access to public transport, jobs, social protection, rising incomes, housing, basic services, and safe communities. Broadening these opportunities requires faster, more inclusive economic growth and higher levels of employment.

When President Zuma accepted the plan, he indicated that there would be a period of consultation within government and in the country before Cabinet finally approves the plan. Then, in his words, all components of government would be expected to implement it with the support and active championing of our citizens. Could we be on the cusp of a new inspirational era?

Anton Eberhard is a professor at the University of Cape Town's Graduate School of Business and is a member of National Planning Commission. The National Development Plan may be accessed at www.npconline.co.za



A POWERFUL NEW STRATEGY TO SHAPE SA'S ENERGY FUTURE

Anton Eberhard

BUSINESS DAY

21 August 2012

The National Development Plan, presented to President Zuma last week in parliament, sets out steps to transform South Africa's energy system. By 2030 coal will contribute proportionately less to primary-energy needs, while gas and renewable energy resources – especially wind, solar and imported hydroelectricity – will play a much larger role.

Energy powers economic development and contributes to the welfare of its citizens. But these benefits need to be accomplished within the bounds of environmental sustainability. Approximately 80 per cent of our greenhouse gas emissions are from the energy sector, mainly electricity production from coal.

Some argue that South Africa will not be able to meet its climate change mitigation commitments if it continues to promote electricity-intensive mining and mineral beneficiation industries. However, modelling by the National Planning Commission (NPC) shows that this is indeed possible. Direct, on-site emissions from these sectors are only five per cent of national carbon emissions and will not grow much in the next 20 years

Two-thirds of the carbon emissions associated with mining and minerals processing derive from the electricity they consume. The current integrated resource plan (IRP) envisages that the carbon-intensity of electricity generation will decline by a third by 2030 as renewable energy technologies are introduced. The structure of our minerals sector will also change as gold mining declines, inefficient smelters are upgraded and aluminium plants close at the end of their favourable Eskom contracts.

The National Development Plan shows that our coal, iron, ferro-alloy and platinum sectors could actually grow and total emissions would be lower in 2030 than they are today.

The core challenge in the coal sector is to ensure that the needs of Eskom's existing power stations are met for the remainder of their design life while valuable export revenues

are maximised. The majority of South Africa's coal reserves are in the underdeveloped Waterberg field in Limpopo. As coal resources in the central Highveld basin – the location of most of Eskom power stations – diminish, a new heavy haul rail corridor to the Waterberg will need to be developed. The coal line to Richard's Bay also needs strengthening to match port export capacity of at least 91 million tons per year by 2020. Expanded coal exports will strengthen our balance of payments and help fund accelerated economic growth.

Coal export markets are changing. We export less to Europe and more to India, which accepts lower ore grades, some of which are needed for Eskom power stations. This has led to calls for restrictions on exports. However, government should be cautious in applying policy measures which might have unintended consequences. For example, banning exports of coal lower than, say, 5500 kcal/kg could dis-incentivise investments in new multi-product mines necessary for supplying future Eskom demand. Instead, a win-win solution between Eskom and the coal miners should be sought. This is entirely possible, as the highest value option for most coal mines is an income stream from both exports and Eskom.

South Africa will need to meet about 29 000 megawatts (MW) of new power demand between 2012 and 2030. A further 10 900MW of old Eskom power stations will be retired. The IRP lays out power generation options in a policy-adjusted scenario that seeks a trade-off between least-cost investment, carbon emissions, technology risks, water-use implications, localisation and regional imports. The plan calls for 21 500MW of new renewable energy capacity to be in place by 2030.

International bidding rounds have already been held to fast-track renewable energy procurement with positive outcomes in terms of falling prices and substantial new private investment which will amount to over R100 billion in the next three years. While the prices of solar energy fell by 40 per cent between the first and second bidding rounds, its cost is still more than three times the current average price of electricity generation in South Africa. Further economic modelling is

necessary to test the short and medium term sustainability of renewable energy investments.

According to the Department of Energy's IRP, more nuclear energy plants will need to be commissioned from 2023. Although nuclear power provides a low-carbon, base-load alternative to coal, South Africa needs a thorough investigation of the implications of nuclear energy, including its costs, financing options, institutional arrangements, safety, environmental costs and benefits, localisation and employment opportunities, and uranium enrichment and fuel-fabrication possibilities.

While many of these issues were investigated in the IRP, and in a recent International Atomic Energy Agency sanctioned Integrated Nuclear Infrastructure Review, a nuclear fleet of power stations will involve a level of investment unprecedented in South Africa. An in-depth investigation into the financial viability of nuclear energy is thus vital. The National Nuclear Energy Executive Coordinating Committee, chaired by the Deputy-President, will have to make a final “*stop-go*” decision on South Africa's nuclear future after actual costs and financing options are revealed.

South Africa needs a “*plan B*” should nuclear energy prove too expensive, sufficient financing be unavailable or timelines too tight. All possible alternatives need to be explored, including regional hydropower and greater use of gas which could provide reliable and flexible power generation through combined-cycle gas turbines. Gas power plants have lower scale-economies and smaller plants can be built quickly to match actual demand growth. While the operational costs of gas power are arguably higher than those of nuclear stations, their unit capital costs are cheaper, they are more easily financed and they are more able to adjust their output to make up the shortfall from variable renewable energy sources.

Gas supplies from Karoo shale, even if economically viable and environmentally acceptable, will take many years to develop. In the meantime, West Coast gas resources should

be developed and investments should be made in liquefied natural gas (LNG) import infrastructure as insurance for the future.

Further refinements and regular updates of the IRP will be necessary to track electricity demand (which could be lower because of energy-efficiency gains, sharp price increases and a sluggish economy, or higher if economic growth accelerates) and to assess whether new generation technologies are delivering timely and affordable power. Planning capability needs to be vested in an independent system and market operator (ISMO) which should be spun out of Eskom. The ISMO will also be responsible for procurement and contracting of new power including independent power producers (IPPs). To ensure a level playing field between Eskom and IPPs, transmission assets and operations should be transferred to the ISMO. Remaining regulatory uncertainties that need to be resolved include the question of IPPs selling to customers other than Eskom, fair access to the electricity grid and rights to trade electricity.

The National Development Plan addresses also other energy policy and planning issues including energy demand management, electricity prices, electrification targets, reliability and maintenance backlogs in municipal electricity distribution networks, the desirability and timing of petroleum refinery investments and modal shifts in transport.

Planning for our energy future is vital for the prosperity and wellbeing of South Africans. It involves complex linkages and alignment with economic, social and environmental policies, strategies, programmes and investments. The National Planning Commission stands ready to support the Department of Energy, other government departments, industry and civil society to ensure that our energy sector delivers reliable, competitively priced and environmentally sustainable services to our economy and society.

Anton Eberhard is a professor at UCT's Graduate School of Business and a member of the National Planning Commission

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The ISMO Bill would have created a level playing field between Eskom and IPPs. These legislative initiatives sought to remove a real conflict of interest. Eskom is the dominant player in the electricity sector, it builds its own generation plant, and it has no incentive to initiate or conclude contracts with independent producers

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FAILED ESKOM RESTRUCTURING THREATENS ELECTRICITY SUPPLY SECURITY

Anton Eberhard

BUSINESS DAY

12 February 2013

Another failed attempt to restructure Eskom will make it more difficult for private investors to build independent power plants (IPPs) and South Africa will face more years of constrained electricity supply.

It has now become apparent that the proposed Independent System and Market Operator (ISMO) Bill will not be voted on by Parliament anytime soon. The ISMO Bill sought to remove electricity market functions - such as planning, procurement and contracting of new power - from Eskom and to place these in an independent state-owned utility, subject to determinations made by the Minister of Energy.

Why is the ISMO important or relevant? Well, quite simply, the ISMO Bill (and proposed amendments to the Electricity Regulation Act) would have created a level playing field between Eskom and IPPs. These legislative initiatives sought to remove a real conflict of interest. Eskom is the dominant player in the electricity sector, it builds its own generation plant, and it has no incentive to initiate or conclude contracts with independent producers.

The consequences of this conflict of interest are plain for all to see: Eskom is unable to meet electricity demand reliably or securely. The reasons are not only because it has not built enough new power generation capacity in time but also because it has missed a number of opportunities to contract additional power from IPPs.

The Energy White Paper, published in 1998, predicted that new power sources would be needed by 2007. Indeed power shortages became apparent from 2006 and in January 2008 Eskom had to shutdown our entire mining industry for a week because it ran out of power. The costs to the economy have run into tens of billions of rand. In November last year, Eskom again declared force majeure and forced large electricity customers to curtail demand by ten percent. Economic

growth is being constrained because of inadequate electricity supplies.

Power shortages will continue for the next three or four years until enough new units of the new mega-coal plants, Medupi and Kusile, are commissioned. But we need not have been in this situation. If Eskom had signed power deals with IPPs, South Africa would now have sufficient energy to support new investment and accelerated economic growth.

For example, developers of the 800MW Kudu gas-to-power project in Namibia have tried on a number of occasions to get Eskom to commit to a long-term contract, but the national utility argued that the prices offered were too high and that it could build power stations more cheaply.

In 2009, the Canadian-financed 1200MW Mmamabula coal project in Botswana offered Eskom a rand denominated long-term contract for 72c/kWh, with below inflation escalation clauses. \$100 million of private risk equity had been spent developing the project. Everything was ready. A Chinese engineering contractor had been signed. Finance was in place. If Eskom had signed a power purchase agreement, the project would now be delivering power. But once again, Eskom argued it could do better. We now know that Medupi and Kusile will cost very much more than either Kudu or Mmamabula.

Eskom has also failed to conclude any of its own IPP procurement initiatives within South Africa. In 2007, it launched a Pilot National Cogeneration Programme (PNCP) and issued an Expression of Interest which subsequently elicited 125 qualifying bids totaling 4900MW. None were contracted.

In 2008, Eskom initiated a Medium Term Power Purchase Programme (MTPPP) for projects that could supply a total of 3000MW of electricity to the grid by 2012. However, decisions

on preferred bidders were delayed repeatedly and by 2011 only 373MW had been contracted.

Eskom also called for an Expression of Interest in April 2008 for potential base-load projects to which 76 companies responded. The utility then issued a Request for Qualification (RFQ) in August 2008 in terms of which 23 local and international developers pre-qualified. However, by 2009 the Base-load Programme, along with the PNCP and MTPPP, had all been suspended by Eskom.

Some have described Eskom's actions as malicious compliance. It has accepted that it is government policy to introduce IPPs. Eskom has gone through the motions, but in each case the initiatives have yielded nothing. Others have suggested that it is unfair to blame Eskom managers who are beholden to the political directions of its shareholder, the Department of Public Enterprise. The real battle, they argue, is who controls big investment decisions.

The Department of Energy's (DoE's) frustration with Eskom came to a head around plans to procure renewable energy IPPs. Originally the plan was to offer a standard feed-in tariff and Eskom was tasked with preparing the contract documents. After two years, and no contracted IPPs, DoE, with the assistance of National Treasury, launched its own renewable energy IPP programme (REIPPP) in 2011. Three rounds of competitive bids have since yielded 64 new projects worth a total of 3920MW and have mobilized R118 billion of private investment.

Given this recent success, some may ask why the ISMO is still important. However, the REIPPP is being managed by more than 100 private consultants under the supervision of less than a handful of key staff at DoE and National Treasury and is vulnerable to changes in personnel in these departments. What is now needed is to institutionalize, consolidate and

strengthen this procurement and contracting capacity within the ISMO and to extend the procurement programme to base-load IPPs.

The serial failures of Eskom to facilitate private investment in power, its inability to meet electricity demand and, by contrast, the success of the REIPPP programme, would appear to make the argument for establishing a neutral, independent institution to procure and contract IPPs obvious.

That makes the shenanigans around the ISMO Bill all the more surprising. President Zuma, in his State of the Nation Address in 2010, committed government to establish the ISMO. The National Planning Commission came out in support of the ISMO. The draft ISMO Bill was approved by Cabinet and was passed by Parliament's Energy Committee in March 2013. It was due for debate in the National Assembly on 12 June but at the last moment was withdrawn and, without any published reasons, was sent back to the Energy Committee which approved it for a second time on 8 November. It was due to be debated on the last day of Parliament last year but on the morning of the debate was moved below the line, effectively meaning that it could not be debated and has thus lapsed. The Bill can now only be resuscitated if it is re-introduced by the Minister. That seems unlikely before the elections.

All this detail may seem a little arcane – but it demonstrates the trench-like warfare by vested interests against the Bill. The issues are now clear: will Eskom be allowed to maintain its monopoly, including being able to frustrate the entry of IPPs, or will government create a professional and independent electricity market operator that will procure and contract new power at competitive prices. It is clear which option is in the national interest. Electricity consumers will watch this space carefully.

Anton Eberhard is a professor at the University of Cape Town's Graduate School of Business and is a National Planning Commissioner

“Future investments in energy technology will dwarf those of other infrastructure sectors such as telecommunications, water and transport. It is vital that we get our decisions on capital expenditure right and that we execute these projects efficiently”

NUCLEAR POWER IS NEITHER NECESSARY NOR COST-EFFECTIVE

Anton Eberhard

BUSINESS DAY
19 March 2013

South Africa can achieve electricity supply security without investing in further nuclear power. Alternative power sources are cheaper and they can also help us achieve our Copenhagen pledges to reduce carbon emissions.

These are the outcomes of modeling recently undertaken by UCT's Energy Research Centre (ERC) that were presented at a high-level international energy meeting convened last week by the Stellenbosch Institute for Advanced Study (STIAS). The modeling takes into account a number of new international developments.

Over the past few years, renewable energy prices have fallen dramatically. In part, this has been due to the subsidies provided in countries such as Germany that have led to higher global production, including in China, and lower costs. In a sweet twist of fate, investment costs of renewable energy technologies in emerging economies are being subsidized by consumers in OECD countries!

South Africa's successful renewable energy independent power procurement programme has also seen prices fall. Between bid rounds one and two, wind prices fell 20 percent and solar 40 percent. South Africa has amongst the most favourable solar resources in the world and is now well positioned to benefit from these global technology developments.

Profound changes are also occurring in the energy intensity of South Africa's economy. This is, in part, due to structural changes in its economy but also due to accelerated energy efficiency investments as electricity prices rise sharply from historical lows.

South Africa's power plan – the IRP 2010 – is out of date. Many of its assumptions no longer apply and its recommended investment decisions are non-optimal. If followed, the existing plan would result in surplus, stranded and expensive generation capacity. Electricity demand growth has been much lower than forecast; it is still below 2007 levels. Less

new generation capacity is now needed over the next 20 years.

It is no longer urgent to make an investment decision on nuclear power, as implied in the old plan. Indeed, new data on nuclear power, and on imported Liquefied Natural Gas (LNG), delivered via purpose-built ships, indicate that gas is a more cost-effective option. There has been an explosion in international gas markets over the past few years. Gas power plants have lower investment costs than nuclear, can also be built more quickly, and complement variable renewable resources. However, imports of gas may adversely affect our balance of payments.

A new, updated power plan would look very different to the existing plan. Forecast electricity demand growth would be lower. No nuclear investments are envisaged. The new power choices gazette by the Minister of Energy at the end of 2012 would not all be needed, at least not yet. In the future there would be more imported LNG. Shale gas is not an option in the next five years until there is more data on economically recoverable resources and environmental impacts in the Karoo. And piped gas, from the large new gas fields discovered off the coast of northern Mozambique, is probably not competitive with LNG sea imports.

The balance of capacity in the new power plan would comprise wind, solar, imported hydro and a modest amount of fluidized-bed coal plant. ERC's model indicates that the appropriate combination of base-load, mid-merit and peaking plant can be provided with this generation mix at a lower electricity price than the existing plan (adjusting for new technology cost assumptions) while still meeting the carbon caps implied in South Africa's Copenhagen pledge.

Uncertainties around the future mean that different investment strategies are needed. Risks can be minimized through investments in a diverse portfolio of modular, less-capital intensive technologies, such as gas and renewables, that can be deployed rapidly and flexibly to meet changing demand patterns. Conversely, investments in lumpy, capital-

intensive technologies, such as large coal or nuclear plants, not only run the risk of cost and time overruns but also of under or over supply, depending on future demand variations. Improving energy end-use efficiencies is the fastest and least expensive method of making existing generation capacity available for new loads. International experience shows that the potentials here are huge and typically not tapped without special attention of governments on appropriate regulations and incentive schemes.

The presentations and discussion at the STIAS meeting serve as a catalyst for new thinking and public discourse on South Africa's electricity future that mirror those advanced by the National Development Plan. They highlight a series of policy and institutional issues that need addressing.

First, there is a need for a unified vision and policy for South Africa's electricity market. Few of the proposals in the Energy Policy White Paper of 1998 have been implemented. A degree of competition for the market is being introduced through the contracting of Independent Power Producers (IPPs). However, this is being done on an ad-hoc basis. What is needed is a clear policy statement on the role of Eskom versus private IPPs in the future of South Africa's power sector. The competitive procurement of renewable energy IPPs has shown that prices can fall, but prices are still above international benchmarks; for example, the prices achieved in Brazil's dynamic reverse auctions for wind energy are at least a third below ours. We need a more purposeful strategy of what we wish to achieve through the introduction of competing private power investors.

Second, the required institutional arrangements are not in place to procure and contract IPPs sustainably. The highly successful Renewable IPP procurement programme – which raised R47 billion in its first round and has resulted in 28 new renewable energy projects breaking ground – was managed largely by outside transaction advisors contracted by the PPP unit in National Treasury. The procurement of new IPPs with dedicated associated fuel sources (such as gas, coal or hydro) will be more complicated. What is needed is an independent

market operator that, at the minimum, has responsibility for planning and procurement of new power. The Independent System and Market Operator Bill before parliament, with necessary amendments, will begin to address these issues. The least risky option going forward would be to combine also power transmission assets with these functions.

Third, as already intimated, a new power plan is urgently required. The Department of Energy needs to request Eskom's planning professionals urgently to revise and update the current plan.

Fourth, there is no national gas strategy to guide the nature, location and sequencing of gas infrastructure. The National Planning Commission is willing to work with the Department of Energy and National Treasury to develop the necessary inputs for such a strategy. This work is urgent given the more prominent role of gas in South Africa's new power plan.

Fifth, regulatory and metering impediments to installing own-generation systems, such as rooftop solar photovoltaic (PV) modules, with reverse grid-feed in, need to be speedily resolved. The dramatic fall in PV prices will lead to a quiet revolution where individuals will make competitive energy choices without expensive subsidies.

Future investments in energy technology will dwarf those of other infrastructure sectors such as telecommunications, water and transport. It is vital that we get our decisions on capital expenditure right and that we execute these projects efficiently. The ERC model outputs do not yet represent a new power plan. The outputs of models depend on input parameters and assumptions and, no doubt, there will be much debate and controversy around them. This is work in progress. But this is exactly what we need: a transparent process to interrogate and improve our confidence in our future power choices.

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URGENT REFORMS ARE NEEDED FOR RESPONSIVE POWER SECTOR

Anton Eberhard

BUSINESS DAY
25 November 2013

“The era of big coal and big nuclear is at an end. What we now need is leadership and active participation of all in the sector to work towards the vision that we all want: a secure, competitively priced, environmentally sustainable and transparent power sector to fuel economic growth and improve the welfare of all our people.”

After Eskom issued an unprecedented warning of strain on the power grid, alerting key industrial customers that they could be required to cut usage by 10%, we are reminded yet again that urgent reforms are needed in South Africa's energy planning and investment paradigm.

We have learned the lesson — through Eskom's new coal power stations, Medupi and Kusile — that what you get with mega plants is not what you planned for: the cost of huge capital infrastructure, especially new, bespoke designs, often ends up being more than double initial estimates and is commissioned long after it is first needed, compromising electricity supply security.

Those who have long enough memories, or have researched Eskom's 1970s and 1980s build programme, know there were time and cost overruns in that programme too. Eskom's mistake was to think big coal would always be less expensive. Its other mistake was to believe it could manage all the risks of mega projects adequately.

The initial investment decision to build a new coal power station, Medupi, was made in 2005. Since then, the price tag has steadily increased to R105bn and is likely to rise to R150bn (including interest during construction and the costs of pollution control), with completion set, rather vaguely, for *"sometime in the second half of 2014"*.

The 1998 Energy Policy White Paper anticipated electricity demand would outstrip supply by 2007 and indeed, the first supply shortages became apparent around that time and culminated catastrophically in the week-long load-shedding in January 2008.

Since then, electricity demand has been flat and we have been spared serious power cuts — until last week, that is. The demand-supply balance will remain precarious — at least for another three years — and it is reasonable to suggest

constrained electricity supply is one of the reasons for South Africa's sclerotic economic growth.

This experience with coal power investments has to make us think more carefully and deeply about our planning processes, how we procure and contract new power and within what kind of market structure.

In the past, Eskom was solely responsible for generation expansion planning. In practical terms it still is, but it produces integrated resource plans (IRPs) under the guidance of the minister of energy, who has statutory responsibility to determine how much new generation is required, from which resources, when, and who should produce it.

The current IRP (IRP2010) is widely regarded as being out of date. Its demand forecasts no longer reflect reality. In addition, options such as gas were not adequately considered, and cost assumptions for nuclear and renewable energy are at variance with revealed market prices. The most recent nuclear power contract — Hinkley Point in the UK — has investment costs of \$7,900/kW. The IRP assumes \$5,800/kW. The most recent renewable energy tenders in South Africa reveal wind prices as low as 66c/kWh, less than half the cost of nuclear energy.

Despite these data deficiencies, there have been repeated statements that IRP2010 is our present plan and South Africa is committed to investing in a fleet of new nuclear power stations, despite international evidence that nuclear is both uncompetitive and unaffordable.

Add to this that the planning and procurement processes around nuclear energy are not transparent. Nuclear is explicitly excluded from new generation regulations under the Electricity Regulation Act and instead has a special Cabinet national nuclear energy executive co-ordinating committee.

Fortunately, we now have an example of a starkly different model of procuring new power — the Renewable Energy

Independent Power Producer Programme (REIPPP). An extraordinary success, the REIPPP has resulted in competitive bids for grid-connected renewable energy in South Africa.

In the space of the past two years, we have gone from fewer than a handful of IPPs to 64 new solar, wind and biomass projects with more than R100bn of private investment committed. This is a significant achievement. Not only have we demonstrated that the private sector is willing to invest in power in South Africa, we have also seen prices fall dramatically. Bid prices for solar photovoltaic power have fallen 68% and wind energy 42% over the past two years. In contrast, the costs of coal and nuclear plants continue to rise.

The experience of procuring renewable energy could not be more different than that for big coal and big nuclear. It's been a transparent and competitive process. We know exactly what the price will be for the next 20 years. The private sector has been willing to invest and it has assumed the major cost risks. If equipment or construction costs rise, the investors take a haircut, not the consumers.

The renewable energy programme has also involved a greater diversity of smaller projects that will be constructed in record time. And there is a pipeline of nearly 50 more projects — totalling 5,000MW — that are ready to be contracted.

Critics will say we cannot compare renewables — with their variable and low capacity factors — with the base load of big coal and big nuclear power. But ultimately, the test is price and reliability of supply. Renewables can be successfully integrated into the grid, especially when complemented by other flexible technologies such as gas power.

It is questionable and, frankly, potentially disastrous for our economy, to rely on an IRP that incorporates wrong cost assumptions and depends on huge, capital-intensive multiyear investments in the context of uncertainty over future demand and costs. We need a new, more responsive

planning and investment paradigm in the face of uncertainty. Several reforms are necessary.

First, we need an updated IRP and, thankfully, the Department of Energy has asked Eskom to do this. We also need to translate planning scenarios, in a less dirigiste manner, into investment decisions. The powers vested in the minister at present are unrealistically ambitious. The IRP or plan should be indicative, rather than mandatory, and the regulator should be allowed to license new IPPs even if they are not in the IRP, provided they contribute to supply security and are cost-competitive.

Third, we need to restructure the electricity supply industry to create a more level playing field between Eskom and private investors, and to create sustainable institutional capacity to plan, procure and contract new power. The proposed Independent System and Market Operator Bill is a step in the right direction, but it would be better if transmission were also separated from Eskom.

The National Development Plan lays out a credible and pragmatic future, suggesting that, in 2030, coal will contribute proportionately less to primary-energy needs, while gas and renewable energy IPPs, and imported hydroelectricity, will play a much larger role. Liquefied natural gas imports, or offshore gas, or shale gas, or pipelines from our neighbours, or a combination of these options, will probably transform our energy sector. The initiative of the Department of Energy to develop a national gas infrastructure master plan is thus welcome.

The era of big coal and big nuclear is at an end. What we now need is leadership and active participation of all in the sector to work towards the vision that we all want: a secure, competitively priced, environmentally sustainable and transparent power sector to fuel economic growth and improve the welfare of all our people.

Eberhard is a professor at the UCT Graduate School of Business.

“A new commission of inquiry will help us understand what has gone wrong. It can build on the restructuring proposals in the 1998 energy policy white paper, which were never implemented, and will provide the basis for opening the sector to investors and operators that can help restore electricity supply security and economic growth.”

COMMISSION OF INQUIRY COULD LEAD US OUT OF POWER CRISIS

Anton Eberhard

BUSINESS DAY
2 October 2014

SA Faces its worse power crisis in 40 years. Eskom has now admitted in Parliament that, after five years of electricity shortages, we are likely to face another five. A decade of power constraints will cripple aspirations to grow our economy.

We need to understand how this crisis emerged and how we can fix it; we need a commission of inquiry to consider deep-rooted power sector reforms.

The realisation that Eskom cannot fix the power crisis anytime soon is dawning slowly on the government. Eskom was seen as integral to the developmental state, and its mega coal power stations — Medupi and Kusile — were meant to power the economy into the future. However, we now know that these new power units are years late, way over budget, and are constraining rather than facilitating economic development.

The 100-day countdown to the synchronisation of the first of six power units at Medupi has begun. But this is mere spin: it will take at least another six months before this unit produces reliable power.

What has not been made public, but has been admitted privately by senior Eskom managers, is that the second Medupi unit - originally due six months after the first - will be commissioned only two and a half years later. Kusile's first unit is due to come online only in 2016, then there is the long, difficult task of commissioning the remaining units. Full power may be available only in 2020.

Ironically, while Eskom battles to add new generation capacity, it has lost the equivalent of a whole power station through breakdowns in its existing power generation fleet. Over a decade, plant availability gradually declined from 90% to 85% but, over the past three years, it has fallen precipitously to 75%. Eskom plans to reverse this trend but scant details are available on how it will accomplish that. By its own admission, 32 of its 87 coal-fired generation units need major surgery and three are in a critical condition.

A third crisis is about to hit Eskom: soon it will not have enough coal to power its generation stations. Adjacent coal mines and long-term supply contracts are reaching their end dates and, within a few years, uncontracted coal for Eskom will reach a staggering 80-million tons a year.

No investments are being made in new coal mines to make up these deficits. There are two major obstacles. Eskom's shareholder ministry declared, with little reference to existing law, that new coal mines supplying Eskom should have majority black ownership. The government has also hinted that coal might be declared a strategic mineral, with price regulation and export restrictions. None of the major international coal mining companies, which supply more than 75% of Eskom's coal, will invest under these conditions. What these proposals ignore is that the economics of many coal mines (especially in the Waterberg) are maximised when supplying coal to export markets and Eskom.

Cost overruns, delayed revenue from Eskom's new power stations, and poor performance of its existing fleet have worsened Eskom's financial crisis. Eskom is not generating enough cash to service the debt it needs to raise to fund the completion of its new power stations. The government has indicated that it plans to inject further equity to strengthen Eskom's balance sheet and that it will sell “nonstrategic” state assets. But privatisation takes time and will be opposed by members of the ruling alliance.

Credit-rating agencies have warned that a downgrading of Eskom's debt to junk is likely unless the government can provide more details on its financial rescue package. That would result in a downward spiral of more expensive loans, more pressure on its balance sheet, and even bigger challenges in the funding of its capital expansion programme.

If Eskom is unable to power economic growth and development in the short to medium term, what are the alternatives?

The past three years have seen an extraordinary response from the private sector. R120bn has been invested in 64 renewable energy independent power projects (IPPs), totalling 3,922MW, through a world-class, transparent and competitive procurement programme run by the Department of Energy with support from the Treasury. Prices for wind energy fell 42% between the first and the third bid rounds — to 66c/kWh — much lower than electricity from Eskom's new power stations. Many of these plants are already supplying the grid.

The renewable energy programme could be expanded but needs to be complemented by gas turbines that fill the supply troughs when the wind does not blow or the sun does not shine. In addition, many industries produce waste heat that can be converted through cogeneration plants to electricity. The department's IPP unit is also planning a competitive procurement for a new private coal-fired plant that will be cheaper, and will be built more quickly, than Eskom's Medupi and Kusile. Energy efficiency investments could also help mitigate supply shortages.

There are long-term plans for the development of gas resources and imports of hydroelectricity, but these will take time, as will nuclear power, which is a dangerous and costly distraction that will contribute nothing to resolving our immediate power crisis.

While alternative power sources provide promise, the institutional framework for securing and expanding their contribution to SA's energy mix is both fragile and restrictive. The IPP unit is an ad hoc arrangement that arose in response to Eskom's failure to procure and contract IPPs. It has too few staff, inadequate governance structures and relies on the expertise of a large (albeit excellent) team of local and international consultants.

Greater policy and regulatory certainty is necessary to free up private sector initiatives and investment. IPPs should

have nondiscriminatory access to the grid and be able to wheel out power to qualifying customers. And electricity planning, procurement and contracting functions need to be embedded in a capable organisation that is independent of Eskom's power generation so that IPPs can compete on an equal footing.

The creation of an independent system and market operator is one option. Another, more elegant, solution would be to spin off Eskom generators into separate companies and for Eskom to retain the grid, system operation and market functions. Eskom is now probably too big to manage effectively; it has 47,000 employees and annual revenues of R140bn. Separating out the potentially competitive generation units from the grid monopoly makes sense and has been accomplished in dozens of countries.

A commission of inquiry will help us understand the origins of the power crisis, the lessons that should be learnt, and could offer policy proposals for reforming the sector. There is a historical precedent for this. When Eskom last embarked on an ambitious programme to build new power stations, it led to a crisis in funding, cost and time overruns, and high tariffs. The recommendations of the 1983 De Villiers commission led to profound changes in Eskom's organisational culture and efficiency.

A new commission of inquiry will help us understand what has gone wrong. It can build on the restructuring proposals in the 1998 energy policy white paper, which were never implemented, and will provide the basis for opening the sector to investors and operators that can help restore electricity supply security and economic growth.

Eberhard is a professor at the University of Cape Town's Graduate School of Business and a member of the National Planning Commission.

“We need to rethink the structure of our electricity market. To oppose private participation is Neanderthal. We have to improve Eskom’s performance while facilitating private investment in supply. If we don’t solve this challenge, the middle and upper classes, and industry and commerce, will defect from national and municipal utilities. They will contract their own power supplies and will become angrier at the failures of Eskom and local government. And that will give rise to an additional challenge — a political one.”

REALISE THAT ESKOM ALONE CANNOT SOLVE OUR POWER CRISIS

Anton Eberhard

BUSINESS DAY

26 November 2014

Most South Africans are better off than they were in 1994. The growing middle class has done well. It is adept at finding alternatives to inadequate state services through private healthcare, schooling and security. Now it faces a new challenge: its electricity supply may not be secure.

Eskom has again embarked on load-shedding as a result of unplanned outages — Orwellian Newspeak for power cuts because power stations are broken. No one likes fumbling in the dark to find candles, or to have their lives disrupted by not being able to watch TV or cook their evening meal. The urban elites are being forced to discover the lives of the urban and rural poor.

It’s a future they do not want. They are becoming angry. They will be even angrier over the next three to five years as power cuts continue.

We now know that Eskom cannot supply all our power needs. Electricity sales are still below 2007 levels. Eskom cannot add enough new power generation capacity in time and the performance of its existing power plants is declining precipitously.

Consider the following: Eskom has about 42GW of installed power generation capacity plus an additional 2GW, or more, of power purchases from independent power producers and imports. Summer demand at present is about 31GW. Hence, we would expect Eskom to meet our demand easily. But it cannot. More than 8GW of power station capacity is unavailable due to equipment failures. Eskom power stations used to generate electricity, on average, more than 90% of the time. That has fallen to less than 75%.

Recently, a silo that feeds coal to Majuba power station collapsed. Eskom staff have managed to rig temporary coal feeders to the power units, but more than half the capacity of the station remains unavailable at a time when it is needed

most. A further 3.5GW need to be reserved for maintenance. The system is on a knife edge.

Eskom is losing capacity faster than it can build new power stations. It began construction of its two coal-fed mega power stations — Medupi and Kusile — as long ago as 2007-08. They should already have been online, adding 9.6GW and preventing power cuts. The latest target date for synchronisation with the grid of the first of six power units at Medupi probably will not be met. Informed sources in Eskom predict that the first 0.8GW unit will produce full power late next year and that the second unit will be commissioned only two years later.

Eskom’s power stations are also costing more than twice originally budgeted. Eskom is facing an unprecedented financial crisis. In its 90-year history it never relied on government funding; it always managed to fund its capital expenditure through raising debt in private capital markets. But in 2008 this changed; for the first time, the government provided a R60bn subordinated loan, the largest yet to a state-owned entity.

This quasi-equity injection has not been sufficient to restore Eskom’s balance sheet and the finance minister has now promised R20bn more. Nevertheless, Moody’s has downgraded Eskom’s credit rating to junk and the other agencies rank it precariously at the bottom of investment grade.

Eskom’s new power stations are late, the performance of its existing power units is declining, its costs are increasing, electricity sales are flat, and its financial and technical performance may continue to deteriorate. Eskom will do its best to keep the lights on. There will be days when there are no power cuts and we may become complacent again. But what is clear as daylight is that blackouts will return. We now have to accept that Eskom alone cannot solve the power crisis.

Marvelously, in this time of crisis, we have seen a successful programme run by the Department of Energy, supported by the Treasury, to procure R120bn in private investment in 64 renewable-energy independent power projects totalling 3.9GW. Between the first and third tender rounds, the tariffs for solar photovoltaic power plants fell 68%, and for wind farms 42%.

These projects are also delivering significant economic benefits. Bids were evaluated 70% in terms of price and 30% in terms of local content, employment, black economic empowerment, and community development criteria. Many of these wind and solar projects are already feeding electricity into the grid. They are all in rural areas and will deliver benefits worth more than R11bn to local communities.

However, some in the African National Congress’s economic transformation committee have argued against the growth of private investment in power production and have disputed the localisation, employment and black empowerment benefits. While the latter concerns need careful interrogation, monitoring and, if necessary, renewed incentives or sanctions, it appears that incorrect data have been peddled and the opposition to private investment is misguided.

Delays in finalising contracts in the present tender rounds are prejudicial for our investment climate and for new electricity supply.

Eskom is broke. It cannot consider any new major investments in power generation. Its focus has to be on completing its new power stations and to turn around the deteriorating performance of its existing fleet. In this context, it is peculiar that the Cabinet’s energy security committee appears to focus only on nuclear power when the update to the Integrated Resource Plan (IRP) suggests that, because of lower electricity demand, this decision is premature. Reliance on the 2010 version of the IRP is now irrational and potentially disastrous for our economy.

If data errors in the updated IRP were to be corrected, the model would recommend that even more wind energy should be built, at the expense of nuclear power. Proponents of nuclear power disagree and make three main arguments: that it is cost-effective, provides base-load power, in contrast to intermittent renewable energy, and that it is essential for meeting our climate change mitigation aspirations. None of these arguments holds up to scrutiny.

Nuclear energy is more expensive than alternatives. Contracted wind energy is now less than 70c/kWh and will soon be less than 60c/kWh. Nuclear power cannot compete.

The intermittent nature of wind and solar energy is easily complemented by load-following gas turbines to meet demand. The weighted average cost of wind plus solar plus gas is below that of nuclear energy and, in combination, these can meet demand as reliably. Further, IRP modelling scenarios show that a diverse mix of options (excluding nuclear, but with renewable energy, gas, hydroelectric imports and replacing some coal power stations) can still meet our climate change mitigation aspirations.

Nuclear power is a dangerous and costly diversion. It will do nothing to resolve our immediate or medium-term electricity security concerns and will most likely saddle us with debt for our children.

We need to rethink the structure of our electricity market. To oppose private participation is Neanderthal. We have to improve Eskom’s performance while facilitating private investment in supply. If we don’t solve this challenge, the middle and upper classes, and industry and commerce, will defect from national and municipal utilities. They will contract their own power supplies and will become angrier at the failures of Eskom and local government. And that will give rise to an additional challenge — a political one.

Eberhard is a professor at the University of Cape Town’s Graduate School of Business.

“It is time for the gloves to come off. The onus is on those who support the procurement of a fleet of new nuclear power stations to demonstrate that this initiative is not corrupt and will not be ruinous for South Africa’s economy.” With a background picture of Koeberg nuclear plant.

NUCLEAR TOO COSTLY AND SA DOES NOT EVEN NEED IT

Anton Eberhard

BUSINESS DAY
29 March 2015

It is time for the gloves to come off. The onus is on those who support the procurement of a fleet of new nuclear power stations to demonstrate that this initiative is not corrupt and will not be ruinous for South Africa’s economy.

We face a possible credit rating downgrade to junk, which will make us all poorer: it will cost a lot more to service our debt, there will be less money for social programmes, the rand will fall even more, and inflation will rise. Yet some still promote a massive nuclear programme that is not needed, that is more expensive and risky than alternative energy sources, that is hard to finance and that will create contingent liabilities for National Treasury when we can least afford them.

We don’t need to procure large chunks of new power now. Electricity demand isn’t growing: it’s falling and is lower than it was a decade ago. Depressed economic activity is partly the reason but it’s not the most important.

Electricity and economic growth data no longer track each other. The size of South Africa’s economy has continued to increase, albeit slowly, but electricity consumption has headed in the opposite direction. South Africa is not unique in this: countries such as Australia have seen a similar decoupling of energy and economic growth.

Could electricity demand in South Africa rebound if economic growth revives? Do we need to cater for depressed electricity demand as a result of Eskom supply constraints? Possibly. But we also need to recognise that there are profound changes to the energy-intensity of our economy as smelters and mines close. The structure of our economy is changing. A fourfold increase in electricity prices over the past decade has accelerated energy efficiency investments and energy conservation.

Official electricity demand forecasts and plans are obsolete. If demand for electricity were to reignite, it would fire off a lower base and the rate of growth would be lower. When we project demand forward to 2030 or beyond, it’s obvious

that we need a lot less power than was forecast in the earlier Integrated Resource Plan of 2010 (the basis for the 9600MW nuclear commitment).

But we also need to replace old coal power plants and compensate for the decline in the performance of Eskom’s existing power stations. I’ve taken all these arguments into account and calculate that we need about 17GW of new electricity generating capacity by 2030. Some may calculate a slightly higher number, others slightly less, but the required capacity will be close to this.

Few appreciate or understand that we have already ordered more power than we need by 2030. The new Eskom Medupi and Kusile coal power stations will add 9.6GW; its Ingula pumped storage scheme another 1.3GW. Two peaking power stations – Desisa and Avon, ordered by the Department of Energy (DOE) - will add a further 1GW. Contracted industrial cogeneration and the DOE’s coal IPPs will each add another 1GW, with plans for more.

In addition, 92 projects, totalling 6347MW, have been contracted over the first four rounds of the DOE’s renewable energy Independent Power Producers (IPP) Programme, amounting to R192bn in private investment. Granted, this is intermittent power and will need to be complemented by gas power plants, which the DOE plans to procure this year. More than 3GWs are in the pipeline. In the meantime, South Africa has negotiated 2.5GW of hydro power from the Inga3 development in the DRC, and is considering further hydro imports from the region.

Together, these power procurements exceed what we need over the next 15 years.

Nuclear energy is also more expensive than alternative power sources and the risks of cost overruns are greater. Eskom argues that its Koeberg nuclear power plant is cheap but this is old, generation II technology and provisions for multi-billion rand decommissioning costs are not fully accounted for.

New power plants will have to incorporate the much more expensive design features of safer generation III+ nuclear technology.

Our cheapest sources of power are now wind and solar energy. The DOE has awarded long term, fixed price contracts for wind energy as low as 57 c/kWh, far below Eskom’s average cost of supply. Renewable energy combined with gas power can offer reliable baseload supply at less than R1/kWh. Imported hydro and coal IPPs will also beat this price. These are facts.

I challenge any nuclear power vendor to sign a long term power contract at less than R1/kWh. Whenever I ask them what nuclear power will cost in South Africa they say “it depends” and “it will need to be negotiated”.

This is the point: nuclear vendors are loathe to submit to a competitive tendering process based on a long term fixed priced contract where they take the risks of construction time and cost overruns. But all the other energy technology providers are prepared to do so. This has been the basis of the success of the DOE’s IPP programme that has delivered such spectacular investment outcomes and price certainty for consumers.

So why would we opt for a nuclear procurement programme that aims only to select a strategic partner, with subsequent price negotiations that have uncertain outcomes?

Nuclear power plants are also hard to finance. A couple of years ago at Davos, the President was asked how 9600MW of nuclear power would be financed. His answer, remarkably, was: “I’ll speak to my Finance Minister”. Well, he would have had that conversation by now and it will be clear that there is no fiscal space to finance a programme that will cost more than a half a trillion rand, when we raise just over a trillion rand annually in taxes to fund all our needs. Debt financing is now the fastest growing component of the national budget and interest payments are more than twice what we spend on higher education.

Our traditional mechanisms for funding power investments are also constrained. Eskom’s balance sheet is stressed and it is struggling to raise sufficient debt on private capital markets to complete Medupi and Kusile. It has no possibility of raising finance for even one nuclear power station.

The private sector will not finance a nuclear plant in South Africa. The only possibility is funding from nuclear vendor countries. France will struggle: its nuclear company, Areva, is technically bankrupt and its latest UK nuclear contract - at £92-50/MWh (R2/kWh) - would be unaffordable for us. Russia will not be able to finance all of its nuclear ambitions. China is a possibility but financing will need to be backed by a long term contract with an agreed electricity tariff and the South African government will have to provide a sovereign guarantee and insurance cover, which will add contingent liabilities to National Treasury that will hasten a credit rating downgrade.

Eskom’s management recently expressed interest in further investments in large coal and nuclear projects. Its big coal, big nuclear and big networks strategy is neanderthal. Similar approaches have bankrupted large utilities, such as E-On and RWE in Germany.

Why would South Africa want to go down this route? It’s irrational. South Africa’s economic situation is precarious. Government now needs to act in concert and remove uncertainty around this nuclear folly. We don’t need it, it is too expensive and we cannot afford it. Those that continue to support the procurement of nuclear power need to demonstrate that it will not be ruinous and that the initiative is not corrupt.

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In South Africa, and indeed in many other emerging economies and developing countries, standard prescriptions of power sector reform, premised on unbundling, privatization and competition, and unconnected to the local political-economy, gain little traction. Instead, reforms need to incorporate effective incremental and pragmatic strategies and mechanisms for dealing with real challenges – such as contracting new power generation investments while improving the financial and technical performance of the off-taker utility.



THE POLITICAL-ECONOMY OF POWER SECTOR REFORM IN SOUTH AFRICA

Anton Eberhard

UNPUBLISHED NOTE

1 April 2016

How do we identify the right political moment for reform?

How do we build effective coalitions for reform?

How do we improve our understanding of the underlying political-economy dynamics of any reform situation and process?

And how do we design better reform proposals and processes?

I shall attempt to answer these questions with reference to my own experience over the past 25 years working as an energy policy researcher and advisor in South Africa. I directed an energy policy research group at the University of Cape Town. For a period, I was also a part-time regulator and, at different times, I have been contracted directly by the government, the utility, private investors and development finance institutions. In one way or another, I have been involved in all the major reform efforts in South Africa's power sector. I currently chair an advisory panel for the Deputy President who has been given overall responsibility for restoring electricity supply security.

South Africa has achieved some successes in electrification and renewable energy IPPs. Unfortunately, we have also had some spectacular failures: the structure of our power sector remains largely unchanged with the national utility, Eskom, now unable to supply sufficient power.

Our first success was an electrification programme that increased the proportion of households with a connection from less than a third to more than 85 percent. The moment was right for this intervention. In the transition from apartheid

to democracy all key stakeholders agreed on the desirability of extending basic services. Our contribution was to work with the new ruling party, the African National Congress, to model what was technically and financially possible and to lift the ambition of Eskom and the Department of Energy's electrification targets.

A more recent success has been the renewable energy IPP programme, which, in less than four years, has contracted 77, mainly solar and wind, projects totaling more than 5000MW and US\$17 billion in private investment, with remarkable price outcomes. Bid prices between rounds 1 and 4 have fallen, in nominal terms, 46 percent for onshore wind and 71 percent for solar PV. Renewable energy prices are now amongst the cheapest in the world, below 5 US cents /kWh for wind and 6.5 US cents/kWh for PV.

How was this accomplished, especially in a country where 90 percent of electricity is produced from coal and where the sector is dominated by the incumbent, state-owned, vertically-integrated utility Eskom?

When I reflect on what happened I conclude that it was, in part, a policy accident and, in part, a fortuitous coming together of a coalition of stakeholders that were able to push through a renewable energy and IPP agenda despite nascent opposition.

The genesis of South Africa's renewable energy programme was the commitment made by President Zuma, at the Copenhagen COP in 2009, that South Africa would reduce its carbon emissions below a business-as-usual trajectory. Although conditional, this commitment was translated into

a new electricity plan that adopted a carbon emissions cap and paved the way for a greater share of renewable energy. All stakeholders agree that the Copenhagen commitment was not pre-planned, was not widely canvassed and prematurely made public carbon-reduction targets that were originally envisaged as a fall-back, negotiating position and were based on a power point slide that crudely summarized a spreadsheet that was based on rather more complex long-term migration scenarios modeled at the University of Cape Town.

More extraordinary is the fact that a substantial grid-connected renewable energy programme followed, in part, simply because the President had made a commitment, but crucially also because a coalition of influence had developed between National Treasury, the Departments of Environment and Energy, private sector investors and, conditionally, also the Department of Trade and Industry. National Treasury's PPP unit was worried about electricity security and Eskom's finances, and wanted to widen the space for private sector investment. With the Department of Energy, it set up an IPP unit that was well resourced and could contract expert transaction advisors. Legislative and regulatory amendments quietly shifted responsibility for generation procurement to the Ministry of Energy. And as South Africa prepared for the Durban COP in 2011, the decision was made to commence procurement of "green technologies" that could potentially tap climate funding.

At the same time, the private sector had begun mobilising around the renewable energy feed-in tariffs published previously by the energy regulator (but which had not yet resulted in any projects). And when the National Treasury and

the Department of Energy drove the shift from feed-in tariffs to competitive tenders, the private sector provided market information that helped the IPP unit design effective auctions.

What's relevant to our debate today is that almost all of this happened without any direct policy advice from outsiders. The World Bank's GEF Renewable Energy Market Transformation Project in South Africa was premised on renewable energy targets set in a previous era and its influence on the design of the auctions was zero. Here was the perfect policy moment, yet it was almost entirely missed by international development finance institutions. 86 percent of finance for the successful bidders was provided by local South African banks. Only later did the IFC and other DFIs start participating. And the World Bank's add-on loan to Eskom for a wind and concentrated solar plant (CSP) has had minimal impact. Eskom has taken twice as long to reach COD than private developers using exactly the same wind turbines, and has been more expensive. And construction on its CSP plant has not even begun while private developers are close to connecting to the grid.

It will be interesting to see whether the coalition of interests that have enabled the success of South Africa's grid-connected renewable energy programme will hold. Black economic empowerment constituencies, endorsed by the ruling party, wonder how they missed rent-seeking opportunities. The auctions were robust, transparent and fair. Criticisms of the distribution of economic development benefits are beginning to emerge despite impressive localization, employment and community development outcomes having been documented. This will be an interesting space to watch: it is not only the development of policy that is important; sustaining policy coalitions and positions is equally challenging.



Identifying the right moment for reform, and understanding emerging coalitions for reform, are clearly critical. Washington-designed technical assistance projects, with lengthy approval processes, and pre-defined multi-year outcomes and milestone indicators, run the risk of irrelevance and missing these key policy reform moments. International technical assistance needs to understand and connect with local policy networks and communities.

Successful policy advice also requires an understanding of underlying political economy dynamics. Tracing the history of power sector reform proposals in South Africa is a depressing story of seemingly sound advice based on international experience coming close to implementation but then being aborted.

I was involved in the preparation of the Energy Policy White Paper published in 1998 that proposed the unbundling of Eskom, competition and private sector participation. It was an exhilarating and heady time. There was space for Intellectual vanguardism as the new South African democratic government was being formed. Policy proposals translated uncomplicatedly into detailed designs of an electricity market not dissimilar to Nordpool or PJM. But political and material interests soon coalesced, including ideological resistance to privatization and a fight-back from Eskom which continues to be a master at *“malicious compliance”* - it professes agreement with government policy positions but quietly and determinedly protects its dominant market position.

Eskom's shareholder Minister eventually pulled the plug on sector reform in 2004 and declared that Eskom would once

again be the supplier of last resort and would invest in mega new coal power stations.

Well, we now know the consequences of that decision: Eskom's new investments are late and way over budget, the performance of its existing fleet of power stations has declined precipitously, and its finances have collapsed. A once proud, investment-grade utility now has junk status, faces a liquidity crunch and cannot keep the lights on. Governance failures, a patronage board and politically-inspired management appointments don't provide much confidence for a turn-around under the prevailing governance and institutional arrangements.

The first set of power sector reform proposals failed to build coalitions for change or to understand the broader political-economy interests within the governing alliance. The current power crisis presents a new opportunity for reform and for being more strategic and prescient around connecting with those constituencies that demand change, such as National Treasury, which is concerned around possible impacts on sovereign debt, and industry, and more broadly electricity consumers and voters, which are now fed-up with power cuts and electricity price hikes.

It is not enough simply to be more sensitive to propitious moments for reform, or to understand coalitions of reform actors and institutions. Political-economy dynamics need to be probed including theoretical and analytical frameworks that facilitate an understanding of underlying material interests as well as the superstructure of institutions and ideology.

In reflecting on this history, I am conscious that key individuals have played seminal roles in successful reform efforts. But as the most famous and controversial of 19th century political-economists wrote: *“Men [and as we would say today, women] make their own history, but they do not make it as they please; they do not make it under self-selected circumstances, but under circumstances existing already”*.

Building on the work on the Nobel Prize winning new institutional economist, Douglass North, Elinor Ostrom and others, ex-World Bank governance expert, Brian Levy, now linked to Johns Hopkins and the University of Cape Town, warns against the brittle foundation of technocratic exhortation and the desperate search for reform champions and big moments of reform. In his recently published book: *“Working with the Grain”*, Levy suggests the focus should be on incremental but real results achieved by working with the critical mass of actors (individuals and institutions) who have standing and have a stake in the policy proposals, and are in a position to support and protect them in the face of opposition.

The emphasis on real and incremental results, made possible within a better understanding of political economy, finds resonance now in our renewed debates around what kind of power sector reform proposals and processes make the most sense.

In South Africa, and indeed in many other emerging economies and developing countries, standard prescriptions of power sector reform, premised on unbundling, privatization and competition, and unconnected to the local

political-economy, gain little traction. Instead, reforms need to incorporate effective incremental and pragmatic strategies and mechanisms for dealing with real challenges – such as contracting new power generation investments while improving the financial and technical performance of the off-taker utility.

Will we have greater policy success? This approach, at least, holds out hope.

Anton Eberhard is a Professor at the University of Cape Town's Graduate School of Business and is Chair of the SA Deputy President's Advisory Panel on Eskom's Turn-Around Strategy and Restoring Electricity Supply Security



SA'S POWER LIES IN BREAKING UP ESKOM'S MONOPOLY MODEL

Anton Eberhard

BUSINESS DAY
21 July 2016

The chairman of Eskom's board has written to the energy minister, stating that Eskom will not sign any further power purchase agreements with independent power producers (IPPs) after the current round of renewable energy projects are concluded. The potential consequences could be devastating for private investment in SA's power sector, and could terminate the government's most successful public-private procurement programme.

In the past five years, the Department of Energy has procured 92 privately funded power projects totalling R193bn. No other government-initiated programme rivals the scale of investment achieved, or the degree of transparency.

Despite intense competition - there were almost 400 bids over multiple tender rounds - there have been no legal challenges or accusations of impropriety or corruption. The programme has also resulted in economic development benefits in domestic manufacture, job creation, ownership and community development.

Eskom argues, misleadingly, that IPPs are too expensive. It quotes first-generation IPP costs and ignores the extraordinary fall in contracted renewable energy prices - more than 70% for solar photovoltaics and almost 50% for wind energy.

The latest IPPs, including bids for coal-powered projects, are now cheaper than Eskom's average cost of supply and about half the cost of its new power stations. And unlike Eskom tariff hikes, contracted IPP prices in the future cannot increase by more than the rate of inflation.

Eskom's opposition to competitive IPPs is not unexpected: that is what dominant incumbent utilities do. But clearly it is not in the public interest when the costs of Eskom's new power stations have doubled and their completion has been

delayed by many years. We forget too easily that these delays resulted in power cuts and a curtailment of economic growth.

Public utilities often argue they can deliver power at a lower cost than private investors. They have access to lower-cost capital, don't they? Eskom's precipitous credit rating decline from investment grade to junk status undermines that claim. And its inefficient deployment of capital has resulted in extra costs being passed on to customers through higher tariffs.

The IPP investment model is exactly the opposite. Private investors take the risk of cost overruns and consumers are protected by fixed-price, long-term supply contracts.

We should protect and accelerate the advances that have been made in opening parallel spaces for private investment in power-generation in SA. This requires regulatory and institutional reforms.

For now, IPPs have been procured by an IPP office that operates outside the department. That procurement function needs to be embedded in a secure institution linked to the heart of the electricity industry - namely the system and transmission operator that balances supply and demand, and connects power generators to distributors and customers.

What we need now is a separate state-owned electricity market and transmission operator that takes care of electricity planning, power procurement and contracting, system operation and balancing, and transmission. Let's call it Gridco. This obviously needs to be independent of Eskom generation and IPPs, so it can procure power at the least cost.

That implies separating Gridco from Eskom's generation business (let's call it Genco) to remove the kinds of conflicts of interest we are now experiencing in which Eskom

frustrates the entry of IPPs and private investment through disingenuous use of facts, political brinkmanship and what lawyers term malicious compliance, through the quiet subversion of government policy by actions such as delayed access or inflated grid-connection costs for IPPs. Breaking up Eskom may seem like a radical restructuring proposal. It is not. It was proposed in the Energy Policy White Paper in 1998.

And it's logical. It separates the potentially competitive elements of the electricity industry - power generation - from the natural monopoly component, transmission. It's also potentially efficient as it creates focused utilities - Genco and Gridco - in two very different kinds of businesses. And it's easily achievable, as evidenced in more than 90 countries around the world.

The first step in this restructuring path is an accounting separation between Eskom's generation and transmission divisions.

The energy regulator, Nersa, already requires this. The next step would be to create separate Eskom Genco and Gridco subsidiaries, again relatively easily achievable within the existing Eskom Holdings company structure. And the logical further step would be to create Genco as a separate state-owned company.

The difficulty in restructuring state-owned utilities is that when they are in crisis, governments are careful not to propose interventions that might further destabilise them. But when the crisis recedes, so does the political imperative for restructuring. When Eskom was load-shedding, the focus was on immediate measures to keep the lights on and on improving its financial viability. It was more difficult to agree on far-reaching reforms that might prevent similar crises occurring in the future.

The restructuring of state-owned enterprises requires vision, leadership and commitment to remove impediments to investment and achieve efficiency improvements that facilitate economic growth. Within the existing contested political terrain, that might be a big ask. But surely individuals and institutions can rise to that challenge?

Instead of Eskom's leadership defending an old, vertically integrated, monopolistic electricity industry model that international experience shows is moribund, would they not want their legacy to be distinguished by a reforming zeal that sets our country on a different path?

This is not a pipe dream. Until recently, SA had no IPPs. But in the past four years, we have attracted more private investment in power than the rest of sub-Saharan Africa has in the past 25 years. We are ranked by Bloomberg New Energy Finance among the top 10 countries

The renewable energy IPP programme is being extended to industrial cogeneration, coal and gas power projects. There are huge opportunities for the advancement of black-owned power companies and the development of local competencies.

It is not naïve to believe SA's electricity sector can, and will, be restructured. I've seen it happen in many countries. Perhaps, SA's fiscal position will have to deteriorate further before we accept we can no longer fully fund our public utilities - we've pumped R83bn into Eskom since 2008 - and that a greater openness to private investment is inevitable. It would be better to start that reform process now, before we are once again in crisis.

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The president, his Cabinet and the ANC's national executive committee have to take responsibility for this failure in governance and economic management. Either they have to demonstrate seriousness in growing our economy and employment or we have to conclude that their failure to make progress in obvious policy areas, or to speak up against blatant corruption, is a tacit acceptance of the inevitable decline of our infrastructure, our economy and, ultimately, of a once principled liberation movement

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NEW LEADERS CAN FIND A PATH OUT OF POLICY FAILURE SWAMP

Anton Eberhard

BUSINESS DAY
09 September 2016

There are now so many instances in which the government is failing to make and implement sensible policy decisions around SA's economic infrastructure, where good governance is being undermined and corruption is out of control, that the country's prospects for economic growth and development are bleak.

Any one instance of government dereliction is troubling, but the cumulative effect of multiple areas of insouciant neglect, or deliberate abuse, of our infrastructure industries indicates a government that has little understanding of the importance of energy, telecommunications, transport and water for economic growth, employment creation and poverty reduction. Or, perhaps, factions within the government and the governing party have ceased caring and are focused only on diverting contracts to the favoured few or to the party itself?

Here's an incomplete catalogue of critical areas where the government is fatally complacent or where there are serious allegations of rent-seeking and corruption.

Amendments to the Mineral and Petroleum Resources Development Bill, first published in 2012, remain unresolved, creating uncertainty not only for mining investments, but also for exploration. Neither the departments of mineral resources nor energy show any urgency in clearing the regulatory, licensing and environmental impediments to onshore and offshore drilling that might prove economic gas resources that could transform our energy economy through a more sustainable energy mix, improved balance of payments and more employment opportunities.

One of the Central Energy Fund's companies recently sold 10-million barrels of our oil reserves at below market prices without the permission of the energy minister or the Treasury. Insufficient funds are available for purchasing new strategic stocks. No one has yet been prosecuted.

Eskom is the largest of SA's state-owned enterprises (SOEs), with annual revenue of R160bn. Its largest expenditure item is coal. Contrary to recent Eskom news statements, its long-term cost-plus contracts with tied coal mines were cheap. But these contracts are not being renewed and are being given to smaller, new mining entrants. That, in itself, is positive for empowerment, but little investment is being made in new mines. Eskom is openly defying the Treasury's scrutiny of these contracts. The government has not acted on the recommendations of the Coal Road Map published in 2013. The risks of coal shortages and steep increases in coal and electricity prices are real.

There are also allegations of corruption in Eskom contracts for the refurbishment of Koeberg nuclear generators and the construction of new power stations, such as Ingula. Eskom tariffs have increased four-fold in a decade, mainly as a result of cost and time overruns on its new power plants. Restructuring recommendations in the energy policy white paper of 1998 have not been implemented and Eskom now frustrates the entry of cheaper private independent power projects through delayed and inflated transmission connection costs and by refusing to sign new power-purchase agreements.

Government interventions in the telecommunications sector have been either misdirected or absent. Telkom was partially privatised while still enjoying a monopoly in fixed-line telephony and other services. The result was price gouging and the frustration of competition. Huge budgets dedicated to universal access were wasted as Telkom lost more phone connections than it gained.

Telkom's control of the "last-mile" in copper and slowness in moving to fibreoptic cables and Broadband Infraco's sclerotic performance, among other policy and regulatory failures, have resulted in South Africans having slower and more limited internet access than many of our African neighbours. SA has missed the deadline set by the International

Telecommunications Union to migrate our television services from analogue to digital signals by 2015, which would have freed up radio spectrum for broadband, but few in the government seem to care. There is no urgency. Instead, the fight seems to be about which companies will benefit from contracts.

While Transnet has produced reasonable financial results, it has come at a cost to the economy. Transnet is unique in being an integrated rail, ports and pipeline utility. Few other transport companies in the world enjoy such protected market share. Unsurprisingly, its ports, general rail freight and pipeline charges are way higher than international norms. It costs more for South African companies to export or import. Treasury studies and proposals for transport reform, dating to 2007, gather dust.

Allegations of corruption in rail tenders abound. Shell companies, with no experience, have been awarded multibillion-rand tenders. It is alleged that some of this money went to the ANC. The transport minister has tried to close down investigations. Like the arms deal, these allegations will not go away. There will be court cases. In the meantime, the renewal of rolling stock and other infrastructure is delayed and is much more expensive than it should have been.

South African Airways lurches from crisis to crisis with naive and crude attempts to extract rent as it desperately seeks new funding. Its management has been decimated and whistle-blowers victimised. Yet the president blocks all attempts to replace its disastrous board.

And the future of water supply to the Johannesburg region is threatened by delays to the second phase of the Lesotho Highlands Water Project caused by political interventions seeking favoured access to contracts for party-connected individuals.

None of the above incidents will be new to readers of this newspaper. However, the full catalogue of indifference or malfeasance reveals the extent to which the governance of key infrastructure ministries and SOEs has broken down. The boards and management openly defy the public enterprises ministry and the Treasury. This would have been inconceivable a few years ago.

None of the recommendations of the Presidential Review Committee on State-owned Entities, tabled in 2013, has been adopted. The one claimed recommendation — that the president chair a new State-Owned Companies Co-ordinating Council — is a fiction and does not augur well given previous actions to undermine good governance in SOEs.

SA needs to enhance economic opportunities for black-owned companies, but rent-seeking attempts are a zero-sum game that will destroy financial sustainability.

The president, his Cabinet and the ANC's national executive committee have to take responsibility for this failure in governance and economic management. Either they have to demonstrate seriousness in growing our economy and employment or we have to conclude that their failure to make progress in obvious economic and infrastructure policy areas, or to speak up against blatant corruption, is a tacit acceptance of the inevitable decline of our infrastructure, our economy and, ultimately, of a once principled liberation movement.

It is entirely possible to make systematic and measurable progress across all these identified infrastructure areas. It would create a positive shift in investment flows and optimism in SA's ability to deal with the challenges of unemployment, poverty and inequality. But that would require a renewal of political leadership.

Eberhard is a professor at the University of Cape Town's Graduate School of Business

“ Eskom is becoming a problem. It sells less electricity than it did 10 years ago, yet it employs 50% more staff, its coal costs have trebled, and its tariffs and revenues have increased fourfold. ”

EXPECT LEGAL CHALLENGE IF ESKOM STICKS WITH ITS IRRATIONAL PROCUREMENT COURSE

Anton Eberhard

BUSINESS DAY

25 October 2016

Eskom is becoming a problem. It sells less electricity than it did 10 years ago, yet it employs 50% more staff, its coal costs have trebled, and its tariffs and revenues have increased fourfold. Now Eskom is blocking cheaper independent power producers (IPPs) and is attempting to mislead the government and the public by arguing that renewable energy and gas power are too expensive, and that Eskom should fast-track the procurement of nuclear energy.

Eskom was a problem before, under apartheid. It served mainly white South Africans. And in the previous century, it was also not particularly efficient in building new power stations. There were cost and time overruns, and there were periods with unacceptably high increases in electricity tariffs.

However, at the birth of our democracy, Eskom was in a sweet spot. It had surplus generation capacity, which was mostly paid for. Its short-run marginal cost of production was low. It entered into two multiyear price compacts with the government to reduce the real price of electricity. Eskom also helped deliver the most successful component of the governing party's reconstruction and development programme by increasing the proportion of South Africans with access to electricity from just more than a third to nearly 90% today.

But between 2006 and 2015, power cuts constrained economic growth, employment and poverty reduction. Now, Eskom is no longer affordable and it is frustrating, rather than supporting, the development of the economy. It is proving to be a poor instrument for a developmental state.

The big questions and decisions for the future are around investment in new power. The electricity business is very capital-intensive, more so than any other infrastructure sector. Eskom's balance sheet, investments and revenues dwarf those in Transnet, Telkom or any other state-owned enterprise. It's crucial that investment decisions are least-cost and that capital is executed efficiently.

It is 12 years since then public enterprises minister Alec Erwin instructed Eskom to build new coal power stations. Medupi and Kusile will be completed only in the early 2020s, nearly 20 years later. Costs have more than doubled and will likely end up being three times over budget.

During this time, Eskom lost its treasured investment-grade rating and its bonds are now rated junk. As a result, Eskom's capital raising is a lot more challenging and expensive. Eskom has 10 times more debt than a decade ago, and its new coal power stations are not even half-built.

Within this context, it is flabbergasting that Eskom now plans to rush the procurement of a fleet of new nuclear power stations. It claims we need large, additional increments of power soon, that nuclear energy is the cheapest option and that it can easily be financed. All of these claims are wrong.

SA's electricity plan - the Integrated Resource Plan (IRP) - is being updated by the Department of Energy. It incorporates new projections on electricity demand and new comparative costs for supply options. As reported in this newspaper, the new plan suggests that nuclear power might only be required in the 2030s - in other words there is no rush to start a procurement process now.

Crucially, the computer model informing the new electricity plan picks nuclear energy only if the contribution of renewable energy is constrained- in other words n any year. Solar and wind energy are now so cheap that the model picks these supply options first, along with gas power to ensure system reliability.

SA has favourable solar and wind resources, and land is not a constraint. Other countries have built much more renewable energy than we plan to. The constraints in the updated plan regarding the contribution of solar and wind, combined with gas, are thus entirely artificial.

The modelling tools used by staff supporting the department are cutting edge and widely used internationally. Research institutions in SA have the same software. They can mimic the demand forecasts and cost assumptions made by the department and can model similar scenarios and energy mixes; indeed they are already doing so. It will not be very clever for the department or Eskom to now amend or subvert the new IRP.

It is crystal-clear: a least-cost electricity plan for SA does not include nuclear energy, even if a carbon emissions cap is imposed that would enable us to meet our climate change mitigation obligations. Thus if Eskom proceeds with the procurement of nuclear power stations, it will not be a rational decision and almost certainly will be subject to legal challenge. The original decision by the Cabinet to opt for nuclear energy is already in the courts.

Eskom executives have launched a concerted public relations campaign in recent months, arguing that renewable energy is too expensive. They are correct that the first generation of IPPs in SA were costly, but they omit the inconvenient fact that prices for solar and wind in the latest IPP bid rounds have dropped dramatically and are now down to 62c/kWh, cheaper than Eskom's average cost of supply at 84c/kWh and far cheaper than Medupi or Kusile or new nuclear power. The recent coal IPPs, announced by the department, are also cheaper than Eskom's new build.

The debates about the relative costs of different power sources can be settled if we look at actual power contracts. Fortunately, the department's IPP programme is transparent: private companies have signed 20-year contracts at fixed prices, adjusted only by inflation. There can thus be no controversy about the costs of solar and wind energy. They are now the cheapest sources of grid power. Perhaps the only question is how cheap. Recent renewable energy auctions in Peru, Mexico, Chile and Dubai have seen prices for wind and solar fall even further.

Of course, we should not consider only the costs of individual supply technologies, but also the system costs to ensure reliable supply. Solar and wind resources are variable and need to be complemented with flexible power sources such as gas, hydroelectricity, storage or demand-side management. The blended cost of solar plus wind plus flexible system resources is cheaper than Eskom coal or nuclear.

Choices about SA's future energy mix are far too important to be made by a couple of Eskom executives. The updated IRP should be published and open to public scrutiny and consultation.

Eskom now needs to be restructured. It is intolerable that it is frustrating the entry of IPPs by delaying and inflating the cost of transmission connections and refusing to sign contracts. Eskom should not be in the position where it is building its own generation capacity and is also the arbiter of whether private investors can enter the market or not. It cannot be a player and a referee.

As I have argued previously on these pages, Eskom's generation business needs to be hived off into a separate state-owned company. Then the residual wires business, along with the system operator, could provide a neutral platform for electricity planning and procurement of least-cost power. This form of power-sector restructuring is the norm in most industrialised countries and emerging economies, and many have undertaken further reforms that allow private generators to trade in power exchanges or directly with customers.

The Cabinet and Eskom's board would do well to pause before rushing into the procurement of new nuclear power plants. South Africans are watching and they will challenge such a decision. Now is the time to reform our power sector.

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POLICY AND GOVERNANCE FAILURES IN KEY INFRASTRUCTURE SECTORS WIDEN SPACE FOR CORRUPTION AND UNDERMINE ECONOMIC GROWTH

Anton Eberhard

MAIL & GAURDIAN

22 November 2016

South Africa may have avoided a technical recession, thanks to better than expected growth in the second quarter, but the country's medium-term prospects for growth and development are bleak.

One would have thought that the most serious election setback since 1994 would have re-energized the ruling party and prompted a commitment to improve government performance, especially around actions that would facilitate economic development and widen opportunities for all South Africans. One would have hoped for a more coordinated and purposeful commitment to implement sensible policy, improve governance of key economic sectors and deal with corruption.

Instead we see a doubling down, a going for broke by factions in the ruling party. Its most obviously manifestation has been the take-over of the Hawks and SARS, to blunt their investigation of those close to the president and the party leadership, and the crude attempts to undermine the integrity of the Finance Minister and the effectiveness of National Treasury.

At a time when South Africa faces its most challenging economic conditions, as unemployment and poverty worsen, these actions are both reckless and shocking. But, we should understand that they are the culmination of years of aggressive rent-seeking, particularly in our large economic infrastructure sectors such as energy, telecommunications, transport and water.

Here's a brief, and incomplete, catalogue of critical areas where government remains fatally complacent around policy implementation, where governance is systematically being undermined and where corruption is out of control.

Amendments to the Minerals and Petroleum Resources Development Bill, first published in 2012, remain unresolved creating uncertainty not only for mining investments but also for exploration for petroleum and gas resources.

A single off-shore well costs upwards of R1.5 billion. Deep, off-shore drilling is also technologically complex. We need large, experienced companies to invest but local antipathy to these international firms, and disputes around government's, take frustrate development.

Neither the Department of Mineral Resources, nor the Department of Energy, show any urgency in clearing the regulatory, licensing and environmental impediments to on- and off-shore drilling that might prove economic gas resources that could transform our energy economy through a more sustainable energy mix, improved balance of payments and more employment opportunities.

One of the Central Energy Fund's companies recently sold 10 million barrels of our oil reserves at below market prices without the permission of the Energy Minister or National Treasury. Insufficient funds are available for purchasing new strategic stocks, which the International Energy Agency recommends should cover 90 days of consumption. No one has yet been prosecuted.

Eskom is the largest of South Africa's state-owned enterprises with annual revenues of R160bn. Its largest expenditure item is coal. Contrary to recent Eskom press statements, its long-term cost-plus contracts with tied coal mines were cheap. But these contracts are not being renewed and are instead being given to smaller, new mining entrants. That, in itself, is positive for empowerment, but little investment is being made in new mines. Eskom has openly defied National Treasury's scrutiny of these contracts.

Government has not acted on the recommendations of the Coal Road Map published in 2013. The risks of coal shortages and steep increases in coal and electricity prices are real.

There are also allegations of corruption in Eskom contracts for the refurbishment of Koeberg nuclear generators and the construction of its new power stations, such as Ingula. Eskom tariffs have increased four-fold in a decade, mainly as a result of cost and time overruns on its new power plants.

Restructuring recommendations in the Energy Policy White paper of 1998 have not been implemented and Eskom now frustrates the entry of cheaper private independent power projects through delayed and inflated transmission connection costs, and through refusing to sign new power purchase agreements.

Certainty around our energy future could be created through publication of an updated Integrated Resource Plan. All the latest models pick renewable energy and gas, not nuclear energy, which is more expensive, has larger risks of

cost-overruns, and is difficult to finance. Yet, government stubbornly sticks to its nuclear plans with its rent-seeking opportunities for a connected few.

Government interventions in the telecommunications sector have been either misdirected or dilatory. Telkom was partially privatized while still enjoying a monopoly in fixed-line telephony and other services. The result was price-gouging and the frustration of competition. Huge budgets dedicated to universal access were wasted as Telkom lost more phone connections than it gained.

Telkom's control of the *"last-mile"* in copper, and slowness in moving to fibre-optic cables, Broadband Infracos's sclerotic performance, amongst other policy and regulatory failures, have resulted in South Africans having slower and more limited internet access than many of our African neighbours. South Africa has missed the deadline set by the International Telecommunications Union to migrate our television services from analogue to digital signals by 2015, which would have freed up radio spectrum for broadband, but few in government seem to care. There is no urgency. Instead, the fight seems to be about which companies will benefit from contracts.

While Transnet has produced reasonable financial results, it has come at a cost to the South African economy. Transnet is unique in being an integrated rail, ports and pipeline utility. Few other transport companies around the world enjoy such protected market share. Unsurprisingly, its ports, general rail freight and pipeline charges are way higher than international



norms. It costs more for South African companies to export or import. National Treasury studies and proposals for transport reform, dating to 2007, gather dust.

Allegations of corruption in rail tenders abound. Shell companies, with no experience, were awarded multi-billion rand tenders. It is alleged that some of this money went to the ANC. The Transport Minister has tried to close down investigations. Like the arms deal, these allegations will not go away. There will be court cases. In the meantime, the renewal of rolling stock, and other infrastructure, is delayed and is much more expensive than it should have been.

South African Airways lurches from crisis to crisis with naïve and crude attempts to extract rent as it desperately seeks new funding. Its management has been decimated and whistle blowers victimized. Yet the President has insisted on retaining its chairperson who was quoted in court papers as saying: “you are not feeding me, when other chief executives are feeding their chairpersons.”

And the future of water supply to the Johannesburg region is now threatened by delays to the second phase of the Lesotho Highlands Water Project caused by political interventions which seek favoured access to contracts for party-connected individuals.

Each of the above incidences of policy inaction, poor governance or potential corruption has been documented. However, the full catalogue of indifference or malfeasance

reveals the extent to which the governance of key infrastructure ministries and state-owned enterprises (SOEs) has broken down. The boards and management of SOEs now openly defy the Public Enterprises Ministry or National Treasury. This would have been inconceivable a few years ago.

None of the recommendations of the Presidential Review Committee on State Owned Entities, which was tabled in 2013, have been adopted. The one recommendation that the President has said he would follow – viz. that the President should chair a new State-Owned Companies Coordinating Council - is in fact a fiction and was not in the Committee’s report. This does not augur well, given previous actions to undermine good governance in SOEs, including blocking the appointments of reputable, principled and experience people to the SOE Boards.

South Africa does need to enhance economic opportunities for black-owned companies, but current rent-seeking attempts are a zero sum game that will destroy the financial sustainability of SOEs and ultimately cripple the economy.

The President, his cabinet and the ANC’s National Executive Committee have to take responsibility for this failure in governance and economic management. Either they have to demonstrate seriousness in growing our economy and employment or we have to conclude that their failure to make progress in obvious economic and

infrastructure policy areas, or to speak up against blatant corruption, is a tacit acceptance of the inevitable decline of our infrastructure, our economy and, ultimately, of a once principled and proud liberation movement.

It’s entirely possible to make systematic and measurable progress across all these identified infrastructure areas. An updated, least-cost electricity plan could be published tomorrow. Irrational plans to procure costly nuclear power that we don’t need could be abandoned. The migration from analogue to digital signals could be expedited and radio frequency allocated for broadband, which has large economic multiplier effects. The Boards of our state-owned enterprises could be renewed to improve governance, root out corruption and improve infrastructure services. And policy recommendations to reform our SOEs could be implemented: pragmatic restructuring options are on the table.

A government that was serious about economic development could initiate credible actions in all of these areas. It would create a positive shift in investment flows and optimism in South Africa’s ability to rise to the challenges of unemployment, poverty and inequality. But that would require a renewal of political leadership.

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“The changes and innovation disrupting the power sector are unprecedented and likely irresistible. In a few years no one will even remember Zuma and Mahlobo’s vain efforts to subvert our energy future”

SA IS SWITCHED ON TO MAHLOBA’S WHEELING AND DEALING IN NUCLEAR BID

Anton Eberhard

BUSINESS DAY

30 November 2017

President Zuma and energy minister Mahlobo are so desperate to restart the process to procure a fleet of new nuclear power stations that they are pressuring the department of energy (DoE) to produce a new “policy adjusted” electricity plan, which overturns the conclusions of its own computer modelers that nuclear energy is not needed.

The outputs from the world-class PLEXOS model, which has been run for the DoE so that it can publish an updated electricity integrated resource plan (IRP), clearly show that the energy mix that will provide South Africa with an adequate and reliable electricity supply, at least-cost, excludes nuclear energy.

Much has changed since the official IRP 2010 was published. Electricity demand is lower than a decade ago and the most optimistic new forecasts assume much slower increases in electricity demand, even if South Africa’s economy expands more rapidly.

The relative costs of power generation options have also changed. The updated model incorporates latest cost data, including actual solar and wind energy prices contracted in South Africa’s renewable energy IPP programme. The costs of nuclear energy used in the model were supplied by the DoE.

I could quibble with the assumptions of the modelers. Their electricity demand forecasts are almost certainly too high, as are their assumptions on renewable energy costs. This year, in a number of emerging economies, we have seen solar and wind auctions deliver prices that are half those registered in the last bid round in South Africa. And the DoE’s nuclear cost assumptions seem much lower than many recent contracts, such as the Rosatom deal in Egypt.

But even with these conservative assumptions, nuclear energy is not picked by the cost-optimising model in any scenario other than one where artificial constraints are placed

on how much solar and wind energy can be built and where additional carbon budget limits are imposed. Even in this extreme scenario, nuclear energy would only be required after 2039.

These outcomes are consistent with the IRP update undertaken by the DoE in 2016 under the direction of the Ministerial Advisory Council on Energy and with the CSIR’s model which used the same PLEXOS software and assumptions as the DoE. A few years back, the National Planning Commission asked South Africa’s premier university-based energy modeling group – the University of Cape Town’s Energy Research Centre – to run an independent model. They too came to the same conclusion.

These outcomes have not been welcomed by the nuclear section in DoE and the modelers were asked to run a scenario where 9600MW of nuclear power stations are “hardwired” or forced into the model. Fortunately the modelers also calculated the additional cost of electricity in this scenario – R800 billion over 20 years!

Originally, nuclear proponents relied on the IRP 2010 for their justification for starting the procurement of new power stations. Now that the updated IRPs indicate we don’t need nuclear energy, they question the usefulness of the models (even with their conservative input assumptions) and instead emphasize the need for a “policy adjusted” plan that takes into account other objectives such as job creation, local manufacture, use of local energy resources and reduced carbon emissions and water usage. But other technologies fare even better than nuclear in meeting these additional criteria.

The optimal, least-cost mix recommended by all these models includes solar and wind energy complemented by pumped storage, hydro-electricity and gas turbines or engines, on top of legacy coal fired powered stations.

But, cry the nuclear advocates, solar and wind energy are intermittent, we cannot rely on them. However, the planning models fully account for their variability and pick a least-cost energy mix that meets specified reliability criteria at any time of the day or year. Indeed, the modelers assert that their optimal mix includes sufficient dispatchable power even when the sun doesn’t shine and the wind doesn’t blow.

But, says our energy minister, our energy security depends on energy diversity – we need to develop all sources of energy. I have never understood this line of argument. What kind of policy suggests that we do everything; use all of the above? Good policy requires informed choices. We clearly need to opt for the lowest cost power sources that offer us adequate power to grow our economy and improve the welfare of all our people whilst still meeting broader social and environmental goals.

As arguments run out for nuclear, the diehard proponents point to Denmark and Germany which have high shares of renewable energy and high electricity prices, conveniently omitting that these countries were early movers when solar and wind energy were still expensive and that they created large new markets which catalysed higher production volumes and innovation which led to price reductions from which we can now benefit.

There are no sound arguments for building nuclear power plants. South Africa does not need them, they are too expensive and they will be difficult to finance. Even if Russia or China offer financing terms, National Treasury will have to provide sovereign guarantees to back the loan agreements and there is no fiscal space left for this. With no arguments left, those who continue to advocate restarting a nuclear programme need to demonstrate that their motives are not corrupt.

A judgement of the Western Cape High Court earlier this year made clear that the energy minister will need to make a new determination in terms of the Electricity Regulation Act if he wishes to procure nuclear energy and this should be based on a valid electricity integrated resource plan. The Minister also needs to obtain the concurrence of the National Energy Regulator of South Africa who in turn has to first undertake a through public consultation process.

The minister is trying to rush through the first step by doctoring the IRP update. He will be challenged in court. The subsequent steps will also be scrutinized by civil society organisations.

Zuma has fired four previous energy minsters for not delivering his nuclear programme. I do not easily celebrate government incompetence, but in this instance I do. Minister Mahlobo is demonstrating more energy and determination, but he too will fail. What a waste of time and resources!

Nevertheless, our energy future is bright. South Africa has amongst the best solar and wind resources in the world. We have the institutions, capabilities, experience and private finance to capitalize on these new low cost technologies, including complementary, flexible power resources, storage and demand-side management. We also have the minerals and rare earths that the manufacture of these technologies require, and South Africa can grow new exports. The changes and innovation disrupting the power sector are unprecedented and likely irresistible. In a few years no one will even remember Zuma and Mahlobo’s vain efforts to subvert our energy future.

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“It’s time to call out the bluff and bluster of those who belligerently oppose private investment in renewable energy independent power projects (IPPs) and the transition away from coal and nuclear set out in South Africa’s new electricity road map, the integrated resources plan (IRP).”

DIRTY BUSINESS AS COALITION OF THE SIDELINED DISTORT THE CLEAN ENERGY DEBATE

Anton Eberhard

SUNDAY TIMES
28 November 2018

It’s time to call out the bluff and bluster of those who belligerently oppose private investment in renewable energy independent power projects (IPPs) and the transition away from coal and nuclear set out in South Africa’s new electricity road map, the integrated resources plan (IRP). It is clear that some are frustrated and angry that they will no longer have access to special deals with Eskom or the coal and nuclear industries, others may feel marginalized and alienated from president Ramaphosa’s new political dispensation – but their false narratives around IPPs and the IRP should not cloud our policy and investment choices.

The contestation around South Africa’s energy future matters. The country can either embrace unprecedented global innovation in new low-cost energy technologies and competitive markets or it can succumb to a parochial and reactionary minority who are more interested in rent-seeking opportunities in old and expensive technologies that will impeded social and economic development. If the outcome is to be supportive of real economic transformation, hard facts and economic analysis must guide public debate and engagement.

Yet, it is a coalition of the recently sidelined and embittered that is currently dominating many public platforms, inciting resentment and opposition. An unlikely coalition, it includes elements from the Zuma-wing of the ANC, the EFF, actors with vested interests in nuclear energy and coal, such as the Nuclear Industry Association of South Africa and the Coal Transporters Forum, unions NUMSA and NUM, a small

minority of energy professionals, including a number of ex-SOE board members and managers accused of corruption, the notorious Black Land First, and a collection of recently-formed advocacy groups such as Transform SA and the South Africa Energy Forum.

Bizarrely, this coalition of the disaffected has been joined on social media by individuals who deny anthropogenic climate change, as well as some members of the Free Market Foundation who apparently believe that government-backed coal and nuclear power projects, with a record of massive cost and time overruns, are preferable to South Africa’s highly competitive IPP programme, which has attracted more than R200 billion in private investment in projects which have been built on-time and on-budget and, in the most recent bid round, without subsidies.

Their anti-IRP and IPP advocacy is founded on four fallacies: they argue that solar and wind energy are unreliable and expensive; that the IRP does not adequately take into account the impacts and costs of renewable IPPs on the rest of the electricity system; that these IPPs don’t promote local industrialisation or create enough jobs; and that the procurement of IPPs has been corrupt and has benefitted the president’s family and a new politically-connected elite. It is easy to demonstrate that each of these arguments is without substance.

While solar and wind energy were costly in the past, innovation and expanding global markets have led to dramatic cost

reductions over the past five years and they are now the cheapest new sources of grid-connected power in South Africa. The most recent bid rounds in the renewable energy IPP procurement programme yielded prices of ZAR 62c/kWh, a third lower than Eskom’s average cost of supply.

International renewable energy auctions are yielding even lower prices. For example, Egypt has solar contracts below US 3c/kWh and Mexico below US 2c/kWh. If South Africa ran a new competitive procurement now, solar and wind energy bids would be below ZAR 50c/kWh, lower than the operating costs of Eskom’s most expensive coal power stations.

Ignoring these facts, some anti-IPP advocates point to countries with large shares of renewable energy where tariffs are high, for example, Germany and Denmark. But these countries were first movers, procuring renewable energy a decade ago when costs were much higher, and a large proportion of their retail tariffs are made up of taxes, making them unfit for comparison.

Others question the costs on the rest of the system for back up and ancillary power services, arguing that the IRP ignores these costs. In fact, the objective function of the IRP is the least-cost mix of power sources which delivers an acceptable reliability standard.

The PLEXOS computer model, which underpins the IRP, is the industry gold standard and is used by a large number of utilities and countries around the world. It shows that gas

(or equivalent flexible resources such as hydro or batteries or demand-side management) complements the variability of solar and wind energy and, together, their weighted average cost is less than either coal or nuclear power stations - despite the fact that the IRP cost assumptions for solar and wind are conservative and those for nuclear optimistic.

The arguments of the anti-IPP coalition against the IRP simply do not stack-up and hence many in this group now abandon any pretence that nuclear or coal are least-cost. They argue instead that industrialisation and jobs should be the deciding factors in our new investment choices, ignoring the impact of electricity prices on affordability or economic growth.

While there is a need for more data and analysis, it is undeniable that renewable energy contributes to local manufacturing and jobs and has potential to do much more. The Department of Energy’s data show that localisation in the renewable energy IPP programme is above 45%. Local equity ownership has exceeded 50% for many projects and more than 80% of debt financing has been from local banks. The benefits to South Africa are clear, though there may be room for policy adjustments that ensure that these benefits are distributed more equitably.

Disingenuously, some in the anti-IPP coalition point to the low number of operating jobs on a solar or wind farm compared to an Eskom power station, ignoring their vastly different energy outputs. On a like-for-like, jobs per kilowatt-hour produced comparison there is incontrovertible evidence



that renewable energy generates more direct jobs than coal or nuclear energy. Another dishonest tactic is to compare the total of direct and indirect jobs of nuclear or coal with only the direct operating jobs of solar or wind farms. These strategies don't wash and are easily exposed.

Jobs in renewable energy involve basic construction, operation and maintenance skills. They are also hi-tech. A solar company in Cape Town not only does the engineering design for its international portfolio of projects (it's market share in solar PV is in the top ten globally) but its staff (nearly all South Africans) monitor and control, via computers and satellite communications, solar farms across the African continent, the Middle East, Asia and South America.

Finally, this disaffected coalition try to spread conspiracy theories that the IPP programme is corrupt and has benefitted members of President Ramaphosa's family and administration. Actually, the IPP procurement programme has been exemplary in its transparency, competitiveness and fairness. There have been more than 400 bids over five procurement rounds where around 92 projects have been awarded under conditions of fierce competition. Bid evaluations and awards have been undertaken under the strictest security, with independent and transparent auditing. The IPP procurement programme is, without doubt, the largest and most corruption-free public-private procurement undertaken by South Africa's government.

The propaganda techniques used by critics of the IPP programme mimic those of Trump. Inconvenient programme attributes are labelled fake news and 'alternative facts', repeated endlessly, seek to sway public opinion. Local use of Trumpian tactics is not coincidental. Members of this coalition have let slip their admiration of Trump's "strong-man" political style while ignoring his dismissal of Africa as "those shithole countries". They admire his promotion of coal and nuclear energy while ignoring data that shows the inexorable decline of both industries within the US. And their hypocrisy is further exposed as they profess adherence to radical economic transformation but fail to acknowledge that Trump's policies benefit the most wealthy one percent.

Facts and logic are probably not sufficient to win the argument for more private investment in renewable energy technologies. Nevertheless, Ramaphosa's administration will certainly need to expose the untruths and cynical agenda of those who seek to make the IRP and IPPs a political football.

It is insufficiently understood or acknowledged that IPPs currently contribute less than 5% of South Africa's electricity. Their costs are transferred directly to consumers through regulated tariffs. Eskom's finances or jobs are not yet directly impacted by IPPs. Neither are jobs in coal mines. But it is coming. Already work has started on a just transition where those most affected by job-losses in

the future might be empowered by opportunities in new energy industries. Now is the time to chart out South Africa's energy path, not only in our energy-mix, but also in Eskom's structure and the growth of competition.

The economics of the energy transition to low-cost renewable energy, private investment and increased competition, are inexorable, just like they were in the ICT revolution in the 1990s. Some South African's may choose to fight this transition. That will set us back for a while. A small, backward-looking, previous political and professional elite may benefit briefly. But their children won't. President Ramaphosa and his Ministers now need to take the lead, and embrace the global tide of innovation in energy markets which can provide competitively priced, reliable and clean energy services for economic growth and prosperity for all our people.

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“South Africa is sleep-walking into further periods of power cuts and darkness.”

THE POWER CUTS WILL CONTINUE UNLESS SA'S NEW ELECTRICITY PLAN TRANSLATES INTO URGENT POWER PROCUREMENTS

Anton Eberhard

DAILY MAVERICK
17 October 2019

South Africa is sleep-walking into further periods of power cuts and darkness. Eskom has reinstituted load-shedding. A fifth of Eskom's power stations are broken at any one time. And no new grid-connected electricity generation capacity has been procured since November 2014, despite Eskom system adequacy reports indicating that we are short of power and risks of further power outages are high.

South Africa's national integrated resource electricity plan (IRP) should have helped us avoid this situation. Its purpose is to model the optimal mix of energy resources and demand-side interventions that would guarantee a reliable, least-cost supply of electricity. Having failed to update the plan since 2011, government this week finally approved a new IRP.

Regrettably, the new electricity plan eschews its primary purpose of defining an optimal mix that could guide sensible investment decisions and instead settles on an uncomfortable compromise: it proposes a *"balanced energy mix"* that purports to offer opportunities and hope to different energy constituencies and interests. However, the political judgements underpinning this approach are misguided and are unlikely to achieve intended outcomes.

It is only a small minority - no-longer politically connected - who believe nuclear reactors are price-competitive in South Africa or, indeed, capable of being financed. All the planning models show nuclear power to be the most expensive option. While there is the interesting possibility of scientific and technological breakthroughs in next generation, small, modular nuclear reactors, none are yet commercially available, let alone for export. South Africa's own attempts at developing such a reactor – the PBMR – ended in failure after many billions of public funds went down the drain without even a prototype being built. It is clearly imprudent to include technologies that have not progressed beyond the research and development stage.

Equally puzzling is the reference to high efficiency, low emissions coal technology. These power stations offer only marginal reductions in carbon dioxide emissions at substantial extra cost. A few years back, the Department of Energy ran a competitive tender for privately funded coal independent power producers (IPPs). Even without incorporation of new low-emission technology, bid prices came in higher than renewable energy alternatives. Banks are withdrawing financing for new coal power stations and it's doubtful these plants will ever reach financial close or commercial operation.

References to new nuclear and coal in the text of the IRP are thus a mirage; a misguided political strategy that will backfire as none of these plants are likely to be built. Failure to make clear, economic choices around South Africa's electricity future also means that the consequences of the inexorable global shift away from coal are ignored and no explicit plans or investments are being made to ease adverse impacts in affected communities, such as coal towns in Mpumalanga, or to fully harness the opportunities that come with more competitive clean energy technologies.

Policy-making should involve explicit choices against clearly defined policy goals or criteria that are supported by sufficient consensus. Despite the fractiousness amongst advocates of different energy industries, there is a remarkable consensus, in most countries, on ultimate policy objectives for the sector: namely, an energy system that can deliver reliable electricity at competitive prices while being environmentally sustainable and also contributing to broader economic objectives such as local investment opportunities, manufacturing and jobs.

But South Africa's new IRP doesn't take this obviously more rigorous and defensible approach where different options are evaluated against the above policy criteria. The draft plan was sent to NEDLAC where the approach of the labour and civil society delegates appeared to be one of negotiating for special interests instead of an informed discussion around the

assumptions and data that underpin the planning models. Subsequently, adjustments have been made to the optimized energy mix without any quantification of the additional costs to society, the economy or the environment.

The consequence of this highly politicized approach to electricity planning is that likely there will be legal challenges, ironically from nuclear or coal lobbies, which government has sought to appease but who feel short-changed, as well as from environmental advocacy groups who are dismayed to see polluting coal plants forced into the plan without any economic justification. These challenges could be disastrous for South Africa's electricity supply security as they will further delay the initiation of new power procurements that are now so urgently required.

Government has three main options to bolster supply security and prevent future power cuts: the performance of Eskom's power stations would need to improve dramatically (unlikely given ageing plant plus inadequate past maintenance); government could initiate a new IPP procurement (which could take time given the requirements for NERSA concurrence and public consultation plus the risk of court cases); or government could free up the distributed generation market so that multiple private project developers and investors can contribute innovative local solutions to meet electricity needs.

While fixing Eskom is hard, and launching a new REIPPP auction may be delayed, lifting regulatory barriers for distributed electricity generation is easy: it simply requires the Energy Minister to gazette an amendment to Schedule 2 of the Electricity Regulation Act that exempts small projects up to 10MW from the requirement to obtain a license from NERSA, and instead require the regulator to institute an efficient, automatic registration system. Government could also issue directives to Eskom and municipalities, backed by more explicit requirements in transmission and distribution

licenses, for grid-connection and wheeling procedures to be expedited. These simple steps will liberate a huge pipeline of investment, contributing not only to electricity supply security but to the President's priority policies aimed at boosting investment, economic growth and jobs.

South Africa should also consider deeper legislative and regulatory reforms to accelerate investment in power generation. The IRP should be indicative, rather than determinative: i.e. it should provide a broad vision and framework for electricity investments, but companies and individuals should be free to innovate and invest in power options for themselves and for trading across the grid. While the Minister should still retain the powers to intervene if supply security is under threat, the regulatory framework should facilitate -not constrain - private initiative and innovation. An independent electricity System Operator should be assigned the responsibility for producing biennial IRP updates so that they become progressively less politically contested over time.

History will judge current Ministers harshly should they fail to act with urgency now to avoid further power cuts. It takes time to plan, design and run competitive procurement processes, and for projects to be constructed and produce electricity. The Energy Minister has statutory responsibility to authorize the next least-cost public power generation procurement (which is already overdue). And the Energy Minister has the statutory powers to ease regulations which could liberate massive investment flows in distributed energy projects. Imagine what energy would be released if government was crystal clear in its vision of the future of the electricity sector in South Africa and required all its ministries, agencies and enterprises to act together with purpose and resolve to facilitate reliable, competitively-priced and sustainable electricity supplies that will power accelerated economic growth and development!

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“The proposals to restructure and reform South Africa’s power sector will be successful not only if there are institutions and programmes that engage stakeholders and manage the process effectively, but also if government moves with urgency to procure new IPPs and to free the market for a multitude of entrepreneurs and consumers to invest in small distributed power systems.”

COMPETITION IS THE BALM TO HEAL ESKOM

Anton Eberhard

BUSINESS DAY
31 October 2019

The debate around whether it makes sense to restructure South Africa’s national power utility, or not, is effectively over. In a special paper released this week, government has outlined a roadmap for Eskom in a reformed electricity supply industry. Eskom, in its current form, will cease to exist. Electricity transmission, system operation and market functions will be transferred into a separate company. Competition in power generation will be encouraged. At a later stage, the division currently responsible for electricity distribution and customer sales will also be moved to a separate entity.

The first milestones have been set: by March 2020 a transitional board for the transmission entity needs to be in place and functional unbundling completed. Legal separation will be completed in 2021. This proposal is not trivial: South Africa often struggles to translate policy ambitions into implementation; now we’ll have an institution incentivized to drive the migration of people, system, assets and debt into this new entity and make the first restructuring step a reality; it will be visible and it will create inexorable momentum for reform.

These restructuring proposals were first made in the 1998 White Paper on Energy Policy but were never implemented. Eskom was not in crisis then and was subsequently ordered to build two mega coal power stations – Medupi and Kusile – which ended up being nearly three times over budget and more than seven years late, resulting in massive debt and the utility’s current insolvency.

Eskom generates less than half the cash it needs to service the principal and interest on its debt. Each year it sells less electricity and the only way to increase revenues has been to hike tariffs, which have increased fourfold, in real terms, since 2006. And the more tariffs go up, the more consumers invest in energy efficiency, use less electricity or defect from the grid. Eskom is now in a classic utility death spiral. Business as usual is no longer an option. Restructuring and the development

of new business models are essential for a more sustainable future.

Eskom restructuring has two broad rationales and potential sets of benefits. First, boards and managers can focus, laser-like, on the three very different elements of Eskom’s value chain. Distribution is a customer-facing business; it requires a different set of capabilities to power generation, which, for now, is mostly around keeping old kit operational and, in the future, being efficient in the face of fierce competition. Transmission, system and market operations are about procuring and purchasing competitively priced, reliable power; balancing supply and demand; and delivering electricity to the main load centers. Dedicated and differentiated skill sets are necessary to reduce costs and improve efficiencies in each of these businesses.

When these functions are all in one company it is easy for the board and management to be consumed by crises in specific areas of the business – as Eskom is currently around the problems with its coal power stations – and to neglect the rest of the divisions, to the detriment of consumers and the nation. Eskom has been diverting capital and maintenance budgets away from transmission and distribution to its problematic generation business, thus threatening the future reliability of the grid.

If these functions are separated, there will also be more transparency around costs and it will be easier for the regulator to establish the prudence of expenditures and to determine cost-reflective tariffs.

There is a second set of benefits that arise from restructuring, namely those that derive from the introduction of competition. Most countries now recognize that there is no intrinsic logic in keeping power generation, transmission and distribution integrated in a single monopolistic utility. While the wires components – transmission and distribution – are

mostly natural monopolies, power generation is potentially competitive; multiple companies can invest, build and operate power stations and compete to sell electricity either to the national transmission, system or market operator, or directly to customers.

Globally, 106 countries have now unbundled their national utilities; South Africa is finally catching up. As President Ramaphosa memorably noted: “currently, all our energy eggs are in Eskom’s basket; when they fall, the economy breaks.... We need more diversity and resilience.”

While government is to be applauded for its commitment to push through the restructuring of Eskom, opposition from some stakeholders – such as organized labour and rent-seekers who benefit from contracts with the national utility – will not easily or quickly melt away.

One of the lessons of the failed attempt to restructure Eskom in 1998 is to understand better the political-economy around Eskom and to engage more effectively with different stakeholders, their material and political interests, their access to formal or informal institutions and networks or de jure and de facto power, and the way they use ideology and knowledge to de-legitimize or legitimize reforms or the status quo.

The Public Enterprises Ministry has its hands full trying to keep South Africa’s major infrastructure state-owned companies afloat. It does not have space or time to manage a dedicated and consistent engagement process with labour unions around Eskom restructuring; neither does the Eskom Board or its senior Executives who are desperately trying to keep the lights on. This is another reason why we need institutional, governance and technical support for driving and managing the restructuring process.

Splitting Eskom into three separate entities is not a sufficient step for fixing Eskom or for securing South Africa’s electricity

supplies at competitive prices, but it is a necessary step to a more sustainable future. The Eskom Roadmap outlines a range of other required interventions such as strengthening the Eskom board, appointing a chief executive who can provide new leadership, building capabilities across the organization, improving the performance and availability of Eskom power stations, compressing costs, reversing the payment arrears of municipalities, and directing the regulator to approve more sustainable tariffs.

Government still has to decide the quantum and mechanism for Eskom debt relief and refinancing. The Minister of Finance appears to be withholding details until seriousness around structural reforms is demonstrated. Government also has to decide whether it will explore the feasibility of reducing the cost of Eskom’s capital through a blended climate-linked finance facility in return for a commitment to accelerate the decommissioning of Eskom’s old, expensive coal power stations. This fund could also provide support for just transition investments to mitigate impacts in coal towns and communities.

The previous attempts to restructure Eskom were halted when government panicked around the risks of power shortages and the urgent need for new investment. We face a similar challenge presently. The difference is that there are now competitive alternatives to Eskom in the form of independent power producers (IPPs) and self-generation. The proposals to restructure and reform South Africa’s power sector will be successful not only if there are institutions and programmes that engage stakeholders and manage the process effectively, but also if government moves with urgency to procure new IPPs and to free the market for a multitude of entrepreneurs and consumers to invest in small distributed power systems.

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“South Africa is beginning to see the consequences of an energy ministry trapped in the past, beholden to interest groups and oblivious to global innovations in energy technologies and markets.”

NEW COAL AND NUCLEAR POWER PROPOSALS UNDERMINE PROSPECTS OF A POST-COVID ECONOMIC RECOVERY

Anton Eberhard

DAILY MAVERICK

17 June 2020

South Africa is beginning to see the consequences of an energy ministry trapped in the past, beholden to interest groups and oblivious to global innovations in energy technologies and markets. Submissions by the Minister and his energy department to parliament in the past month reveal an economically disastrous commitment to policy, procurement and investment options that have no hope of contributing to our post COVID-19 economic recovery.

The Department of Mineral Resources and Energy (DMRE), in its strategic and economic plans, is promoting the role of nuclear energy (mainly small modular reactors) and clean coal (with carbon capture and storage) but these technologies are neither price competitive nor, in the case of small nuclear, are they currently commercially available.

In a presentation to parliament on 26 May 2020, Minister Gwede Mantashe proposed a number of medium term (6-12 month) interventions in response to the economic impacts of the COVID-19 pandemic. Amongst his proposals to boost industrialisation were immediate production of sanitizers, face masks, test kits, antigens and antibodies [sic], ARVs, a multi-purpose reactor and “carbon capture and utilisation as feedstock to the manufacture of chemicals, fuels etc.” Note, these were proposed as DMRE contributions, mostly utilising existing budget allocations. How’s that going, we may ask?

In the same presentation, the Minister proposed a number of interventions (also 6-12 months) to enhance electricity supply security, including acceleration of a nuclear build programme, conversion of Eskom’s diesel-fired turbines to gas and the building of a new oil refinery. None of these, of course, can be accomplished within a year and it’s highly unlikely that even contracts for these projects will be placed anytime soon, if ever. Implementation of South Africa’s IRP electricity plan, which identifies wind, solar and storage as the next least-cost options to ensure electricity supply security, was evidently not

regarded as a priority although - almost as an afterthought - it was offered as a long-term option.

Over the past weekend, DMRE launched a Request for Information (RFI) to commence preparations for a nuclear build programme. Of course, an RFI is non-binding (unlike a Request for Proposals, RFP, in a competitive tender or auction) and participants are perversely incentivized to put forward unrealistically attractive offers and prices which they’ll probably seek to alter when contracts are negotiated. In short, an RFI is not particularly helpful unless you don’t know what you’re doing and want technology and service providers to shape your procurement.

Nuclear power has never been part of any of South Africa’s least-cost IRP electricity plans. In the 2010 IRP it was forced in as a “policy adjustment”. One of the reasons this plan was not updated for nearly a decade was that each time the IRP models were run, nuclear didn’t feature, much to the chagrin of the Zuma administration. In the current 2019 IRP it doesn’t feature in planned capacity additions to 2030. Bizarrely, a sentence was later added to the IRP document saying that preparations would commence for a nuclear build programme.

The Minister has also entertained plans for expanded investment at the Nuclear Energy Corporation of South Africa (NECSA), despite the institution recording unprecedented financial losses. The previous NECSA board chairperson put forward a realistic strategic plan which focused on its profitable medical isotopes business and proposed cuts to its non-viable industrial subsidiaries. The minister rejected these and the chairperson resigned.

Referring to his interactions with the previous NECSA board, the minister was quoted on 20th January 2020: “I grew up in Cala. In that area of the world, when people relate to you as a young boy, you revolt. When people treat you like you are just a

small boy, then you are going to resist that with everything you have. That is how your purported board relates to me. They think I’m just a small boy who does not know what he is doing. Now they are going to get my resistance,” he said.

The minister has now appointed a new board chairperson, the retired nuclear chief officer of Eskom, an ex-British navy nuclear submariner, someone who continues, on social media, to rubbish renewable energy alternatives.

It’s time for a reality check. No country or private company currently offers commercially proven exports of land-based, small modular nuclear reactors. South Africa tried to develop one – the pebble bed modular reactor – but after spending more than R20 billion (in today’s money) the programme was closed after a decade without even a pilot demonstration plant being built. Russia continues to develop small nuclear reactors, mainly for marine purposes, such as ice-breakers and barges. China has demonstration projects under construction and testing and there is renewed research and development activity in the US, Japan, South Korea and the UK, amongst other countries. But these are not yet commercially available, licensed projects for export.

Clean coal is also a mirage. Its supporters argue for high-efficiency, low emission (HELE) power stations. The Kusile coal power plant in Mpumalanga was meant to be an exemplar but we now know how expensive that was, more than double Eskom’s average cost of generation and more than four times the cost of current least-cost generation supply options, as revealed in the latest bids in South Africa’s renewable energy IPP programme and international auctions, which are delivering electricity prices less than USD 2c/kWh.

The only way coal power stations could be regarded as clean is if, in addition to minimizing pollutants such as particulates and sulphur and nitrogen oxides, they scrub climate change

inducing carbon dioxide from combustion gases and store it safely in geological strata. While carbon capture technology has been developed for power plants, less than a handful are operational - for example Petro Nova in the US and Boundary Dam in Canada – and these are expensive. A multi-year research project has revealed that secure and safe sequestration of carbon in South Africa is problematic.

Yet the minister stubbornly promotes “clean” coal. In an interview on 24 February 2020 he stated: “Let us say [another] state-owned company, no names mentioned, comes to us and says: Listen, we want to open a coal-fired power station with carbon capture and storage (CCS), and we can do it in the next two years. And they propose to build it with their own money, operate it to recoup their money, and then transfer it to the state – the Build, Operate, Transfer approach. It is a very attractive proposition, although there has been no decision yet in this direction.”

But this is not how public procurement of infrastructure projects works in South Africa. The constitution and the Public Finance Management Act require fair, equitable, transparent, competitive and cost-effective tendering. Non-transparent, directly negotiated deals are not permissible. Apparently the Minister’s door has been wide open to lobbyists (another recent example is the agreement with Russia’s Gazprom), while his own electricity plan is neglected.

DMRE’s attachment to clean coal and new nuclear as immediate options for a post-COVID economic recovery would be comical if they were not financially ruinous. It’s not only that they are dilatory in their plans to secure a least-cost, secure electricity supply to power economic activity and provide social benefits for households, but their fixation on these non-competitive, non-commercial technologies is now wasting scarce public resources in what have been described by scholars as technologies of prevarication (I would add, also

of obfuscation and rent-seeking). Tragically, a generation of young, talented and skilled professionals will believe there is a still a career in nuclear power or clean coal when economic analyses show the opposite.

Furthermore, new mining and energy entrepreneurs, believing government is charting a future in coal, are likely to be short-changed. International mining majors are selling their coal assets. It is one of the few mining sectors in South Africa that is seeing significant black economic empowerment, but with what consequences? Will these new companies survive declining coal demand and revenues, and contingent environmental liabilities?

Meanwhile, the past few years have seen an upending of comparative power generation costs. I have seen more innovation in energy technologies and markets in the past 7 years than in the previous 35 years of my professional career. Since 2011, solar PV prices have fallen 90% and wind energy 50%. They are now the least-cost electrical energy sources. Of course, a power system depends not only on energy, which, in the case of solar and wind, is variable, but also requires back-up capacity and ancillary services (such as frequency and voltage support, reactive power and inertia) but even with these additional costs renewable energy now provides a supply mix that is cheaper than including new coal and nuclear investments.

And yet the minister remains ultra-sensitive and resistant to any advice on these new options. Speaking at the launch

of the International Energy Agency's Coal Report on 17 December 2019, he referred to the incident in 1865 in South Africa when Nongqawuse, a 15 years old, claimed that the spirits of the ancestors had spoken to her and if the Xhosa people would kill all their cattle and burn their crops, a day would come when they would drive the whites into the sea and gain wealth. *"I am from a province" he said, "in the Eastern Cape one time they were told to kill their cattle because there is wealth coming. That new world has not arrived there, we are starving up to today. So anybody who comes to me and says destroy what you have because there is hope, you will battle, because I am informed by this experience."*

How do we explain the Minister's championing of nuclear and coal and his aversion to even mentioning renewable energy without appending the label *"lobbyist"*. Why is there so little recognition of South Africa's globally recognized, successful renewable energy auctions? Mantashe's history as a miner of gold, copper and coal, rising to general secretary of the National Union of Mineworkers, and his networks with emergent interests in the coal and nuclear industries, provides some insights but one would have thought that his experience of researching a thesis for his Masters degree on the decline of South Africa's gold mining industry would have provided salutary lessons. Employment in gold mining today is a fraction of what it was in the 1980s. There were no effective government interventions to halt this decline.

The demise of the coal mining industry is now equally inexorable. The minister may seek to slow it, but the costs

of coal as a power source, environmental harm, and the availability of lower priced, clean power technologies, mean that we have reached *"peak coal"*: its downhill from here.

Britain, which pioneered the industrial revolution based on coal-fired steam engines and later electricity, will be coal free in the next couple of years. Coal is declining in the US regardless of Trump's policies and slogan that he *"digs coal"*. And despite large coal markets in China and India, these countries are now adding more renewable energy capacity than coal power. A rapid energy transition is underway.

The minister's behaviour may also be explained by the ideological distrust by many in government of the private sector, which has been more embracing of innovation and is willing to invest in competitive, low-cost clean energy technologies. Mantashe has put forward the idea of a new state-owned generation company, unencumbered by Eskom's horrific debt. But why? Most would agree that the state's role should be reserved for correction of market failures, the provision of social services and the protection of the poor, rather than crowding out private capital. It's as though an ideological fog obscures the reality of the present and where an older political generation still wishes a role for the *"developmental state"* in the power sector, despite the spectacular failure of Eskom, which has crippled economic growth through power cuts, charging us prices four times higher than a decade ago, and is draining the fiscus of tens, soon hundreds, of billions at a time when the state can least afford it.

The minister has had responsibility for the energy portfolio for a year now. He was appointed after a period of severe Eskom power cuts and load-shedding. It would be realistic to expect that his top priority would have been rapid investment in new power. Yet the IRP was only finalized 6 months later and ministerial determinations, required by law before initiating new IPP procurements, have still not been gazetted. Neither has there been any meaningful reform of regulations which would ease private investment in small distributed generation.

I once told the minister that history would judge him poorly if load-shedding and power cuts returned. He replied: *I'll blame the public enterprises minister and Eskom*. It was meant as a joke but, in November last year, that's exactly what he did.

The recent cold front that swept South Africa has revealed the fragility of our power system: a 2 gigawatt increase in demand forced extensive use again of expensive back-up diesel-fired turbines, a cold reminder that we do not have enough operational power capacity or a sufficient operating reserve margin to support an economic recovery.

While it's difficult to fix Eskom - we have to try, given its current dominant share of electricity supply - it's not difficult to accelerate private investment in South Africa's power sector. Four actions are urgent. First, the minister needs to demonstrate his commitment to implementing government's optimized IRP through launching planned, least-cost Independent Power Producer (IPP) procurements. It takes time to run tenders or auctions, prepare bids, award

contracts, obtain the required permitting and licensing, secure financing, reach financial close, construct plants and achieve commercial operation. We are already late in guaranteeing planned capacity additions for 2022 and it's almost certain that we'll experience further load-shedding over the next 3 years.

Second, the so-called emergency power procurement, which the minister announced last year, and which was supposed to deliver electricity within 3, 6 or 12 months, should be repurposed and allocated to small distributed generation projects by the private sector. These are the only projects which can deliver new power within these short time frames.

Third, there is a need for regulatory reforms to ease licensing impediments for self-generation and distributed energy resources. At present, only projects below 1MW or off-grid, are exempt from a NERSA licence. There is a huge pipeline of projects in the 1-10MW range, for example rooftop solar PV on commercial and industrial buildings, which face onerous and inefficient licensing requirements and procedures that were designed for power generation plants hundreds of megawatts in size. In addition, there are numerous large industries and mines that wish to build their own generators but also to remain grid-tied: they too face frustrating regulatory barriers. The minister, quoting NERSA, says it's not a problem, that the backlog is not large and the regulator will process licence applications within 120 days. The reality is that NERSA has not issued a new generation licence since 2016 and industry players remain frustrated. It clearly makes sense for the

minister to cut through this red tape by gazetting another amendment to Schedule 2 of the Electricity Regulatory Act with these expanded licence exemptions.

Finally, the minister could make a policy pronouncement that direct power contracts between IPPs and large customers will be encouraged and that barriers to achieving timely and cost-effective connection and wheeling agreements will be overcome.

None of these steps require tax rands or government investments; that's their attraction in a fiscally-constrained post-COVID environment. They are pragmatic, sensible, achievable and they can power economic development and job creation. South Africa needs an energy ministry and department that is focused on the path of economic recovery, not the fantasy of these uneconomic, unrealistic vanity coal and nuclear projects, which are prone to cost and time overruns and rent-seeking. The minister has extraordinary powers. How will he use them? How will he be remembered?

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WILL 2021 BRING MORE POWER CUTS OR PROSPERITY?

Anton Eberhard

BUSINESS DAY

11 December 2020

“There were more power cuts in 2020 in SA, more gigawatt hours of load-shedding, than in any previous year. The next year will likely not be much better. What happens after then depends almost entirely on whether the government rethinks and revises its approach to electricity sector regulation, planning and investment”

There were more power cuts in 2020 in SA, more gigawatt hours of load-shedding, than in any previous year. The next year will likely not be much better. What happens after then depends almost entirely on whether the government rethinks and revises its approach to electricity sector regulation, planning and investment.

As we approach our summer holidays we have had a welcome respite from Eskom supply interruptions, but SA does not have enough operational generation capacity and power cuts will return. Eskom estimates our power gap at about 4GW. The Council for Scientific and Industrial Research thinks it may be closer to 7GW. Essentially, we don't have a large enough reserve margin, and significant, unexpected breakdowns of Eskom power units mean the lights will go off again, as happened in 2007, 2008, 2014, 2015, 2018, 2019 and 2020.

The long-term, declining trend in the performance of Eskom's power stations is inexorable. The average age of its power stations is now approaching 40 years. Old kit is difficult and expensive to maintain. Eventually, power stations have to be closed. SA's Integrated Resource Plan (IRP) for electricity acknowledges that 11GW will need to be decommissioned by 2030.

Eskom is unable to invest in further power generation capacity. It cannot raise the finance. Its debt - close to half a trillion rand - is too high and it generates less than half the cash required to meet interest and principal payments on that debt. The government does not have the fiscal space to invest in energy. SA is now reliant on private investment for its power needs.

The government, and other energy sector stakeholders, will not have missed these messages from Eskom's leadership

over the past months. They are being unusually clear: SA is short of power; existing Eskom power stations are old and it's unrealistic to expect them to perform perfectly; Eskom doesn't have the balance sheet to invest in more power; and SA thus needs to create an enabling environment to accelerate private investment in power.

Has the government heard this message? It appears, not clearly enough. And does the government appreciate the depth of policy interventions, regulatory reforms and investments that are required to secure the country's electricity supplies to power a post-Covid economic recovery? It appears, not yet.

The energy minister believes everything is fine: he's implementing the IRP electricity plan; made a determination to procure 2GW of emergency power, 2.5GW of nuclear and another 11.8GW of solar, wind, storage, gas and coal; and, he says, own-generation is allowed.

But here's the reality. The minister was assigned his portfolio in May 2019, after catastrophic power cuts. Nineteen months later there is not yet a single government-backed contract for new power generation. So-called emergency procurements, which were to be realised in months, have stretched into years. The minister has championed coal and nuclear, which have no chance of being planned and built within five years nor, more likely, 10 years, and he has delayed procurements of short-term, least-cost options in the IRP electricity plan — solar, wind and storage.

And the options that can deliver within a year - electricity customers making their own electricity investment choices - have been frustrated by regulatory barriers that require all grid-tied generators above 1MW to apply to the National Energy Regulator of SA (Nersa) for a licence. Almost every

week I receive calls from enterprises, some of them large international companies with commitments to be 100% renewable energy supplied, exasperated by the dysfunction of the regulatory regime here.

The electricity system in SA is not fine. At least now there is growing consensus on what is needed to fix it. Here are five pragmatic reforms that will help restore energy supply security and sustainability:

First, free up the market for electricity consumers to make their own power generation investment choice by removing the requirement to obtain a licence from Nersa. This is such a simple step. It does not require new legislation, merely a ministerial regulation in terms of the act. There is no objective rationale for Nersa to restrict market access. We need power, and we need it fast. Electricity prosumers (producers/consumers) generate the quickest and most cost-effective solutions. Yet the energy minister was recently quoted as saying there would be chaos if power generation licence exemptions were lifted to 10MW or even 50MW. There is no economic or technical basis for this resistance to regulatory reform.

Second, fast-track public-private partnerships in least-cost power generation procurements. It is unacceptable that despite 10 years of IRP electricity planning models indicating that the least-cost power options are solar and wind energy, and five years since we ran the last renewable energy independent power producer auction, the next bid window is still not open.

Third, update the 2019 IRP electricity plan. Eskom power station performance is lower, demand is lower, renewable energy and storage prices continue to fall, planned investments in

coal fail, the Inga hydroelectric import is at risk, and small modular nuclear options have yet to be commercialised. We need a realistic plan. Better still, the plan should be indicative, not determinative. The government needs to step back and intervene only when there are market failures in electricity security.

Fourth, plans to restructure Eskom need to be accelerated to create an independent transmission and system operator that will create a transparent and fair platform for the planning, procurement and contracting of least-cost power generation.

Fifth, invest in a just energy transition that embraces a blended climate-linked finance facility that reduces Eskom's cost of finance in return for accelerated closure of dirty, old, expensive coal power plant, while creating new economic opportunities for affected workers and communities.

Of course, these reforms are being contested, not least by factions who benefited in the past from captured state-owned enterprises. But our political economy is dynamic. Power cuts mean less economic growth, fewer jobs and more poverty. A coalition for reform is being built, out of necessity and in the certainty of the benefits it will bring. Some of those in leadership positions who still don't understand this inevitable transition will need to step aside. That is the hope for 2021.

Eberhard is professor emeritus and senior scholar at the University of Cape Town's Graduate

MINISTERS ARE HOLDING UP ENERGY REFORMS, KEEPING SA IN THE DARK

Anton Eberhard

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“All eyes are now on these ministers. Will they continue to believe the state should dominate and control the power sector in South Africa or will they be willing to free up space for competition and private investment. The choices are between darkness or light, fruitless protection of an insolvent, dominant incumbent, or democratization of energy. I'm pretty sure what the majority of South Africans would vote for.”

A trillion rand needs to be invested in new power in South Africa over the next decade in order to restore energy security. Opening up the electricity sector to massive flows of private investment in generation capacity is now our only option, but key enabling reforms still have to be enacted.

Regular power cuts in 2018, 2019, 2020, and this year, are indicative not only of Eskom's poor performance, but also of the need for more power. We don't have a large enough reserve margin. And between now and 2030, according to South Africa's official IRP electricity plan, 11 000 MW of coal plant will have to be decommissioned. It's too difficult and expensive to keep old kit going. The average age of Eskom's coal fleet is close to 40 years. Despite renewed maintenance efforts, the declining trend in the availability of Eskom's coal power stations is inexorable.

The IRP indicates that 33 000 MW of new generation capacity has to be added by 2030, most of it from solar and wind energy, complemented by gas and storage.

Eskom is unable to finance and build these new power stations. With debt close to half a trillion rand, no financial institution will provide Eskom with new loans for new power.

Its debt repayments this year on principal and interest amount to R94 billion. Eskom generates less than half the cash it needs to service this debt. It is technically insolvent and survives through government bailouts: R56 billion this year, R31.7 billion next, a total of R220 billion since 2008.

Government is also over-indebted and fiscally constrained. National debt has exploded and will reach 89 per cent of GDP by 2025/6. Government's interest bill amounts to a fifth of gross tax revenue, crowding out much needed additional support for health, education and poverty alleviation.

That leaves the private sector as the only source of finance for new power.

Fortunately, South Africa already has a successful template for attracting private investment into power. The Renewable Energy Independent Power Producer Procurement

Programme (REI4P) ran four large auctions between 2011 and 2014, attracting more than R200 billion into more than 100 projects amounting to 6400 MW of new capacity. Uniquely, the REI4P also resulted in significant social and economic benefits through black economic empowerment, jobs and community development projects, all carefully documented by the Department of Mineral Resources and Energy's IPP Office.

Despite prices dropping around 80 per cent for solar PV, and 50 per cent for wind energy, the programme stalled under the Zuma administration when Eskom refused to sign power purchase agreements with winning bidders. A four year hiatus ended when President Ramaphosa came to power, but his energy minister has been painfully slow in getting the fifth REI4P auction going.

The minister, instead, has chosen to focus on a so-called emergency procurement of 2000MW. He hoped to have projects on the ground within months, but timelines have stretched to years. Winning bids have yet to be announced. Not a single contract for new power has been signed in the two years he has held the energy portfolio.

The minister has also announced his determination to procure 2500MW of nuclear power, despite this technology not featuring in any of South Africa's least cost IRP scenarios; it has always been forced in as a “*policy adjustment*”. The CSIR has calculated that for nuclear to break even with cheaper alternatives, investment costs would need to be lower than R3400/kW. Nuclear bids received by Eskom in 2007 were double this value, and nuclear costs globally have increased since then.

The National Energy Regulator of South Africa (NERSA) is required in law to concur with this section 34 determination. NERSA held public hearings this past week. They were revealing of the political-economy of power in South Africa and how different interest groups, with varying degrees of access to power, employing networks, knowledge and ideology, seek to influence policy and investment decisions. And behind these processes is the fact that large, centralized

nuclear procurements, with non-transparent government to government deals, lend themselves to rent-seeking, which, in contrast, is much more difficult in multiple, small, incremental, distributed, highly competitive renewable energy projects.

NERSA members repeatedly said their decisions would be evidence-based, but they seemed unswayed by either the IRP or the scientific, peer-reviewed evidence presented by the CSIR, Meridian Economics and others. They asked repeatedly, why Eskom spends billions on IPPs which appear to produce so little, ignoring that IPPs are only five per cent of supply and that these costs include expensive diesel peakers, procured with limited competition, and solar, procured ten years ago. Investment decisions for the next megawatt need to be informed by current market prices. They also seemed unconvinced that renewable energy, coupled with gas and storage, can power an industrialized economy reliably.

Seven NERSA Board members now hold the balance of power around South Africa's nuclear future. Should they demur, the minister will not be able to proceed with a procurement. If they concur, their decision will almost certainly be challenged in the courts as being irrational and not in the public interest.

Innovation in energy technologies and institutional and business models are disrupting power markets. To succeed, South Africa needs to embrace these innovations. Thousands of megawatts of solar and wind energy, and utility-scale battery storage, need to be procured every year going forward. The private sector has shown they are ready. Large industrial and mining power users are enthusiastic. There are no costs for the state or taxpayers. Regular procurements will also stimulate local manufacturing and job creation.

President Ramaphosa has acknowledged that all our energy eggs have been in Eskom's basket and if they fall the economy breaks. Diversifying our power sector is no longer a choice, it's an imperative, not only through utility-scale REI4P auctions but also in distributed energy generation. We need to abolish Kafkaesque licence requirement for small and medium sized power projects, freeing consumers to invest in their own power solutions or to contract private providers directly. The

grid also has to be unbundled from Eskom, creating a fair and a transparent platform for competitive, least-cost procurement and delivery of power.

The President has committed government to these reforms. But the responsible ministers are not out front leading. There has been not a word from Minister Mantashe since Ramaphosa said licensing exemptions would be extended to projects up to 50 MW. Minister Gordhan rarely talks about Eskom restructuring and no hard deadline has been set for the second stage of taking the Independent Transmission System and Market Operator subsidiary company completely out of Eskom as a separate state-owned entity. Where is the legislation that will enable this? Why is he sitting on a radical paper by Eskom that proposes solutions to South Africa's power crisis?

The president relies on these ministers politically, but they appear to be less than enthusiastic supporters of a reform agenda that could free up a trillion in private investment. They seem lost in an ideological fog that prevents them seeing the problem clearly or responding with the purpose and leadership South Africa so desperately needs. They hope Eskom can still be a key instrument in the developmental state, when plainly it is not, with power cuts crippling our economy and bailouts draining the fiscus. Even the CEO of Eskom now admits publicly that the utility cannot resolve South Africa's power shortages on its own and that we need to accelerate private investment.

All eyes are now on these ministers. Will they continue to believe the state should dominate and control the power sector in South Africa or will they be willing to free up space for competition and private investment. The choices are between darkness or light, fruitless protection of an insolvent, dominant incumbent, or democratization of energy. I'm pretty sure what the majority of South Africans would vote for.

Eberhard holds the posts of Professor Emeritus and Senior Scholar at the Power Futures Lab at the University of Cape Town's Graduate School of Business

POWER MARKET AND REGULATORY REFORMS CAN AND MUST IMPROVE SUPPLY

Anton Eberhard

BUSINESS DAY
21 August 2021

“By Gwede Mantashe’s own admission, President Ramaphosa had to twist his arm and the energy minister has now gazetted an amended schedule to the Electricity Regulation Act that exempts projects up to 100MW from the requirement of applying to the National Energy Regulator of South Africa (NERSA) for a licence”

The next steps in President Ramaphosa’s structural reform agenda are being taken with power market and regulatory reforms that will facilitate increased private investment and competition. The implications are potentially far reaching and will contribute to the fundamental restructuring of the electricity supply industry in South Africa, and hopefully an end to load-shedding and power cuts.

By Gwede Mantashe’s own admission, President Ramaphosa had to twist his arm and the energy minister has now gazetted an amended schedule to the Electricity Regulation Act that exempts projects up to 100MW from the requirement of applying to the National Energy Regulator of South Africa (NERSA) for a licence.

This is a significant development. The energy minister and NERSA were acting as gatekeepers to market access, and new private power projects faced onerous, uncertain and lengthy delays. The previous licence exemptions were only for off-grid, self-generators and for projects less than 1MW. Now the new exemptions apply to much larger investments and will enable grid-connected projects to sell to multiple customers, even wheeling across the grid to distant electricity users who no longer have to rely solely on Eskom or municipal supply.

These licence exemptions will prompt further reforms: NERSA project registration processes, and grid connection and use of system agreements, will need to be streamlined if investments are to flow.

These reforms are not being driven by any ideological commitment to privatization, but rather as a pragmatic response to South Africa’s power shortages. The president uses the metaphor of all our energy eggs being in Eskom’s basket and if they fall the economy breaks. In his State of the Nation Addresses he has called for diversification of the country’s electricity supplies and for more competition.

Over the past year, we have also seen Eskom’s CEO call for more private investment to relieve a constrained power system

so that the utility has more space to undertake reliability maintenance. Eskom is technically insolvent and survives through government bailouts, a total of R220 billion since 2008. It generates less than half the cash it needs to service principal and interest payments on its debt. No-one will lend Eskom money for new power generation until its balance sheet is deleveraged. The state has no fiscal space to invest in power and its ability to provide sovereign guarantees is evaporating. That leaves the private sector as the main source of funding for new power.

There is now an unprecedented and growing consensus among government, business and labour around power sector and regulatory reforms. When the previous attempt was made - after the publication of the 1998 Energy White Paper - to restructure Eskom, and to introduce competition and private sector participation, energy intensive users and business associations were ambivalent; they enjoyed low electricity prices and reliable supply at the time.

Now the material conditions are different: there has been load-shedding in 2007, 2008, 2014, 2015 and every year since 2018, and electricity tariffs have increased four-fold. Business organisations now advocate for power market reform. Even COSATU issued a statement supporting the licensing exemptions.

However, easing investments in distributed energy generation alone will not solve South Africa’s power crisis; we also need to restructure Eskom and expand utility-scale procurements.

Two years after the President’s Eskom Sustainability Task Team submitted its recommendations, Eskom’s debt overhang remains unresolved. But another recommendation, on Eskom unbundling, has made progress, mainly as a result of the leadership of its new CEO.

A separate transmission subsidiary company will be established by the end of the year. The next step will be taking that subsidiary out of Eskom as an Independent Transmission

System and Market Operator (ITSMO), which has responsibility for least-cost and reliable power planning, procurement, contracting, dispatch, system balancing and transmission.

Eskom unbundling will facilitate further utility-scale procurements. Least cost generation sources are now mainly distributed solar and wind energy, and considerable investments – Eskom estimates R180 billion - will need to be made in strengthening and reconfiguring the grid. The ITSMO will start migrating to investment grade and will be able to access competitively priced capital. An ITSMO will also facilitate direct contracting between generators and large electricity users, reducing the need for sovereign backed security measures and guarantees.

And, as the number of private, renewable energy IPPs and traders on the grid grows, the capabilities of the system and market operator will need to be enhanced. A balancing market will need to be created as well as competitive procurement of flexible resources to complement the variability of solar and wind, and ancillary services to maintain system quality, strength and reliability.

While South Africa’s power market is slowly heading in the right direction, it is not clear whether government fully appreciates yet the enormity of the challenges ahead. Eskom’s CEO has said that 22,000 MW of old coal plant will need to be decommissioned over the next 15 years. That amounts to nearly 50% of Eskom total current generation capacity. South Africa’s electricity plan – the IRP2019 – envisages that 33 GW of new capacity, mostly solar, wind, gas and batteries, will need to be built by 2030, implying total new investment of over a trillion rand.

Current efforts are far from adequate. The energy minister is wasting time trumpeting coal and nuclear power – neither of which can be built in time. They will be difficult to finance and, like the poorly designed Risk Mitigation IPP programme, will be costly.

With all the competitive options available to South Africa, it is almost unbelievable that 20 year contracts are now being considered for emergency power ships and for solar/wind/battery hybrid projects that, because of flaws in the procurement specifications, will involve massive curtailment and wasted energy. None of these projects reached financial close by the required date but, strangely, DMRE has chosen to extend deadlines rather than cancel or redesign these disastrous contracts.

After an hiatus dating back to 2014, the fifth auction round of the renewable Energy IPP programme is finally underway, with bids submitted last Monday. International solar and wind prices continue to plummet, with most auction bids now below US 2c/kWh and the world record down to 1c. With its higher cost of capital, and economic development add-ons, South Africa will likely see prices below ZAR 50c/kWh and maybe even below 40c, a third of Eskom’s average cost of supply. One concern, however, is that competition may have been limited by Eskom not providing in time to all bidders the grid cost estimation letters that were required as part of the bids.

If power cuts are to be relegated to the past, South Africa will need to continue to ease the path for private investment in power, both at utility-scale and distributed energy generation. It is hoped that Operation Vulindlela, in the President’s office, will continue to push for further power market and regulatory reforms, and for massively expanded bi-annual, IPP Office or ITSMO-led power auctions. Anything less, will mean continued load-shedding and constrained economic growth and development.

Eberhard holds the positions professor emeritus and senior scholar at the Power Futures Lab at UCT’s Graduate School of Business.



“We need to double the power procurement from the IPP office and continue with regulatory and structural reforms that facilitate private and consumer ownership in power.”

SOUTH AFRICA STILL NEEDS ESKOM BUT ALSO NEW GENERATION CAPACITY

Anton Eberhard

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Intermittent power cuts in 2007, 2008, 2014, 2015 and every year since 2018 have propelled South Africans into periodic bouts of frustration and depression. And then the lights come on again, spirits lift, and pressure on the government to fix the power system subsides somewhat.

Is there now complacency in the government that load-shedding is the new normal? Where is the urgency to resolve this crisis?

The average energy availability factor of Eskom's generators has declined inexorably from above 90% 20 years ago to just more than 60% now, with two exceptions when it rose in Eskom's financial years 2017 and 2018. The average age of the utility's coal power stations is more than 40 years, and some are beyond their design life.

The obvious solution has been to exhort Eskom to buck the long-term trend of declining performance, to catch up on maintenance, and to emulate the so-called miracle performance of Brian Molefe, who was briefly Eskom's CEO sometime in 2015 and 2016, and Matshela Koko, who was subsequently an acting CEO, also briefly. Surely, it's possible to raise the performance of Eskom's generators back up to 75%, as they apparently did?

Simple. Except it's not so simple. Any scientist looking at a 20-year set of data with a consistent long-term downward trend would question the two obvious data outliers in 2017 and 2018. A credible explanation would need to be offered why performance was extraordinary in those years. Among Eskom's 47,000 employees at the time, one man, Molefe, without any prior experience in the power sector, evidently created a miracle.

Actually, the first thing Molefe did was ban the publication of weekly generation performance data. And, in a fit of hubris, he threatened to close down a publication that raised critical questions. Koko, who had worked at Eskom since 1996, argued that it was his *"practice note"* in 2016 that did the trick. Rather, it seems, his authoritarian red-card system struck the fear of god into his power station managers who, anticipating sanctions and dismissals, might have underreported power station problems, and ran them much harder than experience would regard as prudent. The consequences are now clear.

Eskom's energy availability factor is a ratio. Decreasing the denominator — the number below the line — and increasing the numerator (above the line) has the effect of increasing the availability factor. There are many ways in which this could be done. The same logic applies to maintenance data, unplanned outages and plant utilisation factors.

Koko argues that this is impossible. Eskom auditors signed off on his data, and he won (sort of) an appeal to the Press Council against an article that claimed he had cooked the data. I would argue that neither Eskom's annual auditing process, nor the judge presiding over the press case, were sufficiently exposed to technical details that showed how manipulation of the data might have been possible.

From 2007, as the generation reserve margin declined, Eskom managers were constrained in their ability to set aside sufficient plant for maintenance. The mantra leading up to the 2010 Fifa Soccer World Cup in SA, was *"Keep the lights on at all costs"*. And now we bear those costs. Since then Eskom's executives have sought to recover the generation fleet and invest more in maintenance. Some have done more, others

less, but all have seen this as a priority, as does the current Eskom leadership.

So what is it that Molefe and Koko can offer in the current crisis: dancing E-band managers in Megawatt Park, whose numbers and salaries were hiked overnight? Molefe resigned after being exposed by the public protector's report on state capture. And, as the realities of Koko's management style took effect, 220 (yes, that many!) senior Eskom managers signed a letter to President Cyril Ramaphosa expressing their concerns. Koko resigned soon after, ducking a disciplinary process.

I believe most of the current Eskom employees are trying their best, but that the goal of reversing the long-term decline in Eskom's ageing power stations is really difficult, expensive and no longer the only, nor indeed the optimal, solution. We need to add new generation capacity as fast as possible to restore supply security. There has been rapid innovation in the power sector, which now sees renewable energy plus storage and flexible resources as the cheapest options to provide reliable electricity. And consumers are becoming producers of electricity — in effect *"prosumers"*.

One of the first things the Ramaphosa administration did was unblock the renewable energy independent power producer (IPP) projects, whose power purchase agreements Molefe and Koko refused to sign. Unfortunately, the hiatus in IPP procurements between 2015 and 2019 has meant sufficient new generation capacity has not come online in time to avoid load-shedding.

While we need to continue efforts to improve Eskom's performance, we also need more urgency in accelerating new

investments in power generation. It's intolerable that SA's mineral resources & energy minister, in office since 2019 and with extraordinary statutory powers, thinks he is doing his job by implementing an out-of-date Integrated Resource Plan, behind time and without him causing a single new publicly procured megawatt to come onto the grid, as yet.

The government needs to show more urgency. It could do so by making decisions more quickly. Existing IPPs should be allowed to export more electricity to the grid. They can, and they are willing. I don't know too much about Eskom's standard offer to small generators, but it seems worth exploring. Why has the government not signed off on this, six months after approval by Eskom's board?

We need to double the power procurement from the IPP office and continue with regulatory and structural reforms that facilitate private and consumer ownership in power. Ultimately, the government needs to provide greater leadership in making the independent transmission system and market operator a reality, a transparent and fair platform for contracting and trading new power, and providing non-discriminatory access to the grid with enough capacity to deliver adequate power to consumers who increasingly want choice, competitive prices and a reliable electricity supply.

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MOVING ESKOM TO MANTASHE'S MINISTRY WOULD BE A SERIOUS MISTAKE

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It would be a grave mistake to shift the governance of Eskom to the Department of Mineral Resources and Energy (DMRE), as recently proposed by the ANC's National Conference. The consequence will almost certainly be ongoing power cuts as Eskom is protected at the expense of private investment.

There is an obvious conflict of interest for a government ministry being a shareholder of a state-owned enterprise (SOE) whilst also having responsibility for sector competition and regulation. This conflict is even more evident when a vertically-integrated SOE like Eskom owns and operates close to 90% of electricity generation as well as access to the national grid.

One of the reasons South Africa currently experiences load-shedding is that government previously sanctioned Eskom's refusal to sign contracts with independent power projects, resulting in a seven year hiatus between the opening of bid windows four and five for new power generation capacity.

International best practice demonstrates that shareholder ministries or agencies should concentrate mainly on the financial sustainability of their SOEs so that they can deliver effective services. Separate ministries are charged with formulating sector policy and providing regulatory oversight. Indeed, South Africa's Electricity Regulation Act enjoins the Minister to *"promote competitiveness and end user choice"* in the electricity sector. How will ministers do that if they are also responsible for Eskom?

What's extraordinary in this latest proposal is the collective amnesia around the policy work that that has been done on the governance of SOEs. In 2013, the Presidential Review Committee on State-owned Entities recommended that *"government must delineate the separate roles of owner,*

policymaker, regulator and implementer". The current Presidential State-Owned Enterprises Council has made similar recommendations. Both build on the benchmark OECD Guidelines on Corporate Governance of State-Owned Enterprises.

Eskom's power stations are failing at a faster rate than new power generation capacity is being added. The solution is thus twofold: Eskom has to perform better but South Africa also needs accelerated investments in new power. Fixing Eskom is really difficult. I believe the current management and staff have tried their best. It's hard to see how one or two new executives can make a revolutionary difference in a huge organisation struggling with historically poorly maintained and ageing power stations, and not enough reserve capacity or funds to undertake the required maintenance.

Sensible strategies to end load-shedding thus have to include also massive new power generation investments. Government doesn't have the fiscal space to fund this. Eskom's indebtedness and insolvency restrict its ability to raise new finance. In contrast, the private sector, and to a lesser extent development finance institutions, have demonstrated their ability and willingness to fund required investments. These investments require a level playing field, non-discriminatory access to the grid and a competitive market, unconstrained by the dominant incumbent.

Yet South Africa's energy minister believes in only one side of this equation. He has made this absolutely clear in his numerous public utterances. *"Load-shedding is a result of 22,000 MW of idle Eskom generation capacity"*, he says. And every time he announces the award of new renewable energy generation contracts, he states they will not resolve load-shedding.

In fact, in Gwede Mantashe's nearly four years as energy minister, he has not yet delivered a single publicly procured megawatt that is yet generating electricity for the grid. The earlier renewable energy IPP auctions, initiated between 2011 and 2014, delivered around 100 projects totalling more than 6400 megawatts and R200 billion in private investment. In contrast, Mantashe has presided over a disastrous emergency procurement, and renewable energy IPP bid windows five and six, initiated far too late, and that have signed only a few hundred megawatts, in contrast to the thousands that were on offer. No procurements have been launched for gas or storage.

One of my meetings with Mantashe was in 2019, soon after his portfolio was expanded to include energy, and I warned that his legacy would be *"Mr Load-Shedding"* unless he acted with ambition and urgency in procuring new power, enabled by the extraordinary powers granted him in legislation. He tucked my briefing note into his jacket pocket and said he would blame Pravin Gordhan when the lights went off. And subsequently he did exactly that. My encounter was not unique. Others have reported similar experiences.

Later he accused Eskom's chief executive, Andre de Ruyter, of undermining the state through loadshedding and acting as a policeman against corruption. And now Mantashe has his wish: de Ruyter, who was accused of *"not enabling the comrades to eat"*, is departing Eskom. Good luck to managing the criminal syndicates that have been identified in Eskom and are currently causing power plants to fail, with unprecedented levels of load-shedding, until, they hope, law enforcement eases.

Another impediment to new power generation investments is inadequate transmission. The Just Energy Transition

Investment Plan of US\$ 8.5 billion has allocated significant international funds for relieving these constraints. Yet, Mantashe, in a recent meeting with one of the funding partners, railed against *"colonial agendas"* prompting the visiting deputy head of state to suggest they could direct their funding elsewhere. Who then actually is undermining South Africa's attempts to restore energy security?

This raises another conflict of interest. Mantashe is clearly ill-equipped to resolve South Africa's power crisis. He is backward looking rather than embracing the energy transition. His position also as minister of mineral resources, plus his ideological necrophilia, makes him focus on Eskom coal power stations almost exclusively, rather than also how to mobilise the R1 trillion that Eskom itself estimates is necessary in new generation capacity by 2030.

But this is not ultimately a question about personalities; it's about structural reforms. If the oversight of Eskom is transferred to the DMRE, or even a separate energy ministry, the restructuring and unbundling of the utility will falter, inadequate transmission investments will be made to connect private generators, plans for a power exchange and market will be abandoned, South Africa will not have enough new power generation capacity to meet demand, and power cuts will persist, crippling economic development. Let's hope the President makes the right decision.

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MINISTERS HAVE LET RAMAPHOSA DOWN ON UNBUNDLING ESKOM

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Inadequate electricity transmission grids, rather than the availability of finance, is the main impediment now for new power generation investments and South Africa's attempts to end load-shedding. Transmission constraints to connect new wind and solar generation are ultimately a consequence of not yet unbundling an independent transmission system and market operator from Eskom.

The core diagnosis of South Africa's power crisis is captured in a single phrase: new power generation is not being added to the grid at a fast enough rate to compensate for the worsening failure of Eskom's coal plant. Of course, Eskom should do everything it can to improve the availability of its power stations. Equally, we need massive investments in new power generation, to create the space for adequate maintenance, but also to replace ageing power stations that have reached the end of their design life. Eskom estimates more than R1 trillion is needed by 2030.

Government's main strategy for adding new power generation capacity has been procurements by the Independent Power Producers Office. The first four auctions, run between 2011 and 2014, resulted in around 100 renewable energy projects, totaling more than 6400MW and R200 billion in private investment. But then Eskom refused to sign power purchase agreements, with an hiatus of seven years before the fifth bid window was finally opened in 2021.

The fifth auction revealed that transmission grid capacity in the Northern Cape, which has the best solar resource, had been fully utilized. The sixth auction, in 2022, demonstrated that there is insufficient grid capacity in the Western Cape and Eastern Cape, which also have the best wind resources, to connect planned power. Of the 5200MW on offer, only 860MW could be allocated – 5 projects out of 56 that bid.

South Africa's high voltage grid was designed to transmit power from the coal fields in Mpumalanga and the North West to the rest of the country. Now, the cheapest power resources – solar and wind energy – are in the Northern, Western and Eastern Capes. The grid needs to be reconfigured and requires an investment of at least R180 billion.

But the transmission grid is currently part of, and is controlled by, Eskom, a vertically-integrated utility, which includes generation, transmission and distribution. Investments have been skewed mostly to power generation. Transmission comprises only around ten percent of Eskom's assets. Despite excellent transmission grid plans, timely and adequate investments have not been made.

A different scenario would have materialized if a separate company was responsible for transmission, fulfilling a mandate to provide a backbone for all power producers and users. Transmission is a well understood business that can produce a predictable, regulated revenue. Investors consider it a quasi-bond, an attractive, relatively low-risk investment. An independent transmission company would achieve progressively higher credit ratings and would be able to access competitively priced capital, thus funding its investment plans.

In December 2018, President Ramaphosa appointed me to chair his Eskom Sustainability Task Team with a remit to respond not only to immediate challenges but also to propose measures to secure the long-term sustainability of the electricity sector. Our recommendations on structural and regulatory reforms, accelerated private investment, and the energy transition, were accepted by the president. We resurrected the proposals of the 1998 Energy Policy to unbundle Eskom. In the president's February 2019 State of

the Nation Address he announced that Eskom would be split into three separate entities: generation, transmission and distribution, with the initial focus on taking transmission out of Eskom.

Public Enterprises Minister Pravin Gordhan, as the shareholder of Eskom, was given the responsibility for driving this restructuring. Four years later, inadequate progress has been made. Transmission is still a division of Eskom. It does not have its own Board. It does not have its own licence from NERSA. What has gone wrong?

Our task team proposed that a restructuring czar be appointed, supported by a technical and financial team, to drive the restructuring. Gordhan subverted this proposal by appointing a temporary Chief Restructuring Officer whose mandate was restricted to proposals on restructuring Eskom's debt. His report has never been published, let alone acted on, and Eskom remains technically insolvent.

The President initially invested a great deal of political capital in persuading social partners that unbundling of Eskom made economic sense. Yet Gordhan has always been less than enthusiastic, preferring a band-aid to Eskom's ill-health rather than radical surgery, his ideological worldview clouding a clear appreciation of the benefits of structural reforms.

He has been reluctant to appoint a separate board for the transmission company, even though Eskom ran a highly professional and thorough advertising, head-hunting and interview process, and recommended experienced and respected candidates, some of them ex-CEOs of international transmission companies. A separate board would at least have provided leadership in driving the separation of transmission, but independent agency is not something Gordhan has ever been comfortable with.

Inaction has consequences. Here we are, four years later, actually 25 years later, if we consider the 1998 Energy White Paper, and Eskom is in meltdown. It remains a vertically-integrated, dominant state-owned power utility, an anomaly in a world where 110 countries have unbundled transmission.

Structural reforms may seem arcane or theoretical to some, and not the immediate challenge. But if implemented they can result in profound and lasting changes in incentives and performance. There is no more important reform in South Africa's power sector than taking transmission, system and market operation out of Eskom. It will create a financially viable company, insulated from the financial woes of Eskom generation, able to access competitively priced capital to invest in required transmission capacity. It will guarantee non-discriminatory access to the grid, and provide a fair and transparent platform for electricity trading.

President Ramaphosa quickly understood and accepted the need for these reforms. He has been let down by his ministers. If he doesn't now appoint new, young, competent and reform minded ministers, who are far-sighted and brave enough to restructure Eskom, South Africans will remain in the dark ages, literally.

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“South Africa can end load-shedding, not by costly hubris around a coal revival, but by embracing a clean energy transition that incorporates an increasing share of low-cost, quickly mobilised solar and wind energy, enabled by transmission access, and backed by flexible power.”

PLANS TO CLOSE OLD COAL POWER STATIONS ARE CRAZY, RIGHT? NOT SO FAST

Anton Eberhard

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For many, rhetorical questions may seem to have obvious answers. Nevertheless, an implied answer may be misleading or premised on inadequate information and analysis. South Africa's new electricity minister, Dr Kgosientsho Ramokgoba, has questioned why there are plans to close old coal power stations whilst the country experiences power cuts. The point he makes – that it's ill-advised – seems obvious. It is not.

After visiting Eskom's 14 remaining coal power stations, and recognizing that the utility is technically insolvent, Ramokgoba announced that he would recommend to Cabinet that National Treasury should raise more sovereign debt to fund the refurbishment of old kit. In an extended media briefing, which at times seemed like an *'Eskom for Beginners'* lecture, he pointed out that Komati had been decommissioned, along with three power units at Grootvlei and four at Hendrina, and that, together with Camden, Arnot and Tutuka, these power stations are all due for complete closure by 2030.

Ramokgoba says his proposal is justified because South Africa's economy would be saved from the cost of loadshedding, and subsequent economic growth would deliver higher tax revenues, which would offset the initial fiscal transfers to Eskom. Obvious, right? Not so fast. There are a lot of assumptions and chinks in his argument.

Ramokgoba provides no budget estimates for his proposal. How much would it cost to refurbish Eskom coal power stations that have exceeded their design life of 50 years? Are alternative new generation options not cheaper? Are the original equipment manufacturers, from the previous century, still able to provide all the replacement parts? How long will it take? Are the requisite skills still available? Where's the space, in an already stressed electricity system, plagued by near constant power cuts, to take additional units offline for multi-month refurbishments?

National Treasury has provided R220 billion in Eskom bailouts since 2008 and recently committed a further R254 billion of conditional debt relief over the next four years to deleverage the utility's balance sheet, with prohibitions on incurring new debt. Ramokgoba will not get the hundreds of billions more he wants for old Eskom coal power stations. Treasury will wait for a more considered assessment by the independent team appointed to review Eskom's operations.

Recent times have seen a worrying trend in the governing party - the emergence of superficial, apparent answers to enduring power sector problems that have already been analysed within government, with solutions formulated and agreed, but never fully implemented. The proposal from the ANC's December Conference to transfer Eskom to the Department of Mineral Resources and Energy comes to mind, a proposal that completely ignored the conclusions of two presidential commissions on state-owned enterprises, which pointed out the inherent conflicts of interest of a single ministry being responsible for a dominant, state-owned utility, whilst also having a policy and regulatory mandate for encouraging private sector investment and competition in the sector.

And then there's the fiasco of the short-lived State of Disaster, advocated because the party needed to be seen to be acting urgently to resolve the power crisis but which ignored previous advice to the Presidency that promulgating a power emergency would face legal challenges. Notable, in Ramokgoba's recent media briefing were his protestations that emergency regulations were not needed to end load-shedding. I agree with him but I suspect that he does not yet appreciate the perverse irony that they indeed may be necessary for his proposals to keep old coal plants running.

None of Eskom's power stations meet the minimum emission standards of South Africa's Air Quality Act and temporary postponements expire finally in two years. Compliance will require hundreds of billions of rands and many years to accomplish. Operating old, dirty coal power stations beyond 2025 will require either amendments to the Act or its temporary circumvention through another attempt to gazette emergency regulations.

So, what's the alternative? How can South Africa end load-shedding? To his credit, Ramokgoba acknowledges that the mandate given him by President Ramaphosa to implement the sensible Energy Action Plan of the National Electricity Crisis Committee includes investment in new power generation and the restructuring of Eskom to create a sustainable electricity supply industry.

Ramokgoba's initial focus is fixing Eskom but he will soon appreciate its difficulty. The fleeting improvements in power station performance, which coincidentally followed his Eskom visits, have evaporated and power cuts are back with a vengeance.

Eskom power stations have been failing at a rate faster than new generation is being added. Eskom has estimated that more than a trillion rand needs to be invested in up to 60 gigawatts of renewable energy, storage and gas power by 2030. Government has yet to demonstrate its appreciation of the scale and urgency of this investment challenge.

South Africa can end load-shedding, not by costly hubris around a coal revival, but by embracing a clean energy transition that incorporates an increasing share of low-cost, quickly mobilised solar and wind energy, enabled by transmission access, and backed by flexible power.

Ramokgoba posed another rhetorical question: can renewable energy reduce loadshedding when solar cannot contribute to early morning and late evening electricity demand peaks? No, he implied but, once again, what might seem like an obvious answer is simply wrong. South Africa experiences load-shedding all the time. Any new energy is helpful, especially during the day when most businesses operate. And, as any electricity system operator will confirm, energy security is achieved by balancing and optimizing the system as a whole. More solar and wind on the grid provides additional energy to replenish pumped storage and diesel, which enables peak electricity demand to be met.

The inexcusable delay in unbundling an independent transmission, system and market operator from Eskom has constrained investment in transmission, which is now the most serious obstacle to connecting new generation capacity. Postponing the planned decommissioning of old, expensive, dirty coal power stations will also torpedo South Africa's best opportunity for financing the expansion of the grid – the US\$ 8.5 billion Just Energy Transition Investment Plan.

Let's hope that our new electricity minister asks a new, better informed, rhetorical question: can we end load-shedding without urgently restructuring the power sector and mobilising massive new investments? This time the answer is indeed obvious.

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ENERGY SECURITY : OVERCOMING THE ENERGY CRISIS IN THE COUNTRY

Anton Eberhard

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“Minister Mantashe has struggled with all of these myths and continues to argue that Eskom has idle capacity, solar and wind cannot make a difference and are expensive. The consequence of these false beliefs has meant that not a single kWh of electricity has yet been added to the grid commercially from any project that he has procured in the more than four years he has been energy minister.....This is a devastating indictment, as the economy reels from loadshedding



Energy Security clearly has to be South Africa's priority, given the enormous cost of loadshedding to the economy and to welfare.

That doesn't mean that the other two dimension of the energy trilemma – namely affordable energy and environmental sustainability – are irrelevant. As shall become clear, our future energy path is one where all three policy objectives – clean energy, competitive prices and security can be achieved because the least-cost energy path is also one which involves climate-friendly technologies and where an adequate and reliable supply can be achieved.

The contrast of our current situation to these laudable goals and outcomes could not be starker. We have zero energy security with almost daily power cuts. Electricity prices have increased nearly 600% in nominal terms over the past 15 years. And we have the most carbon-intensive power sector in the world (expressed as carbon dioxide emitted per kWh produced), which will likely begin to threaten the competitiveness of our exports through the imposition of the carbon border adjustment and other mechanisms.

But back to electricity security. What are the prospects of load-shedding or power cuts ending soon? In answering this question, I first want to deal with a number of myths or false narratives; second, I'll assess current initiatives; and then third, I'll refer to the energy transition and how it can help us restore electricity security.

The first myth or false hope is that Eskom sits with 15,000 MW of idle generation capacity that is not being used and that all that is needed is to bring this back online – as energy Minister Mantashe has said on a number of occasions.

Of course, that statement ignores that that this kit is broken, and continues to break, and that for more than 12 years now, successive Eskom management teams, supported by their technical staff, have tried their best to reduce these unplanned capacity loss factors, but the energy availability factor (EAF) of

Eskom power stations has continued to decline, from above 90% in the early 2000s to below 60% today, and at times even below 50%.

Comments on idle capacity, and even the perennial expressions of hope that Eskom coal power station performance can or will soon improve significantly, are irresponsible, fact free, and fly in the face of actual performance data over time, and mostly do not engage in any useful detail on what Eskom should be doing, that has not already been tried, to reverse the situation.

A fundamental step in solving South Africa's loadshedding crisis is a recognition and acceptance that it is very difficult, and very expensive, to reverse Eskom's long-term decline in the electricity available from its coal fleet. Eskom does not have the reserve margin or space to take sufficient plant offline to fully catch up on maintenance, especially for lengthy refurbishments. It does not have sufficient free cash flows to fully fund the required maintenance. And it has lost far too much of the experience and skills to undertake effective, comprehensive maintenance of the scale and quality required. Most commentators on Eskom completely underestimate the complexity and gravity of these challenges.

Yes – there have been some recent gains. There is a greater openness to accepting private sector advice. Some experienced grey beards have been brought back, although not nearly enough. National Treasury's debt relief package has meant that Eskom now spends less repaying principal and interest on its loans and more cash is potentially available for maintenance. Procurement processes for parts and engaging Original Equipment Manufacturers have been streamlined. Incidents of sabotage have reduced.

Six power stations are being focused on and have been targeted for major improvements. But weren't they also in the past? Much is being made of the fact that newly appointed generation and power station managers are making a difference. But overall, Eskom governance is in disarray with

the recent resignations of the Chair of the Board and no permanent Chief Executive. Indeed, Eskom is being run by its CFO, who has indicated that he wants to leave.

Electricity Minister Ramokgoba has made a real contribution to increasing the transparency and regularity of information flows on what is being done to reduce Eskom loadshedding. Politically, he is under immense pressure to show results and often that unfortunately leads to optimistic interpretations of the data. Yes – there will be times when things improve but data on the long-term deterioration in Eskom performance is clear. I can guarantee you that there will be more plant breakages and that there will once again be periods of severe loadshedding in the short to medium term.

The second myth is that renewable energy cannot make a difference to load-shedding. The sun doesn't always shine, or the wind doesn't always blow, so how can they make a difference? Again, this is a false narrative that is fact free. Eskom's own daily data releases show that wind energy alone frequently contributes more than 2000MW, importantly also helping to meet peak demand. In other words, if those wind farms were not there, we would be having up to two stages of worse loadshedding at times. And similarly with solar.

But here is some solid science that shows the potential impact of solar and wind energy in reducing loadshedding. As an exercise, hourly Eskom data over a year was taken for electricity demand and supply and loadshedding, and the impact was modeled of adding 5000MW extra solar and wind energy, producing in exactly the same profile as existing solar and wind plant. So, these are not theoretical, ivory tower academic assumptions. The analysis is based on actual Eskom data. The results are truly extraordinary. If we had done this - i.e. added an extra 5000MW - in 2021 there would have been 96% less load shedding, and last year 70% less.

How is this possible, you might say. Solar and wind are intermittent, variable. Well, the answer is twofold. First, there

is the direct effect: in the context of ongoing loadshedding, any new electrons on the system make a difference and reduce loadshedding. But, second, there is an important indirect effect. Extra energy on the system creates the space for replenishing diesel and pumped storage. Much of our current loadshedding is when these resources have been overused and when Eskom does not have enough energy or space to replenish storage.

I'm using the examples of solar and wind energy because it has now been proved, in the auctions run by the IPP Office, that they are the cheapest new sources of energy - around 50c/kWh compared to Eskom's average cost of supply which now exceeds R1.40 c/kWh.

And this highlights the third false narrative, namely that renewables are expensive. Yes – they were, a decade ago, but no longer. The costs of solar PV fell by 90% and for wind energy 70%. We're interested in the future, i.e. the cost of the next kWh to be added to the grid.

Minister Mantashe has struggled with all of these myths and continues to argue that Eskom has idle capacity, solar and wind cannot make a difference and are expensive. The consequence of these false beliefs has meant that not a single kWh of electricity has yet been added to the grid commercially from any project that he has procured in the more than four years he has been energy minister, despite him having extraordinary powers under the Electricity Regulation Act to determine how much electricity, from which sources, should be procured by whom and by when. This is a devastating indictment, as the economy reels from loadshedding.

And so, we cannot restore energy or electricity security purely by thinking we can fix Eskom. We need also to add new power generation.

There is a fundamental conceptual issue here which has huge strategic and investment implications. Eskom has been

failing at a rate faster than we have added new capacity. That departure point has to be accepted in order to solve load-shedding.

We might wish it were different. We may wish to hold on to the belief that Eskom can be a key instrument in the developmental state when patently it is not – it's crippling the economy and draining the fiscus. But we have to deal with reality and act on that, including acceptance that more and more Eskom power stations will begin exceeding their design life of 50 years – indeed Eskom envisaged that 11GW would need to be retired by 2030 and a total of 22GW by 2035. That programme may be slowed, but the fact is that South Africa will continue to lose Eskom generation capacity.

And Eskom itself estimates we need 50-60GW of new solar and wind energy by 2030, complemented by storage, gas and new grids, amounting to investments totaling R1.2 trillion

Electricity Minister Ramokgoba has bravely acknowledged these facts and was reported this week as saying: “ An overreliance on these [Eskom coal] units I think is particularly unhelpful; it's the height of folly to suggest that, going into the future, this is going to resolve our problems . . . and that's why we must accelerate the onboarding of new generation”.

So what can we say around government's response to our power crisis. Clearly, it has not done enough. We had loadshedding in 2007 & 8, 2014 & 15 and every year since 2018, and every year it has got worse.

Nevertheless, the President's Energy Action Plan, announced in July last year, is broadly well conceived and framed. It comprises three broad goals and desired outcomes: first, every effort needs to be made to at least stabilise Eskom (we rely still for 85% of electricity from the utility); second, investment in new private generation needs to be accelerated; and third, the power sector needs to be restructured sustainably.

The National Electricity Crisis Committee, supported by Business, is making steady progress in these three areas. I'm not going to report on the detailed work that is being done but let me make an overall assessment.

While the interventions at Eskom are positive, I think there is still not sufficient appreciation of the complexity of turning Eskom around and rebuilding the knowledge, skills, systems, processes, practices and budgets to make a step change in the performance of old, long-neglected plant.

And we are moving far too slowly on new power generation. Again, I don't believe there is sufficient appreciation of the scale of what is required: more than one trillion rand, 60GW by 2030. We are way off those targets.

While we are seeing encouraging signs of private investment in generation – for example Eskom estimates close to 5GW private solar generators have recently been added (these are mainly rooftop and non-REIPPP investments) there are still major obstacles in connecting new power, not least transmission constraints.

And, we have to admit now, the IPP Office, under the DMRE, is a failure. A hard thing to say, I know, but let's look at the facts. Between 2011-14 the IPP Office ran four auctions that delivered 6300MW of solar and wind projects that are now all delivering electrons to the grid. Then Messrs Molefe and Koko, with the support of President Zuma, blocked the programme, the contract of the well regarded and internationally lauded head of the IPP Office was terminated, and frankly we now have a team there that have failed to deliver.

The so-called emergency procurement has so far delivered a measly 150MW (a solar and battery installation that is currently being commissioned). Bid windows 5 and 6 have not yet delivered a single electron. Bid window 7 seems to be disarray. The battery and gas bid windows keep on being delayed.

Within the context of severe loadshedding, and its cost to the economy, this is completely unacceptable. In addition to private sector initiatives, we need to keep these public procurement initiatives going. International experience demonstrates that centrally run auctions deliver big megawatts at competitive prices.

South Africa has the capabilities to do this well. The world applauded our Bid windows 1, 2, 3 and 4. We need to bring those skills and capabilities back into the IPP Office, which needs to be removed from the DMRE and institutionally located in the new Central Purchasing Agency of the Transmission System Operator (TSO).

Which brings me to Eskom restructuring. Again, progress has been far, far too slow. The Energy Policy White Paper of 1998 said we would separate out transmission from Eskom. When I chaired President Ramaphosa's Eskom Sustainability Task Team in late 2018 and early 2019, we resurrected these proposals. The president accepted our recommendations and announced in Parliament that Eskom would be split into three, starting with Transmission.

Four years later, Minister Gordhan as not yet fully implemented this decision. Now there is a mad scramble to get the Electricity Regulation Act Amendment Bill through this parliament, which will enable the establishment of the TSO.

These restructuring proposals may seem arcane and not terribly important, but they are fundamental. A Transmission System Operator, separate from Eskom, removes Eskom's conflict of interest as both a generator and also the owner of the monopoly transmission grid. It creates a fair and transparent platform for contracting and trading with multiple generators, thus accelerating private investment and competition.

And, critically, an independent TSO will be insulated from Eskom's financial woes. It will generate predictable, regulated

revenues, start migrating back to investment grade, thus accessing competitively priced capital markets, and investing in strengthening and expanding the grid to connect new power generation.

This could have happened already, if the decisions of the President and Cabinet had been implemented. Even the simple step of appointing a board for Eskom's transmission subsidiary has not been done, despite excellent and knowledgeable candidates being identified in the search and interview process. Doing nothing has consequences.

During the parliamentary consultation on the ERA Amendment Bill, concerns have been expressed that the creation of the TSO implies privatisation. We need to explain that categorically it does not. Privatisation means the sale of state assets. The TSO will remain publicly owned and will be a state-owned enterprise.

Let me end with the reminder that we're in the midst of a radical energy transition. I often say to my students that I have seen more change and innovation in the power sector in past five years than in the previous 35 years of my professional career. Disruptive innovation in enabling technologies and business models is having a major impact on power markets, system operation and regulation.

These innovations include the break-through of unsubsidized solar and wind which are now the cheapest sources of new power. As their share on the grid grows, they need to be complemented by flexible resources such as hydro, pumped storage (operated in innovative ways), batteries and also gas.

Balancing and capacity markets will need refining, including the competitive procurement of flexible ancillary services to maintain system voltage, frequency, strength and inertia. The capabilities of system operators, including at the regional level, will need to be enhanced.

Demand-side resources and virtual power stations will be critical parts of the system.

The distributed nature of renewable energy means distributed energy resources will multiply with implications for transmission and distribution planning, business models, and distribution system operation. City utilities will once again become key elements of the power system.

More electricity consumers will become producers as well (i.e. prosumers). As power flows both ways, advanced net metering and payment systems will provide greater spatial and temporal granularity. The internet of things, artificial intelligence, big data, and blockchain, mobile money and pay-as-you-go models will change the way in which electricity is controlled and paid for. Peer to peer trading between prosumers will begin, also across connected mini-grids. There will be more aggregators and traders in the market. Community-ownership models could arise alongside utilities.

All this means that we need informed and far-sighted leadership in the sector, especially in managing the energy transition which unfortunately is increasingly contested. Of course, some constituencies will be adversely impacted by the energy transition and we need to ensure that it is just – that workers and communities in the coal belt are supported through re-skilling and focused investments.

Again, I need to call out responsible Ministers. It is irresponsible to continually questioning the role of clean energy technologies and the just energy transition. And once again we need to challenge false narratives. It is now almost a mantra that Germany tells us to go green but has massively increased its imports of our coal. While it is true that the disruption of Russian gas supplies resulted in Germany temporarily using more coal, the data now clearly shows that European imports of South African coal have declined, as has electricity produced from coal. In the first 9 months

of 2023, coal contributed 26% of Germany's electricity, while renewable energy contributed 60%. Facts. Not myths.

Indeed, it is amazing that South Africa has placed so many obstacles in liberating the US\$ 8.5 billion just energy transition fund promised at COP26. I have spoken to senior people in these development finance institutions who have expressed frustration that despite being willing and ready to disburse these funds they are unable to get any traction or make any progress with the South African government.

We know that a large chunk of this money could be directed to strengthening and expanding our transmission system that would connect new power generation to alleviate load shedding. Yet, individuals choose to criticize these funding opportunities as a new kind of colonialism.

In conclusion, South Africa can end loadshedding and restore electricity supply security but it's going to take time. There may be short periods of respite but loadshedding will return as old Eskom generation plant continues to deteriorate and older plant exceed their design lives.

We need to massively scale up investment in power generation and restructure the power sector to facilitate private participation, investment and competition.

Ultimately, we need a forward-looking Energy Ministry. One that appreciates better its legislated mandate to restore energy security and one which understands and supports the inexorable energy transition that is sweeping the world. We need to be ahead of the game, not dragging our feet. And embracing the energy transition will help us to restore energy security earlier, rather than later.

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